Estimating Mean-Field Games with a Payoff Threshold

Candidates and Party Lists in Brazil

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Abstract

Estimating a discrete-choice game of incomplete information means evaluating the joint action profile, which forecloses the settings such games describe: a Brazilian municipality fielding fifty-five council candidates across five lists has more than 10^{38} profiles. We change the equilibrium concept rather than the algorithm. Anonymity makes the fixed point one dimension per list and none per candidate, and a discontinuous winner rule turns existence into a regime-consistency problem over proposed clearing sets, which we show can have no solution. Because a threshold pays several alternatives at once, two can be compared inside one market, and three log-odds contrasts identify the parameters with no probability of abstention anywhere in the estimator. At municipal scale the likelihood attenuates the coefficient on the anticipated chance of a seat, because a list's share is counted from a few dozen candidates, and a split-sample instrument on an unbiased reciprocal of that share recovers nine-tenths of it in simulation. On 926,900 Brazilian council candidacies the corrected coefficient is positive for non-incumbents and for all candidates, for every one of thirty-two demographic types, and with and without a control for local party strength; its magnitude under that control is not stable. Candidates with a university degree respond more to the chance of a seat than candidates without one, a difference found among non-incumbents and not among incumbents.

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1. Introduction

Half a million Brazilians stand for municipal council every four years, and each of them must first choose a party list. The choice is strategic. In the first round of the seat allocation a list takes one seat for each electoral quotient its votes reach, the quotient being the valid vote divided by the number of council seats. Votes are cast for candidates and pooled within their list, so the votes a candidate brings count toward the list's quotient; the seats the list wins then go to whichever of its candidates draw the most personal votes, so every candidate who joins is also one more claimant on them. Whether to stand with a large party or a small one is a decision about an opportunity that the deciding itself dilutes.

The object chosen is an electoral list, and it is not the same object in our two cycles. In 2016 several parties could register one joint list and pool their votes against the quotient, and four lists in five were such coalitions. A constitutional amendment abolished them, and in 2020 every list is a single party.

We study the choice as a discrete-choice game of incomplete information among the registered candidates of a municipality. A candidate's utility from a list has two parts. *List appeal* is what the party's name and organisation are worth to a candidate, and it may differ according to whether the list is paid. The *seat payoff* is the list's *seat opportunity*, its anticipated seats per candidate, multiplied by the candidate's utility weight on it, the *seat-payoff coefficient*; it is paid only if the list's share of the vote reaches the quotient's share, one over the number of seats. A list's vote share is its share of the candidates weighted by the votes each of its candidates draws, which we treat as an attribute of the list, so candidates' choices move a list across the threshold through its share of the candidates. Candidates affect one another only through those shares and never through who in particular stands where, which is the anonymous, or mean-field, formulation of Weintraub, Benkard, and Van Roy (2008) and Adlakha, Johari, and Weintraub (2015).

The model's threshold is a simplification of the law's. Seats left over after the quotient round are allocated by highest average, and from 2020 that round is open to lists below the quotient. The lists that took a seat and the lists that reached the quotient differ by 2 of 25,653 in 2016 and by 4,691 of 40,117 in 2020. The equilibrium results, and every fixed-point computation in the paper, are for the quotient rule; they describe that threshold and not the 2020 allocation (Section 6.1). The identification results are statements about log-odds given the branch each list is on, and their algebra does not use the rule that assigns lists to branches. What they ask of the data is that the recorded assignment be the one candidates' payoffs followed: in the model the set of paid lists is known to candidates when they choose, and the estimates take the published outcome of the election to be that set (Assumption 2). They define the paid lists as those that took a seat, which is what each cycle's law implemented, and report the quotient definition beside it (1.33 against 1.41 for the baseline coefficient).

The paper makes three contributions. It characterises equilibrium when a payoff is switched on by a threshold on the equilibrium share itself. It shows which comparisons of candidates' choices identify the seat payoff separately from list appeal, and that differences in responsiveness across types of candidate are identified under weaker assumptions than its level. And it takes the model to 926,900 candidacies in 11,129 municipality-cycles.

Findings. Candidates respond to seat opportunity, and the estimate of how much depends on correcting an error of measurement. The likelihood of Section 4 divides a list's anticipated seats by a share counted from the market's candidates, a few dozen of them in a median municipality, and in a Monte Carlo design at the market sizes of our estimation sample it returns three-fifths of a known coefficient with nominal 95% intervals that

cover it 13% of the time. We correct it in two steps: the reciprocal of the counted share is replaced by one that is unbiased, and the seat opportunity built from a random half of each market's candidates is instrumented by the one built from the other half. In the same design the corrected estimator returns 91% of the coefficient, its standard error matches the spread of its estimates, and the weights on candidate characteristics are unchanged (Section 5). Every estimate in the application is the corrected one.

With one coefficient for all types, the estimate is 1.33 among non-incumbent candidates, with a standard error of 0.08, and 1.92 among all candidates. It is 0.72 and 1.20 without the correction, ratios of 0.54 and 0.63 against the three-fifths the simulation predicts. The coefficient is positive on either definition of a paid list, in each cycle, and for every one of thirty-two demographic types (Section 6.4).

The level rests on an assumption the data do not let us relax cleanly: that a list's appeal, beyond an effect for its party within its state, is unrelated to its seat opportunity. Local party strength is the leading way that can fail. A design that separates the two through the size of the council leaves the coefficient positive in every sample, between 0.57 and 4.44, but does not move it in one direction: among non-incumbents on the two widest samples the control lowers it, to 0.57 on all markets and 1.00 on the markets of sixty candidates or more, and elsewhere it raises it. We report 1.33, give 1.00 and 0.57 as what a local attraction of the estimated size would leave of it, and claim the sign and not a bound on the confound (Section 6.5).

The seat-payoff coefficient is larger for candidates with a university degree than for candidates without one. This is a comparison between types, the object the model identifies under its weaker assumption. With a separate coefficient for each type and each type weighted equally, the mean coefficient of the sixteen types with a degree exceeds the mean of the sixteen without by 0.80 on the logit-utility scale, with a standard error of 0.13. The gap belongs to non-incumbent candidates: among their types it is 1.18 (0.06), and among incumbent types 0.41 (0.25). An estimator with a common effect for every list in every market, which absorbs list appeal whatever its local component and leaves only differences across types to be estimated, gives the same pattern and a larger gap, 2.60 (0.17) among non-incumbent types and 0.60 (0.61) among incumbent types; in placebo draws in which no type responds more than another it returns no gap. Local list appeal is therefore not the source of the education gap, and the two estimates bracket its size. The gender, incumbency and age patterns were sensitive to how seat opportunity is measured, and we report them without interpretation (Section 6.4).

Adding the seat payoff improves the model's fit to a cycle it was not estimated on, in a conditional exercise. We estimate the full model, and a benchmark without the seat payoff, on 2016 alone, including the first stage behind seat opportunity, and hold every parameter fixed; a 2020 list takes its party's 2016 effect. The exercise takes the 2020 lists, council sizes and 2016 vote shares as given, uses the votes per candidate realised in 2020, and finds each market's fixed point by iteration from the observed 2020 shares, so it is not a forecast available before the election, and where a market has several fixed points the starting point takes part in choosing among them. Measured as one minus the ratio of squared error to that of equal shares within each market, the fixed point of the full model reaches 0.290 against 0.267 for the benchmark, a gain of 0.023; in sample the gain is 0.056. The out-of-sample gain is the same when the parameters are estimated on the quotient definition of a paid list, the rule the fixed point uses, because the two definitions coincide in 2016. The result is a modestly better conditional fit under these inputs and this solution rule. It does not separate a response to seat opportunity from local party strength that persists between the cycles, because the seat payoff is the only channel through which a party's local standing enters the model (Section 6.8).

Equilibrium under a threshold. In the models of social interactions and of sorting with congestion that we build on, an alternative's payoff moves continuously with the share of agents choosing it. Brock and Durlauf (2001) put a logit choice into such a payoff, solved the fixed point and identified the parameters from within-group log-odds; Bayer and Timmins (2007) and Timmins and Murdock (2007) estimate logit models in which an alternative becomes less attractive as more agents choose it. Our model has the same choice structure and the same congestion: a clearing list's seat opportunity is shared among its candidates. What it adds is the threshold, which makes the payoff discontinuous in the equilibrium share.

The discontinuity changes the equilibrium problem in three steps. Given a proposed set of clearing lists, the game has a continuous conditional counterpart, and that counterpart has a fixed point. The fixed point is an equilibrium only if the proposal is *self-fulfilling*: every list it names clears at the fixed point and no other list does. And a self-fulfilling set may fail to exist, or there may be several. We exhibit primitives, robust to perturbation, at which none exists, so existence is a property to be checked and not a consequence of the assumptions (Section 2.4, Section A). These are results about the mean-field equilibrium, which approximates the finite-player game under a separation condition (Section B). In the 9,125 markets with two to eight lists, checking all 2^J proposed sets at the estimated parameters finds at least one self-fulfilling set in every market and more than one in many of the larger ones. That is a count of clearing sets and not of equilibria, since a set may have more than one fixed point (Section 6.7).

Identification. The payoff has two branches, and three comparisons of one type's choices across a pair of lists in the same market use them. Under the logit structure, with list appeal carried by party effects, the log-odds between two clearing lists depend on the difference in their seat opportunities and identify the seat-payoff coefficient; the log-odds between two non-clearing lists identify the relative appeal of the two parties; and the log-odds between a clearing and a non-clearing list identify the weights on candidate characteristics, which enter the clearing branch only (Section 3). The comparison between two clearing lists exists because the quotient rule pays several lists in one market and pays them different amounts: at least two lists clear in 99% of the 11,129 municipality-cycles in our data, on either definition of a paid list (Table 5). A rule that pays one alternative per market offers no comparison between two paid alternatives with different prizes. Non-clearing lists serve as the observed reference, so no outside option has to be constructed.

List appeal is correlated with a list's share by construction, since an attractive list attracts candidates. Differencing a second time, across two types on the same pair of lists, removes list appeal under one assumption: that whatever a list is worth beyond its seats, it is worth the same to every type. Given that assumption, the logit structure and two clearing lists with different seat opportunities on which both types stand, the difference between the two types' coefficients is identified with no restriction on the level of list appeal, on its distribution, or on its correlation with the equilibrium share. The level of the coefficient needs more: list appeal must split into a party effect the data can absorb and a local residual unrelated to seat opportunity. Local party strength is local appeal the party effect does not absorb, and it is related to seat opportunity, so it is the leading way that requirement can fail (Section 3.5).

The result is pairwise. The education gradient is an equal-weighted contrast of thirty-two type coefficients, and it inherits the result only where each type stands on clearing lists with different seat opportunities alongside the others; in the estimation sample all thirty-one differences from a reference type are estimable. The estimate

of the gradient with an effect for every list in every market rests on the double difference. Every level we report comes from a likelihood with party-by-state effects, which leaves the local component of list appeal in the error.

Scale. Anonymity does two different things for computation. Estimation does not solve for equilibrium: it conditions on the observed shares and the observed paid lists, so it is a conditional logit on counts of candidates by market, type and list, whose cost is in the number of such cells and not in the number of candidates (Section 4). Prediction and counterfactuals do solve for equilibrium, and there anonymity makes the unknown a vector of list shares. A median municipality fields fifty-five candidates across five lists; the fixed point for a given clearing set has five dimensions, where the joint action space has more than 10^{38} elements. The remaining cost is the number of proposed clearing sets, 2^J without pruning. Full-solution and pseudo-likelihood estimators for games (Su and Judd 2012; Su 2014; Aguirregabiria and Mira 2007; Pesendorfer and Schmidt-Dengler 2008) and methods that set-identify games with multiple equilibria or weaken informational assumptions (Tamer 2003; Ciliberto and Tamer 2009; de Paula and Tang 2012; Magnolfi and Roncoroni 2023) are formulated on the players' joint action profile, and their applications to entry involve a handful of players per market (Berry 1992; Vitorino 2012).

Related work. The static large-game approximation is Bodoh-Creed (2013); it requires payoffs continuous in the aggregate, which is why Section B proves the finite-player reading directly.[1] The aggregative-games perspective is Jensen (2010) and Acemoglu and Jensen (2013). Inference for large Bayesian games is developed by Kojevnikov and Song (2023), and Menzel (2016) shows that in anonymous many-player games estimation can condition on the realised equilibrium without a selection assumption, which is what we do. Self-fulfilling clearing sets stand beside the viable-candidate-set equilibria of Cox and Shugart (1996), Myerson and Weber (1993), Cox (1994) and Fey (1997). There, voters coordinate on which candidates or lists are viable; here, candidates sort across lists, and whether a list clears is the outcome of their sorting. Brazilian open-list proportional representation and its consequences for party fragmentation are treated by Mainwaring (1991), Nicolau (2007) and Calvo, Guarnieri, and Limongi (2015), and candidates' career incentives by Samuels (2003).

Scope. Two boundaries shape how the counterfactual and the estimates can be used. Equilibrium selection is unresolved: stability, history and an extremal rule each fail to select among self-fulfilling clearing sets, so the counterfactual of Section 6.9 compares two fixed points, the baseline and the counterfactual, each reached by the same iteration started at the market's observed shares, and is local in that sense. And entry, screening and the supply of lists are taken as given. Parties screen applicants and the registration cap binds (Section 6.6), so the model explains how a given pool of candidates sorts across a given set of lists and does not predict who stands or how parties would change their lists. That candidates choose individually, and are not placed in blocs, is a maintained assumption (Section 2.5).

Section 2 sets up the game, Section 3 gives the identification results, Section 4 the estimator and Section 5 the Monte Carlo, and Section 6 reports the application. A further test, on the 2024 cycle, in which the rule that determines which lists are paid changed, is specified in advance in Section G, in the spirit of Pathak and Shi (2021); this paper does not report its outcome. The appendices contain the equilibrium results, a worked three-list example (Section E) and supporting analyses for the application (Section F).

[1] The same move — replacing a combinatorial object by a limit in which only aggregates matter — carries large matching markets (Menzel 2015; Galichon and Salanié 2022) and network formation (Mele 2017).

2. The Model

Candidates choose simultaneously which party list to stand on. A candidate's payoff depends on the others' choices only through the list shares, the share of the market's candidates standing on each list. Each candidate treats those shares as deterministic and unaffected by their own choice. The model is in this sense a *mean-field* model (Weintraub, Benkard, and Van Roy 2008; Adlakha, Johari, and Weintraub 2015): equilibrium is a fixed point in the list shares rather than in a profile of individual strategies. The finite-candidate Bayesian game, in which a candidate does move the list they join and faces a random composition, is a separate object, and Section 2.6 states when the mean-field model approximates it.

2.1. Markets

The unit of equilibrium is the *market*: a municipality in an electoral cycle, indexed $m = 1, \ldots, M$. Markets do not interact. Each has its own set of lists $\mathcal{J}_m = \{1, \ldots, J_m\}$, its own council of C_m seats, and its own electoral quotient. A candidate's action is a list in $\mathcal{J}_m$.

No outside option. The model conditions on registered candidacies, and whether to run is not modelled. The conditional-logit estimator of Section 4.1 has the same choice set, and Corollary 1 states what is identified without an outside option. Counterfactuals hold the candidate pool fixed: when Section 6.9 enlarges the council, nobody new enters.

Parameters are shared across markets and the choice problem is not. Types are defined identically in every market, so a type's coefficients are common to all of them, and a party's effect is common to the markets that share its label; the set of lists and their seat opportunities differ from market to market. Section 3 says which of that variation identifies which parameter.

Timing and information. Within a market, the quotient q_m, the council size C_m and the council's rules are fixed first. Each candidate then chooses a list, simultaneously with every other candidate; the resulting shares determine which lists clear the quotient; payoffs are realised. The public primitives of the market are common knowledge among its candidates: the quotient and the council size, the type distribution, and for every list its prize $B_{m,j}$, its vote productivity $\rho_{m,j}$ and its appeal $\xi_{m,j}$ and $\delta_{m,j}$, all defined below. A candidate's own type and preference shock are private, and other candidates' types are known only through the type distribution.

What the researcher measures. List appeal is known to candidates and unobserved by us. The prize is an anticipated object, which Section 6.3 constructs from information that predates registration. Vote productivity is measured by its realised value, a proxy for what candidates anticipated at registration; Section 6.8 reports what the fixed point does when the realised value is replaced by one built from the previous cycle.

2.2. Players, types and actions

Market m contains I_m agents. Each has a type $\tau \in \mathcal{T}$, where $\mathcal{T}$ is finite with $|\mathcal{T}| = K$ and is the *same set in every market*; the type vector $X(\tau) \in \mathbb{R}^{n_X}$ collects observable characteristics. Empirically a type is a combination of

gender, education, age band and incumbency. The type distribution $\mu_m(\tau)$ is observable and sums to one within each market.

Each agent draws an idiosyncratic preference shock $\boldsymbol{\varepsilon}_i = (\varepsilon_i(j))_{j \in \mathcal{J}_m}$, privately observed, with components iid Type I extreme value with unit scale and independent across agents. The unit scale fixes the scale of utility; its location is fixed by a convention given below.

2.3. Shares, the quotient, and payoffs

Let $s_m = (s_{m,1}, \ldots, s_{m,J_m}) \in \Delta^{J_m - 1}$ be the composition of market m, where

$$s_{m,j} \;=\; \sum_{\tau \in \mathcal{T}} \mu_m(\tau)\, \sigma_m(j \mid \tau) \tag{1}$$

and $\sigma_m(j \mid \tau)$ is the probability that a type-τ agent in market m stands on list j. The vector s_m is the only feature of the action profile that enters payoffs, which is the anonymity of Jensen (2010) and Acemoglu and Jensen (2013).

The share $s_{m,j}$ is a population object, generated by choice probabilities. The realised share is

$$\hat{s}_{m,j} \;=\; n_{m,j}/I_m, \tag{2}$$

with $n_{m,j}$ the number of market m's I_m candidates who stand on list j. Equilibrium below is a fixed point in s_m. The estimator of Section 4 makes two substitutions, and they are different in kind. It puts the realised share $\hat{s}_m$ in place of s_m, which is a finite-sample approximation whose cost Section 5 measures. And it puts the observed seat-winning set, the lists that took at least one seat, in place of the clearing set defined next; the two sets differ where a list takes a seat without reaching the quotient, which is a difference between the model's rule and the law's and not sampling error (Section 6.1).

A list becomes eligible for the model's seat prize when its share of the *vote* reaches the electoral quotient. Candidates are not interchangeable across lists in the eyes of voters, and lists differ in how many votes each of their candidates draws: write $\rho_{m,j} > 0$ for the votes per candidate on list j. With q_m the quotient expressed as a share of the vote, the *clearing set* is

$$W(s_m) \;=\; \left\{ j \in \mathcal{J}_m : \frac{\rho_{m,j}\, s_{m,j}}{\sum_{k \in \mathcal{J}_m} \rho_{m,k}\, s_{m,k}} \;\geq\; q_m \right\}. \tag{3}$$

A list's vote share is its candidate share only when every list in the market draws votes per candidate identically, $\rho_{m,j} \equiv \rho$, in which case (3) reduces to $s_{m,j} \geq q_m$.[2] The primitives do not require that homogeneity, and Section 6.7 reports what imposing it costs in self-fulfilment, so vote productivity is allowed to vary across lists.

Unlike an arg max rule the clearing set is typically not a singleton, and Section 6.1 reports that several lists clear in almost every municipality. That plurality is what makes the identification of Section 3 possible, and it is the substantive difference from a winner-take-all contest.

[2]The reduction also uses the absence of an outside option, under which candidate shares sum to one. With homogeneous vote productivity and an outside option of share $s_{m,0}$ the condition would be $s_{m,j} \geq q_m(1 - s_{m,0})$.

Payoffs are

$$\pi(j, s_m, \tau) = \begin{cases} X(\tau)'\beta + \alpha(\tau)\,\dfrac{B_{m,j}}{s_{m,j}} + \xi_{m,j} & j \in W(s_m), \\ \delta_{m,j} & j \notin W(s_m). \end{cases} \tag{4}$$

Here $\alpha(\tau)$ is the type-specific marginal utility of the prize and β are common weights on observable characteristics. The term $X(\tau)'\beta$ is a branch-specific utility shifter: it captures how candidate characteristics alter the value of standing on a clearing rather than a non-clearing list, and it is common across the clearing lists of a market.

$B_{m,j}$ is list j's *anticipated* electoral prize, normalised by the number of registered candidates I_m in market m: it is built from information available before registration and stands for the seat opportunity a candidate expects the list to control if it clears the quotient. The ratio $B_{m,j}/s_{m,j}$ is therefore proportional to the anticipated prize per candidate on list j. It is a congestion index rather than an individual election probability, because open-list seats are allocated by personal votes rather than equally across list members. Section 6.3 constructs the index, reports how unequally personal votes are distributed within a clearing list — the candidates who take its seats are 16.7% of its candidates and hold 44.5% of its votes — and states what the approximation costs. A candidate-level prize would require modelling personal votes and is beyond the scope of this paper.

The remaining terms are what a list is worth beyond its seats. A party's name, its local organisation, whatever else makes one list more attractive than another at an equal prize: $\xi_{m,j}$ where the list clears and $\delta_{m,j}$ where it does not, each decomposing as

$$\xi_{m,j} = \xi_{p(j)} + \nu_{m,j}, \qquad \delta_{m,j} = \delta_{p(j)} + \nu'_{m,j},$$

into a party component shared by every market that carries the label $p(j)$ and a market-specific component. The label is a party within a state in the estimates of Section 6.4 and a party in the fixed-point exercises, which need an effect for every list in every market. A coalition list has several parties, and its party term is the average of its members' effects weighted by each member's share of the list's registered candidacies, $\xi_{p(j)} = \sum_p w_{j,p}\,\xi_p$ and likewise for δ; the weights are the registered composition, a primitive of the market that is held fixed in every fixed-point computation. Neither term carries τ: a list's appeal enters every type's payoff by the same amount, which is what Proposition 3 uses.

The two branches carry separate party effects because party appeal need not be worth the same when a list clears the quotient as when it does not. The threshold therefore does more than switch the prize on. Crossing it changes a type-τ candidate's payoff by

$$X(\tau)'\beta + \alpha(\tau)\,\frac{B_{m,j}}{s_{m,j}} + \xi_{m,j} - \delta_{m,j}, \tag{5}$$

of which only the term in $\alpha(\tau)$ is the seat payoff; the rest is the difference in what the list is otherwise worth on the two branches, and nothing restricts its sign.

Why non-clearing lists are not interchangeable. Setting $\delta_{m,j}$ to a common constant would say that a candidate on a non-clearing list receives the same payoff whichever list it is. Non-clearing lists differ in what they offer beyond seats: ballot exposure, the support of a local organisation, a claim on a candidacy in the next cycle; and

different parties supply different amounts of each. We therefore allow non-clearing lists to have list-specific values $\delta_{m,j}$. Section 6.8 describes the dispersion of non-clearing shares that a common δ would have to be reconciled with.

Normalisation. Adding one constant to every ξ_p and every δ_p changes no choice probability, so one location is fixed by convention: the δ of a reference party is set to zero. Where the party labels fall into groups that never share a market, as labels defined within a state do, there is one such shift for each group, and the levels of the party effects are comparable within a group and not across groups (Section 3.4). A second convention would be needed if the type design spanned a constant, that is, if some linear combination of the columns of $X(\tau)$ equalled one at every type in $\mathcal{T}$, since a common shift of the clearing branch could then be moved between β and the ξ_p. The design estimated here does not span a constant (Remark C.1 checks its four columns), so one convention suffices, and $X(\tau)'\beta$, $\alpha(\tau)$ and every difference of party effects, within a branch and across the two, are unaffected by it (Proposition C.1, Corollary C.2). The convention fixes the location of the parameter and nothing else: whether the parameters are identified from the data is the question of Section 3, and the split of a list's appeal into a party component and a local residual is not settled by any convention (Section 3.5).

The full random utility is $u_i(j, s_m, \tau) = \pi(j, s_m, \tau) + \varepsilon_i(j)$.

2.4. Best response and equilibrium

Agents form rational beliefs about the composition through s_m. Under the Type I extreme value assumption (McFadden 1974; Train 2009) the type-conditional best response is

$$\sigma_m(j \mid \tau) \;=\; \frac{\exp\big(\pi(j, s_m, \tau)\big)}{\sum_{l \in \mathcal{J}_m} \exp\big(\pi(l, s_m, \tau)\big)}. \tag{6}$$

No expectation over rivals' actions appears: the candidate takes the aggregate s_m as given.

DEFINITION 1 (Mean-field equilibrium). *A pair $(\sigma_m^\star, s_m^\star)$ is a* mean-field equilibrium *of market m if $\sigma_m^\star$ satisfies (6) given $s_m = s_m^\star$, and $s_m^\star$ satisfies (1) given $\sigma_m = \sigma_m^\star$.*

ASSUMPTION 1 (Primitives). *In every market m: $\mathcal{T}$ is finite and $\mu_m(\tau) \geq 0$ sums to one; $q_m \in (0,1)$; the vote weights satisfy $0 < \underline{\rho}_m \leq \rho_{m,j} < \infty$ for every list, with $\underline{\rho}_m = \min_k \rho_{m,k}$; $|\alpha(\tau)| < \infty$ for every type and $0 < B_{m,j} < \infty$ for every list; β is finite and $X(\tau)$ is bounded on $\mathcal{T}$; and each $\xi_{m,j}$ and each $\delta_{m,j}$ is finite.*

The assumption bounds $\alpha(\tau)$ and does not restrict its sign, which matters because two of the type-specific estimates in Section 6.4 are negative. Section A.1 restates it in the market-free notation the appendix works in.

The truncated operator. Write Φ for the map that takes a share vector to the shares it induces, $\Phi(s_m)_j = \sum_\tau \mu_m(\tau)\, \sigma_m(j \mid \tau)$ with σ_m given by (6) at s_m; a mean-field equilibrium is a fixed point of Φ. The clearing rule makes π, and with it Φ, in general discontinuous where a list crosses the quotient, so a single Brouwer argument is not available. Fix instead a proposed clearing set $A \subseteq \mathcal{J}_m$. A list that clears satisfies $\rho_{m,j} s_{m,j} \geq q_m \sum_k \rho_{m,k} s_{m,k} \geq q_m \underline{\rho}_m$, because shares sum to one, so its share is at least

$$\underline{s}_{m,j} \;=\; q_m\, \underline{\rho}_m / \rho_{m,j} \;>\; 0, \qquad \underline{\rho}_m = \min_k \rho_{m,k}. \tag{7}$$

The truncated payoff assigns the clearing branch to the members of A whatever their shares, with the share in the prize's denominator bounded below by that floor, and the non-clearing branch to every other list:

$$\pi^A(j, s_m, \tau) = \begin{cases} X(\tau)'\beta + \alpha(\tau)\dfrac{B_{m,j}}{\max\{s_{m,j},\, \underline{s}_{m,j}\}} + \xi_{m,j} & j \in A, \\ \delta_{m,j} & j \notin A, \end{cases} \tag{8}$$

and Φ^A is Φ with π^A in place of π. The branch no longer depends on s_m and the prize term no longer diverges as a share goes to zero, so Φ^A is continuous on the whole simplex. Wherever the lists that clear are exactly those in A, every member of A has a share at or above its floor, so the truncation is inactive and $\pi^A = \pi$; the floor places no restriction on the lists outside A.

PROPOSITION 1 (Regime fixed points, and when they are equilibria). *Under Assumption 1, for every proposed clearing set $A \subseteq \mathcal{J}_m$ the set $\mathcal{S}^A$ of fixed points of the truncated operator Φ^A is non-empty. A fixed point $s^A \in \mathcal{S}^A$ is* self-fulfilling *if every member of A reaches the quotient at s^A and no non-member does; a self-fulfilling s^A is a mean-field equilibrium with $W(s^A) = A$. A mean-field equilibrium exists if and only if some A has a self-fulfilling fixed point.*

The existence of a regime fixed point is Proposition A.1, and the equivalence between self-fulfilment and a genuine equilibrium is Proposition A.2; both are proved in Section A. Corollary A.1 turns the self-fulfilment condition into a closed form in this paper's choice set, which is the version the application uses.

Existence. The last sentence of Proposition 1 is an equivalence, and the condition can fail. Section A.4 exhibits primitives satisfying Assumption 1 at which no proposed clearing set has a self-fulfilling fixed point, so that no mean-field equilibrium exists, and shows that the failure survives perturbation. The counterexample uses the possibility that the payoff falls at the crossing: by (5) a list can be *less* attractive on its clearing branch than on its non-clearing branch, for every type, when $\delta_{m,j}$ exceeds the largest clearing payoff the list can offer. In the empirical enumeration every market examined has at least one self-fulfilling clearing set (Section 6.7).

Multiplicity within a regime. For a given A the set $\mathcal{S}^A$ may contain more than one fixed point, and self-fulfilment must be checked at each. A count of self-fulfilling clearing sets is therefore a lower bound on the number of equilibria. In the application, re-solving each regime from several starting points returns the same fixed point (Section 6.7); the search establishes no uniqueness.

Multiplicity across regimes. Several clearing sets can be self-fulfilling at once. Section A.5 characterises which are, Section E exhibits a three-list market with two, and Section 6.7 finds several in many markets, more as the number of lists rises. The conditional estimation step takes the observed set as given and so needs no rule for selecting among clearing sets (Section 3.6); the counterfactuals do need one, and Section 6.7 reports what is available.

2.5. Mean-field representation

The fixed point of Definition 1 has J_m dimensions whatever I_m may be, where the joint action space of the finite game has $J_m^{I_m}$ elements, and estimators formulated on that space (for instance Bajari et al. 2010) have been applied to markets with a handful of players. What the representation sets aside is the candidate's own effect on the list they join and the randomness of the composition they face; Section 2.6 bounds what that costs.

Individual choice. The model treats each registered candidate as a decision-maker. If list assignment were instead made by parties or recruiting blocs, the number of independent choices in a market would be closer to the number of blocs than to I_m. Observed demographics explain little of list choice within a municipality-cycle, and larger demographic cells are rarely unanimous in their list choice (Section 6.2), which is inconsistent with a rule that assigns candidates to lists by those characteristics alone; it does not exclude assignment that is random or uses other inputs. We maintain individual choice as an assumption, and Section 7 records the evidence that would bear on it.

2.6. Finite-player approximation

We interpret Definition 1 as a large-I_m approximation to a corresponding finite-player game. Static large-game results such as Bodoh-Creed (2013) require payoffs to be continuous in the aggregate, and (4) is not: crossing the quotient switches the prize on. Payoffs are continuous within each regime cell of Section A.2, but continuity within a fixed regime is insufficient, because a unilateral deviation can move a list across a regime boundary and the approximation must also control the payoff change at such deviations.

One agent moving between lists moves a list's share by $1/I_m$ and its vote share by an amount of the same order, which Lemma B.1 bounds. The realised composition generated by the other agents' choices, when each chooses independently with the mean-field probabilities, departs from the mean-field share by an amount of order $I_m^{-1/2}$, which is larger; near the quotient the realised and the mean-field share can lie on opposite sides of it. And when either movement carries a list across the quotient, the payoff changes by the branch difference (5), which need not vanish as I_m grows. Ruling out that one agent crosses a list on their own is therefore not enough: the list must sit far enough from its quotient that the probability of the random composition crossing it, and with it the expected payoff effect of a crossing, is small.

The finite-market claim is accordingly conditional on a margin, the quotient margin $\eta_m^\star$ of Definition B.1: the smallest distance between a list's vote share and the quotient, evaluated at the mean-field equilibrium share $s_m^\star$ and not at a realised share. Proposition B.1 states that the mean-field strategy $\sigma_m^\star$ is an ϵ-equilibrium of the finite-I_m Bayesian game in the interim sense: when every other agent plays $\sigma_m^\star$, no agent, of any type and at any realisation of their own shock, can raise their expected payoff by more than ϵ_m by departing from it, the expectation being over the others' choices. The bound on ϵ_m is a within-regime term plus a regime-crossing term that falls in the margin, and Corollary B.1 recovers $\epsilon_m = O(1/\sqrt{I_m})$ under the *strong separation* condition (A25). Without separation no such rate follows. The rate describes a sequence in which I_m grows with every primitive of Assumption 1 held fixed: the number of lists, the quotient, the vote productivities, the appeals, and the prize $B_{m,j}$, which is normalised by I_m. That is a replication of the market at larger candidate counts, and it need not agree with the relation the institution of Section 6.1 imposes among seats, quotient and prize; the bound

of Proposition B.1 itself is a statement about one market at its own I_m and needs no sequence. The constant, which Section B gives, rises with the prize and falls with the quotient.

The margins in the data, which are computed at the observed shares in place of $s_m^\star$, are often large enough for the no-crossing condition to be informative. Measured against Lemma B.2's sharp no-crossing threshold, 93.0% of market-list pairs clear the margin and 69.6% of multi-list markets clear it on every list at once; those fractions are upper bounds on the fraction satisfying separation, which adds a cushion the proof needs. Section B proves both results and states the constants and the cushion, and Section 5 reports what the approximation does to the estimator at this scale.

3. Identification

This section derives the log-odds identities the payoff (4) implies and asks what they identify. The parameters fall into two groups. Differences in the prize coefficient across types, and the coefficients on candidate characteristics, are identified from comparisons across types on the same lists. Those comparisons allow an arbitrary effect for each list in each market, provided it is common to the types, and they use the logit structure and variation in the prize index. The level of the prize coefficient and the party effects are not identified when that list effect is unrestricted, and identifying them takes a restriction on its local component. Throughout, identification is from population choice probabilities; what the estimator does with counts is the subject of Section 4.

3.1. Setup

Markets, types and shares are those of Section 2: $\mu_m(\tau)$ is the prevalence of type τ in market m, $\sigma_m(j \mid \tau)$ the probability that a type-τ candidate stands on list j, and $s_{m,j}$ the population share of list j. In the notation of this section the payoff (4) is

$$\pi(j, s_m, \tau) = \begin{cases} X(\tau)'\beta + \alpha(\tau)\, P_{m,j} + \xi_{m,j} & j \in W_m, \\ \delta_{m,j} & j \notin W_m, \end{cases} \tag{9}$$

with $P_{m,j} = B_{m,j}/s_{m,j}$ the prize index of Section 2.3, a function of the market and the list and not of the type, and W_m the clearing set (3). The list-level terms are $\xi_{m,j} = \xi_{p(j)} + \nu_{m,j}$ and $\delta_{m,j} = \delta_{p(j)} + \nu'_{m,j}$. Two features carry the argument: $X(\tau)'\beta$ has no j subscript, so it is common to the clearing lists of a market, and ξ and δ have no τ, so a list's appeal enters every type's payoff by the same amount. Nothing is assumed about ν or ν' until Section 3.4.

Payoffs are finite under Assumption 1, so a type present in a market, $\mu_m(\tau) > 0$, chooses every list with positive probability and every log-odds below is defined. A sample cell can be empty; how the estimator handles empty cells is a matter for Section 4.

The prize index is treated as known. Its numerator $B_{m,j}$ is a forecast built from pre-registration information (Section 6.3); its denominator is the equilibrium share, which list appeal moves, so $P_{m,j}$ cannot be treated as exogenous to list appeal.

ASSUMPTION 2 (Observed regime). *The clearing set $W_m = W(s_m)$ of the equilibrium the market is in is known.*

Assumption 2 belongs to the population identification exercise: it says which branch of (9) each list's payoff is on. Three sets have to be kept apart. The model's clearing set is defined at the population share s_m. The data record the set of lists whose realised votes reached the quotient, and the set of lists that took a seat, which is larger because a list can take a seat by highest average: the two recorded sets differ for 2 of the 25,653 lists in the 2016 registry and for 4,691 of the 40,117 in the 2020 registry, 11.7% (Section 6.1). The application uses a recorded set in place of the model's, and Section 4 discusses that substitution. The identities below hold for the assignment of lists to branches that payoffs actually follow. Section 6.4 reports the estimates under both recorded sets and says which one the headline uses.

3.2. Log-odds identities

PROPOSITION 2 (Three contrasts). *Fix a market m and a type τ with $\mu_m(\tau) > 0$.*

(i) For two clearing lists $j, k \in W_m$,

$$\log \frac{\sigma_m(j \mid \tau)}{\sigma_m(k \mid \tau)} \;=\; \alpha(\tau)\big(P_{m,j} - P_{m,k}\big) \;+\; \big(\xi_{m,j} - \xi_{m,k}\big). \tag{10}$$

(ii) For two non-clearing lists $j, k \notin W_m$,

$$\log \frac{\sigma_m(j \mid \tau)}{\sigma_m(k \mid \tau)} \;=\; \delta_{m,j} - \delta_{m,k}. \tag{11}$$

(iii) For a clearing list $j \in W_m$ and a non-clearing list $k \notin W_m$,

$$\log \frac{\sigma_m(j \mid \tau)}{\sigma_m(k \mid \tau)} \;=\; X(\tau)'\beta \;+\; \alpha(\tau)\, P_{m,j} \;+\; \xi_{m,j} \;-\; \delta_{m,k}. \tag{12}$$

PROOF. Under the logit best response (6) the log-odds of two lists equal the difference of their payoffs. In (i) both lists take the clearing branch of (9) and $X(\tau)'\beta$ differences out; in (ii) both take the non-clearing branch; in (iii) the lists take different branches and nothing further cancels. □

Each identity has an unknown list-level term on its right-hand side. In (10) that term is $\xi_{m,j} - \xi_{m,k}$, and the prize difference is not exogenous to that appeal difference, the demand-side endogeneity of Berry, Levinsohn, and Pakes (1995); the identity determines $\alpha(\tau)$ only once something is known about the appeal difference. No identity involves a probability of not standing, because the model conditions on registered candidacies (Section 2.1).

What the threshold rule adds is (10). A rule that pays one list per market has no pair of paid lists in the same market, so it offers no comparison in which $X(\tau)'\beta$ and the value of an unpaid list are both absent.

3.3. What comparisons across types identify

PROPOSITION 3 (Differences in the prize coefficient). *Fix a market m, two lists $j, k \in W_m$, and two types τ, τ' with $\mu_m(\tau) > 0$ and $\mu_m(\tau') > 0$. Then*

$$\log \frac{\sigma_m(j \mid \tau)}{\sigma_m(k \mid \tau)} \;-\; \log \frac{\sigma_m(j \mid \tau')}{\sigma_m(k \mid \tau')} \;=\; \big(\alpha(\tau) - \alpha(\tau')\big)\big(P_{m,j} - P_{m,k}\big). \tag{13}$$

If $P_{m,j} \neq P_{m,k}$, the difference $\alpha(\tau) - \alpha(\tau')$ is identified.

PROOF. Apply (10) to each type and subtract. Neither the prize difference nor $\xi_{m,j} - \xi_{m,k}$ depends on the type, so the appeal difference cancels. With $P_{m,j} \neq P_{m,k}$ the left-hand side, a function of population choice probabilities, divided by the known prize difference gives $\alpha(\tau) - \alpha(\tau')$. □

PROPOSITION 4 (Coefficients on characteristics). *Fix a market m, a clearing list $j \in W_m$, a non-clearing list $k \notin W_m$, and two types τ, τ' present in m. Then*

$$\log \frac{\sigma_m(j \mid \tau)}{\sigma_m(k \mid \tau)} \; - \; \log \frac{\sigma_m(j \mid \tau')}{\sigma_m(k \mid \tau')} \;=\; \big(X(\tau) - X(\tau')\big)'\beta \;+\; \big(\alpha(\tau) - \alpha(\tau')\big)P_{m,j}. \tag{14}$$

If $\alpha(\tau) - \alpha(\tau')$ is identified, $(X(\tau) - X(\tau'))'\beta$ is identified; and β is identified if the vectors $X(\tau) - X(\tau')$ over such pairs of types span $\mathbb{R}^{n_X}$.

PROOF. Apply (12) to each type and subtract. Both $\xi_{m,j}$ and $\delta_{m,k}$ are common to the two types and cancel. The prize index is known, so with the coefficient difference identified the remaining term is $(X(\tau) - X(\tau'))'\beta$, and a spanning set of such differences determines β. □

Both results allow list appeal to be any finite effect of the market and the list that is common to the types. They need no decomposition of it into a party effect and a residual, no restriction on its distribution across markets, and no orthogonality between it and the prize index, because the additive list effect through which the prize index is endogenous cancels in the difference. Differencing across types also removes any intercept in $X(\tau)'\beta$, so Proposition 4 identifies slopes; the design estimated here has no intercept (Remark C.1). Proposition 4 carries one further maintained restriction, that $\delta_{m,j}$ does not vary with the type; Section A.5 reports a test of it that cannot distinguish a constant δ from a mildly type-varying one.

One pair of types in one market identifies one difference. Call two types *linked* if some market holds both and has two clearing lists with different prize indices. Every difference $\alpha(\tau) - \alpha(\tau')$ is identified if the graph on $\mathcal{T}$ with those links is connected, since a difference between two types never linked directly is the sum of the differences along a path.

The level of α is a different matter, and the reason is the same list effect that the two propositions difference away.

PROPOSITION 5 (The level is not identified with a free list effect). *Let $\xi_{m,j} = \xi_{p(j)} + \nu_{m,j}$ with $\nu_{m,j}$ unrestricted. For any constant c, the parameter values (α, ν) and $(\tilde{\alpha}, \tilde{\nu})$ with*

$$\tilde{\alpha}(\tau) \;=\; \alpha(\tau) + c, \qquad \tilde{\nu}_{m,j} \;=\; \nu_{m,j} - c\,P_{m,j} \quad \text{for } j \in W_m, \tag{15}$$

and every other parameter unchanged, generate the same choice probabilities and the same shares in every market.

PROOF. The clearing-branch payoff $X(\tau)'\beta + \alpha(\tau)P_{m,j} + \xi_{p(j)} + \nu_{m,j}$ is the same under the two parameter values for every type and every clearing list, and the non-clearing branch is untouched. Payoffs determine choice probabilities, and choice probabilities determine shares. □

The transformation changes neither $\alpha(\tau) - \alpha(\tau')$ nor β, in agreement with Propositions 3 and 4, and it leaves the shares a fixed point of the best-response map, so imposing the equilibrium condition does not exclude it

(Remark C.2). The level of α therefore rests on a restriction on ν. The two groups of results correspond to two sets of estimates in Section 6.4. Those with an effect for every list in every market report differences in α across types and the coefficients β, on the basis of Propositions 3 and 4; the coefficient of their reference type is a normalisation and not an estimate. Those with an effect for each party within a state also report the level, on the basis of the restriction that the next subsection states.

3.4. The level and the party effects

The benchmark restriction sets the local components to zero, $\nu_{m,j} = \nu'_{m,j} = 0$, so that a list's appeal is its party term. For a list with several parties that term is the composition-weighted average $w'_j\xi$ on the clearing branch and $w'_j\delta$ on the other (Section 2.3), with weights that sum to one; a single-label list has a unit weight on its party. The payoff index is then linear in $\theta = (\{\alpha(\tau)\}, \beta, \{\xi_p\}, \{\delta_p\})$.

Fix the population shares s_m, the clearing sets W_m and the list attributes of every market, as the observed choice probabilities imply them. Let Z be the matrix with one row for each market, type present in it and list, and with entries $\mathbf{1}\{j \in W_m\}P_{m,j}$ in the column of $\alpha(\tau)$, $\mathbf{1}\{j \in W_m\}X(\tau)'$ in the columns of β, $\mathbf{1}\{j \in W_m\}w_{j,p}$ in the column of ξ_p and $\mathbf{1}\{j \notin W_m\}w_{j,p}$ in the column of δ_p; and let Q subtract from each row the mean of the rows of its market and type.

PROPOSITION 6 (Rank condition). *Let $\nu_{m,j} = \nu'_{m,j} = 0$, and let Z be built at the given shares and clearing sets. Two parameter values θ and θ' generate the same choice probabilities at those shares and clearing sets, in every market, if and only if $QZ(\theta - \theta') = 0$. A linear function $c'\theta$ is identified if and only if c is orthogonal to the null space of QZ. That null space always contains the common shift of every ξ_p and every δ_p by one constant.*

PROOF. The payoff index of list j for type τ in market m is the corresponding row of $Z\theta$. By Lemma C.1, two parameter values give the same choice probabilities for a market and type if and only if their indices differ by a constant across the lists of that market, that is, if and only if the rows of $Z(\theta - \theta')$ for that market and type are equal, which is $QZ(\theta - \theta') = 0$ on those rows. The statement about linear functions is the standard one for a linear system. A common shift η adds η to every list's index on either branch, because the weights sum to one. □

The proposition characterises the parameter values that support the observed equilibrium. It says nothing about the other equilibria those parameter values admit. It covers coalition lists, whose composition weights are columns of Z.

For single-label lists the null space can be read off the pattern of which parties meet in which markets, and the corollary gives sufficient conditions in those terms.

COROLLARY 1 (What is identified when list appeal is a party effect). *Suppose every list carries a single party label and that:*

- *(i)* prize variation across markets*: for every type τ there are parties $p \neq p'$ and markets $m \neq m'$, both holding τ, in which a list of p and a list of p' clear, with the two prize gaps different;*
- *(ii)* no local component*: $\nu_{m,j} = \nu'_{m,j} = 0$, so that the list-level terms are the party effects $\xi_{p(j)}$ and $\delta_{p(j)}$;*
- *(iii)* connectedness*: the bipartite graph joining parties to markets is connected through clearing appearances and, separately, through non-clearing appearances;*

(iv) mixed reach*: every type is present in some market that holds both a clearing and a non-clearing list;*

(v) rank*: $X(\tau)$ has full column rank on $\mathcal{T}$.*

Conditions (i), (iii), (iv) and (v) are (D1)–(D4) of Assumption C.1. Then (10), (11) and (12) identify $\{\alpha(\tau)\}$, the differences $\{\xi_p - \xi_{p'}\}$ and $\{\delta_p - \delta_{p'}\}$, and the index $X(\tau)'\beta$, up to the invariance group of Proposition C.1. If in addition $X(\tau)$ does not span a constant on $\mathcal{T}$, that group is the common shift $\xi_p \mapsto \xi_p + \eta$, $\delta_p \mapsto \delta_p + \eta$, which leaves $\alpha(\tau)$, β and every difference of party effects unchanged, and β is identified.

The proof is in Section C. Its first step is the one with economic content: under (ii) a pair of parties that clear together in two markets has the same appeal difference in both, so two markets with different prize gaps separate $\alpha(\tau)$ from that difference, as a slope from an intercept. The conditions are sufficient. A repeated pair is the most easily checked way of ensuring that variation in prize gaps is not absorbed by contrasts of party effects, and other configurations do the same, for instance three parties that clear pairwise in three different markets with prize gaps that do not sum to zero; the general statement is Proposition 6. Condition (ii) is the substantive one, and Section 3.5 takes it up.

The conditions on the panel. Section F.9 reports the supports and ranks in full; three results matter here. First, the cross-type results have support: on the markets of sixty or more candidates, 5,088 of 5,089 markets have two clearing lists with different anticipated prize indices and the type link graph is connected; 4,251 of them also hold a non-clearing list, and the differences $X(\tau) - X(\tau')$ among the types present there have rank four, the number of characteristics. Second, the rank condition can be checked on the design the estimator uses, with coalition lists and their composition weights, both cycles and party-within-state labels. With the δ of one party fixed, QZ has 1,534 columns and rank 1,509 when each type has its own prize coefficient. No party label is shared between the 26 states, and the 25 unidentified directions are exactly the common shifts of the party effects of each of the other 25 states: each such shift is annihilated by QZ, the 25 of them are linearly independent, and no unidentified direction has a component on any $\alpha(\tau)$ or on β. Under condition (ii), therefore, every $\alpha(\tau)$ and β are identified in the estimated design, and differences of party effects are identified within a state and not across states. Third, none of these checks bears on condition (ii), which is an assumption about list appeal.

3.5. The local residual

Proposition 5 leaves the level of α to whatever is assumed about the local component $\nu_{m,j}$, a party's standing in a market relative to its label's average.

Party effects and what the likelihood assumes. The estimates of Section 6.4 that report a level absorb $\xi_{p(j)}$ and $\delta_{p(j)}$ with effects for each party within a state, and their likelihood is the model with $\nu = 0$ beyond those effects. A local component related to the prize index biases the level, and local party strength is the leading case: it is local appeal that the party effects do not absorb and that moves with the prize. A weaker restriction than $\nu = 0$, that the difference in local components between two lists is mean-independent of their prize difference given the party effects, would identify α in the linear population equation (10). It would not make a logit likelihood that omits ν correctly specified, and we have no result on how far the maximum-likelihood estimate then departs from α.

A restriction across branches. ξ_p and δ_p are the value of one label on the two branches. Setting $\xi_p = \delta_p$

would tie the party component of the clearing branch to the non-clearing contrasts, which carry no prize term. It is a substantive restriction, since a list that will not clear may be attractive for different reasons from one that will, and it does nothing about ν. We do not impose it.

Instruments. The registration cap is a function of council size, which is a step function of population, and it binds on some lists and not on others. Section 6.6 reports what that variation can and cannot support, and records two exposures built from pre-period levels that failed an exclusion placebo.

The equilibrium restriction. Requiring the shares to be a fixed point of the best-response map in every market does not exclude the transformation (15), which leaves every payoff at the observed shares unchanged. Solving the model inside the estimator therefore adds no information about the level while ν is free.

Section 6.5 reports what can be said about the local component that the party effects leave: the council-size design cannot bound α away from zero, and controlling for local party strength reverses the sign among non-incumbents, on both the all-markets sample and the markets of sixty candidates or more.

3.6. Equilibrium selection

A market can have several self-fulfilling clearing sets (Section A.5); the enumeration of Section 6.7 finds one on average in two-list markets and 6.4 in eight-list markets, which are lower bounds on the number of mean-field equilibria.

PROPOSITION 7 (The identities hold at every equilibrium). *Let $(\sigma_m^\star, s_m^\star)$ be any mean-field equilibrium of market m, with clearing set $W(s_m^\star)$. Then (10), (11), (12), (13) and (14) hold at $(\sigma_m^\star, s_m^\star)$, whatever other equilibria the market has.*

PROOF. Each identity follows from the best-response equation (6) at a given share vector and a given assignment of lists to branches, and every equilibrium satisfies that equation at its own shares and clearing set. □

The identifying equations are written at the equilibrium the market is in, with the clearing set of Assumption 2, so they need no rule for selecting among equilibria. That is the device of Menzel (2016). The proposition concerns the population equations; whether an estimator built on realised shares and on a recorded set is consistent, and whether its sampling distribution depends on which equilibrium the market is in, are questions for Section 4 and Section 5. A counterfactual reported as a point needs a selection convention, the alternative being to report a set; Section 6.9 reports differences between fixed points reached from the market's observed share vector, and Section 6.7 records the selection rules that were tried.

4. Estimation

4.1. Step one: market-level empirical objects

For each market m and type τ, let $n_{m,\tau,j}$ count the registered candidates of type τ on list j, $N_{m,\tau} = \sum_j n_{m,\tau,j}$ the candidates of that type, $n_{m,j} = \sum_\tau n_{m,\tau,j}$ the candidates on list j, and $N_m = \sum_j n_{m,j}$ the registered candidates in the market, which is the model's I_m. Shares and the type distribution are the raw proportions,

$$\hat{s}_{m,j} = \frac{n_{m,j}}{N_m}, \qquad \hat{\mu}_m(\tau) = \frac{N_{m,\tau}}{N_m},$$

so $\hat{s}_m$ lies on the simplex Δ^{J_m-1} of Section 2.3. Choice probabilities carry a Jeffreys correction against small cells,

$$\hat{\sigma}_m(j \mid \tau) \;=\; \frac{n_{m,\tau,j} + \varsigma}{N_{m,\tau} + \varsigma J_m}, \qquad \varsigma = 0.5, \tag{16}$$

which adds ς to each of the J_m lists and to nothing else, there being no outside option, so that $\sum_{j \in \mathcal{J}_m} \hat{\sigma}_m(j \mid \tau) = 1$. The correction enters only the log-odds outcome of the contrast estimator in Section 4.2, where the denominator of (16) cancels within a cell. The likelihood of Section 4.3 works from the counts and does not use it.

The clearing branch is read off the registry rather than estimated, and the set we read is not the one (3) defines. The model's W_m is the quotient-clearing set. The estimates are computed on the set of lists the seat allocation paid,

$$\hat{W}_m \;=\; \big\{\, j : \text{list } j \text{ took at least one council seat in market } m \,\big\},$$

one rule for both cycles, and published (Assumption 2). The quotient-clearing set and $\hat{W}_m$ differ wherever a list took a seat in the highest-average round without reaching the quotient: 2 of the 25,653 lists in the 2016 registry and 4,691 of the 40,117 in the 2020 registry (Section 6.1). We use $\hat{W}_m$ because the clearing branch of (4) is the branch that carries the prize, and a list that took seats cannot be placed on the branch that carries none. The quotient-clearing set is the named sensitivity, and Section 6.4 reports the estimates under both sets.

No arbitrary prize normalisation is needed. Section 2.3 defines $B_{m,j}$ as list j's anticipated prize normalised by the number of registered candidates I_m, so that $B_{m,j}/s_{m,j}$ is the anticipated prize per candidate on list j. Write $S_{m,j}$ for whichever seat count stands in for the seats candidates anticipated. Then $B_{m,j} = S_{m,j}/N_m$ and $s_{m,j} = n_{m,j}/N_m$, so

$$\frac{B_{m,j}}{s_{m,j}} \;=\; \frac{S_{m,j}}{n_{m,j}} \;\equiv\; P_{m,j}, \tag{17}$$

the list's anticipated seats per candidate. N_m cancels, so the prize term needs no count beyond the registered candidates on the list, and no pool of potential candidates has to be constructed — which, by Corollary 1, no part of the estimator requires. The single-market estimator had to set $B_{j^\star} = 1$ because the prize was unobserved. Here the prize is measured in seats and its scale is not the researcher's to choose. What remains to be chosen is $S_{m,j}$.

Anticipated versus realised seats. Candidates register months before the count, so $S_{m,j}$ must be the seats they anticipated and not the seats realised. Three candidates for $S_{m,j}$ appear in this paper, and they do not have equal standing.

Realised seats give the benchmark

$$B^{\text{real}}_{m,j} \;=\; \frac{\text{seats}_{m,j}}{N_m}, \qquad P^{\text{real}}_{m,j} \;=\; \frac{B^{\text{real}}_{m,j}}{s_{m,j}} \;=\; \frac{\text{seats}_{m,j}}{n_{m,j}}.$$

We use $P^{\text{real}}_{m,j}$ only as an ex post descriptive benchmark. It is not a valid measure of the anticipated prize, because seats are determined jointly with the candidate choices whose sorting across lists we seek to explain; Section 4.2 shows the leak and what it does to the estimate. The list's seats in the previous cycle are fixed before registration,

but they have almost no power on this panel. The main specification therefore forecasts a list's seat opportunity from information available before registration, the party's lagged vote share and the council's size, and (22) below is that forecast. Section 6.3 reports all three and the sensitivity across them.

4.2. Step two: three blocks of contrasts

Index a *contrast* by $c = (m, \tau, j, k)$ with both cells non-empty, and write

$$y_c = \log \frac{\hat{\sigma}_m(j \mid \tau)}{\hat{\sigma}_m(k \mid \tau)}, \qquad \Delta P_c = P^{\text{lo}}_{m,\tau,j} - P^{\text{lo}}_{m,\tau,k}, \tag{18}$$

where $P^{\text{lo}}_{m,\tau,j}$ is the anticipated prize with the cell's own members left out of its denominator, defined in (22) below. The three parts of Proposition 2 give three estimating equations in the notation of (18), one per kind of pair, and they stack:

$$\text{both clear:} \quad y_c = \sum_{\tau'} \alpha(\tau')\, \mathbf{1}\{\tau = \tau'\}\, \Delta P_c + \left(\xi_{p(j)} - \xi_{p(k)}\right) + u_c, \tag{19}$$

$$\text{neither clears:} \quad y_c = \delta_{p(j)} - \delta_{p(k)} + u_c, \tag{20}$$

$$\text{mixed, } j \text{ clears:} \quad y_c = X(\tau)'\beta + \sum_{\tau'} \alpha(\tau')\, \mathbf{1}\{\tau = \tau'\}\, P^{\text{lo}}_{m,\tau,j} + \xi_{p(j)} - \delta_{p(k)} + u_c. \tag{21}$$

The residual u_c is not an ordinary regression error. It contains the local list appeal the party effects do not absorb — $\nu_{m,j} - \nu_{m,k}$ in the clearing block, $\nu'_{m,j} - \nu'_{m,k}$ in the non-clearing block and $\nu_{m,j} - \nu'_{m,k}$ in the mixed block — together with the sampling error in y_c and any specification error. It need not be orthogonal to the prize index, and that is the local-residual confounding of Section 3.5. Least squares on the stack therefore does not by itself deliver a consistent estimate of the level of $\alpha(\tau)$; it delivers what Section 3.5 says is identified. Estimation is one least-squares solve on the three stacked together, which imposes the cross-block restrictions — a common $\alpha(\tau)$ across (19) and (21), and a common δ_p across (20) and (21) — rather than estimating the blocks separately and reconciling afterwards.

The common level of ξ and δ is not identified, and not because only their within-branch differences appear: (21) carries $\xi_{p(j)} - \delta_{p(k)}$ across the two branches. The group of transformations that leaves the three blocks unchanged is one-dimensional here (Proposition C.1(ii), Remark C.1, and condition (D1) of Assumption C.1, verified on this panel rather than assumed, Section F.9), so one normalisation is needed and a second would be a binding restriction rather than a convention. That count is for party labels shared across the country. With the party-within-state labels of Section 6.4 no label is shared between states, and the invariance has a common shift of the party effects for each of the 26 states; the estimator fixes one δ and leaves the other 25 directions at the representative its minimum-norm steps select, so differences of party effects are interpretable within a state and not across states (Section F.9). We set $\delta_{p_0} = 0$ at a reference party p_0 and leave every ξ_p free. The restriction touches neither $\alpha(\tau)$ nor any component of β; what it fixes is the common level of the two blocks, and that is the whole of what it costs (Corollaries 1, C.1 and C.2).

The outside option is absent. No equation above contains $\hat{\sigma}_m(0 \mid \tau)$, and none of the three needs a potential-candidate pool to be defined. That matters because where the only alternative to the prize-paying one is abstention,

the outside option sits in the denominator of every observation and has to be constructed from a pool the data rarely record. Here the non-clearing lists supply an observed reference in its place.

Three further features are worth stating because each is a restriction the single-market specification could not impose.

No intercept in the first two blocks. $X(\tau)'\beta$ differences out of (19) and never enters (20), so both run through the origin by construction. It does enter (21), which is what identifies it. Adding a constant to either of the first two and testing it against zero is a specification check with content: a non-zero intercept says some list-level component is not captured by the party effects. $X(\tau)$ itself carries no constant either — four columns, no intercept, age as a single standardised score — and that is not a convenience but the fact that makes the invariance group one-dimensional and one normalisation sufficient (Remark C.1).

Party effects enter as differences within each block, and across the branches in the mixed one. In (19) only $\xi_{p(j)} - \xi_{p(k)}$ appears, so the design matrix carries $+1$ in the column of j's party and -1 in k's, and the same holds of δ in (20). Equation (21) is not of that form: its $+1$ sits in j's ξ column and its -1 in k's δ column, and the row also carries $X(\tau)$. So the rank deficiency of the three blocks stacked is the invariance group of Proposition C.1 and not an artefact of the ± 1 pattern. Here that group is the single common shift of the two levels: the stacked design drops exactly one column, a δ base, and every ξ column stays free, which leaves the reference level $\xi_{p_0} - \delta_{p_0}$ an estimable coefficient and β separable from it across types because $X(\tau)$ spans no constant and condition (D1) holds on this panel, verified rather than assumed (Corollary C.1, Remark C.1, and Section F.9). This absorbs the national component of the list-level unobservable of Section 3.5 and leaves the local residual $\nu_{m,j} - \nu_{m,k}$ inside u_c, which is the honest position of that section rather than a solution to it.

The prize regressor varies. ΔP_c differs across markets and, within a market, across pairs of clearing lists. How it is built decides whether that variation measures anything: the realised-seat benchmark fails that test and the anticipated prize is built to pass it.

The realised-seat benchmark leaks through both numerator and denominator. $P^{\text{real}}_{m,j} = \text{seats}_{m,j}/n_{m,j}$ contains the contrasting cell's own members twice, and the two leaks run in opposite directions.

The denominator is the familiar half. The prize has $n_{m,j}$ below the line while y_c has the same count above it, so the raw version carries a mechanical negative correlation; replacing $n_{m,j}$ by $n_{m,j} - n_{m,\tau,j}$ removes the cell entirely and what remains does not vary with the cell's own choice. In the mean-field limit an agent's own contribution to the aggregate is negligible, so the correction is asymptotically innocuous. It is not innocuous in finite samples: without it the estimate is -0.27 and with it $+0.33$.

The numerator is the half we missed. Under the realised prize the numerator is $\text{seats}_{m,j}$, and a list wins seats partly because this cell's members won them, so the prize is mechanically *higher* exactly where the cell is present. In the vereador panel a cell's own members hold 5.6% of the seats that define their list's prize, and removing them moves the estimate by far more than 5.6%: from $+0.32$ to -0.083. That starting figure and the $+0.33$ of the previous paragraph are one quantity measured on two contrast panels, and both are the leak rather than a measurement. The panels differ twice over. The previous paragraph's is the clearing block alone, over the markets that seated a council; Section 4.3's is all three blocks with no restriction on the council at all, so its 357,081 rows include 707 from the seven contests that seated none, which Section 6.4 excludes throughout. Either difference on its own would move a figure in the second decimal.

Removing it does not repair the estimator, because what is subtracted is the cell's *realised* success: a cell that placed many members on a list and won there has the most taken away, which is a downward mechanical link replacing an upward one. The defect is upstream of any leave-out. $\text{seats}_{m,j}$ is an outcome of the same election in which the candidate chose, and no function of it is clean.

The anticipated prize avoids the numerator leak and leaves the cell out of the denominator. We build the anticipated prize from objects that exist before anyone registers,

$$P^{\text{lo}}_{m,\tau,j} \;=\; \frac{C_m \cdot \widehat{\text{seatshare}}\big(v^{\text{lag}}_{m,p(j)},\, C_m\big)}{n_{m,j} - n_{m,\tau,j}}, \tag{22}$$

where C_m is the size of the council and $v^{\text{lag}}_{m,p}$ is party p's share of the municipal vote in the previous cycle. In the notation of (17) the numerator is $S_{m,j}$. It contains no member of the current cell, and the forecast in it uses no outcome of this election, so it needs no leave-out; only the denominator does. Where a cell is the whole of its list the denominator is set to one.

The council's size is fixed before registration, by population band within the art. 29 ceiling. The statutory size is not in the registry, so we proxy it with the seats the market filled. The statutory size and the filled count differ wherever a seat went unfilled, and there the scale C_m carries something from this election even though the forecast does not; Section 6.3 reports how often that happens.

The superscript and the τ subscript are not decoration, and they do not make the prize type-specific. The model's prize $P_{m,j}$ carries no type: it is the same object for everyone on the list. Leaving the cell out of the denominator is a finite-I_m correction to it, made because a cell's own members are part of the share they respond to, and it introduces a type index the population object does not have. $P^{\text{lo}}_{m,\tau,j}$ and $P_{m,j}$ coincide as I_m grows.

The forecast deliberately omits $n_{m,j}$, which would recontaminate it. The resulting measure correlates with realised seats at 0.55 and its lagged vote share is positive for 86% of lists, against 0.08 and 29% for lagged *seats*, which have no leak and no power either.

The alternative worth naming is the one this construction avoids. Where the prize enters through a single market-level number it makes a constant column, and the coefficient on a constant is the regression's intercept rescaled: an identity rather than a measurement. The prize has to vary across alternatives within a market, and (22) is how we make it do so.

Which contrasts to use, and how to pick the reference. A cell touching c lists of one kind admits $\binom{c}{2}$ pairs, but only $c - 1$ are independent: any pair decomposes as $y_{(j,k)} = y_{(j,l)} - y_{(k,l)}$. We therefore form contrasts against a reference list within each cell and kind.

The reference must be drawn at random. Choosing it on list size — the largest clearing list, say — selects on a variable inverse to the prize, since every construction of the prize has the list's candidate count in its denominator, so every contrast is taken against a systematically low-prize list and both the intercept and the slope inherit it. In our data that one choice moved the intercept from −0.0002 (0.0008), $t = -0.3$, to −0.0359 (0.0008), $t = -45.0$: from a figure indistinguishable from zero to one that would have been read as an omitted list-level component, and is an artefact of the reference rule. Both come from the same panel of clearing contrasts, the 216,967 counted below, so the reference rule is the only thing that differs between them.

Counts are 216,967 clearing rows, 83,334 non-clearing rows and 56,073 mixed rows — one per cell, since Section F.9 shows a cell supplies exactly one independent mixed contrast however many lists of each kind it holds.

4.3. The likelihood, and why it is the main specification

Equation (19) is the linear form of a conditional logit over the alternatives available within each market. Writing the indirect utility of list j to a candidate of type τ as

$$V_{m,\tau,j} \;=\; \mathbf{1}\{j \in \hat{W}_m\}\left[X(\tau)'\beta + \alpha(\tau)P^{\text{lo}}_{m,\tau,j} + \xi_{p(j)}\right] \;+\; \mathbf{1}\{j \notin \hat{W}_m\}\,\delta_{p(j)} \tag{23}$$

reproduces all three contrasts of Section 4.2 as differences, with the same parameters and the same single normalisation (Corollary C.1). The log-likelihood is

$$\mathcal{L} \;=\; \sum_{m,\tau,j} n_{m,\tau,j}\, V_{m,\tau,j} \;-\; \sum_{m,\tau} N_{m,\tau}\, \log \sum_{k \in \mathcal{J}_m} \exp V_{m,\tau,k}. \tag{24}$$

The prize in (23) is the leave-cell-out anticipated prize of (22), the same regressor the contrasts use, and the branch is the $\hat{W}_m$ of Section 4.1. The type index on $P^{\text{lo}}_{m,\tau,j}$ is the finite-market leave-cell-out correction of Section 4.2 and not a type-specific structural prize. On a list the cell did not enter, $n_{m,\tau,j} = 0$ and $P^{\text{lo}}_{m,\tau,j} = P_{m,j}$, so the correction is vacuous on exactly the zero cells that the likelihood adds to the contrast sample.

We estimate (24) and report (19) as a robustness check, rather than the other way round, for a reason that is about identification and not efficiency. *The contrast cannot use a zero.* y_c is a difference of logarithms, so a cell that placed nobody on a list supplies no row, and on the full vereador panel 55.6% of (m, τ, j) triples are exactly that — 56.6% on the markets of sixty candidates or more that Section 6.4 estimates on. A type that avoids a low-prize list entirely is evidence about α, and it is the evidence the log-odds form is structurally unable to read.

The magnitudes make the point. Under the anticipated prize (22), moving from the linear contrast to the likelihood one difference at a time gives

specification	$\hat{\alpha}$
linear contrast, Jeffreys shares, unweighted	0.098
+ raw shares (smoothing off)	0.130
+ precision weights	0.161
likelihood, entered lists only	0.171
likelihood, full choice set	**0.876**

Every row is the common-α specification, one coefficient for all types, which is used for this comparison of estimators only; the estimates of Section 6.4 let α vary by type. The three contrast rows are fitted on the same list pairs, and every pair has a positive observed count on both sides, since Section 4.2 forms a contrast only between non-empty cells. "Raw shares" therefore sets $\varsigma = 0$ on that common support and takes no logarithm of zero. The precision weight of a pair is $\left(1/n_{m,\tau,j} + 1/n_{m,\tau,k}\right)^{-1}$.

Smoothing, weighting and functional form together account for a movement from 0.098 to 0.171. Restoring the discarded zeros accounts for the rest, which is 91% of the gap between 0.098 and 0.876. The Jeffreys

correction of (16), which the likelihood does not need because it works from counts, moves the estimate by 0.032, against 0.705 for the zeros.

One inconsistency in this ladder we record rather than repair. The two likelihood rows impose the positive-council restriction Section 6.1 applies. The three contrast rows are formed on the 357,081-row contrast panel, which does not, so they carry 707 rows from the seven markets whose councils filled no seats. Every list in those markets is on the non-clearing branch, so the 707 rows carry no prize term.

4.4. Inference

The likelihood. The reported inference for the main specification is a sandwich variance clustered on the municipality, with the two cycles pooled into one cluster. Write $z_{m,\tau,j}$ for the row of the design that gives $V_{m,\tau,j} = z'_{m,\tau,j}\theta$, and $\sigma_{m,\tau,j}(\theta)$ for the logit probability. The score of municipality g sums the cell scores of its markets,

$$\psi_g(\theta) \;=\; \sum_{m\in g}\sum_{\tau}\sum_{j\in\mathcal{J}_m} z_{m,\tau,j}\,\big(n_{m,\tau,j} - N_{m,\tau}\,\sigma_{m,\tau,j}(\theta)\big),$$

and with $\hat{H}$ the observed information of (24) at $\hat{\theta}$ and G the number of municipalities,

$$\hat{V} \;=\; \frac{G}{G-1}\,\hat{H}^{-1}\Big(\sum_{g=1}^{G}\psi_g(\hat{\theta})\,\psi_g(\hat{\theta})'\Big)\hat{H}^{-1}. \tag{25}$$

Clustering on the municipality allows arbitrary dependence among the types and lists of a market, which share one share vector and one council, and between the two cycles of the same town, whose unobservables plausibly persist.

The variance (25) treats the prize as given. The seat-share forecast in the numerator of (22) is fitted on the same panel and its sampling error is not carried into $\hat{V}$: there is no generated-regressor correction, and we have not computed what one would add.

The split-sample estimator and its standard error. The estimates of Section 6 do not come from (24) with the prize (22) as it stands. Section 5 shows that likelihood recovering about three-fifths of the prize coefficient at municipal scale, because the share in the prize's denominator is counted from a few dozen candidates, and Section 5.7 constructs the estimator the application reports: the same likelihood, with the prize built from one random half of a market's candidacies on an unbiased reciprocal, and the residual r of its within-cell regression on the other half's prize added as a control with coefficient ρ. Write π for the coefficient of that first stage, $d_{m,\tau,j}$ for the instrument's within-cell deviation, and W for the block-diagonal matrix of multinomial weights $N_{m,\tau}(\operatorname{diag}\sigma - \sigma\sigma')$. The control depends on $\hat{\pi}$, so the score of the second stage does, with derivative $J = \rho\, Z'Wd$, and $\hat{\pi} - \pi = \sum_g \phi_g$ with $\phi_g = \sum_{(m,\tau,j)\in g} d_{m,\tau,j}\, r_{m,\tau,j} / \sum d^2$. Municipality g's influence on $\hat{\theta}$ is therefore

$$\iota_g \;=\; \hat{H}^{-1}\big(\psi_g(\hat{\theta}) + J\,\phi_g\big), \tag{26}$$

and the variance is $\frac{G}{G-1}\sum_g \bar{\iota}_g\bar{\iota}'_g$, where $\bar{\iota}_g$ averages ι_g over the two orientations of the halves and over the random splits. Averaging the influences before forming the variance matters as much as the first-stage term: the

orientations share every observation, and averaging their separate standard errors overstates the standard error of their mean. Section 5.7 checks the result against the standard deviation of the estimates across replications.

The contrast estimator. Two sources of dependence rule out treating the 217,000 clearing contrasts as independent observations.

Within a market, everything is shared. All contrasts drawn from market m involve the same share vector s_m, the same quotient, and overlapping lists; contrasts from the same cell share a reference list on one side. We therefore cluster at the municipality level here as well, pooling the two cycles. That gives roughly 5,500 clusters against the nominal 217,000 rows, and it is the cluster count rather than the row count that governs precision.

Step one is estimated. y_c is a difference of logs of estimated proportions, so its sampling variance falls with cell size and the second-stage standard errors must account for first-stage noise. We apply the structural delta-method correction of Pesendorfer and Schmidt-Dengler (2008), with the multinomial margin for a difference of two log-proportions in the same cell,

$$\operatorname{Var}(y_c) \;\approx\; \frac{1}{N_{m,\tau}} \left(\frac{1}{\hat{\sigma}_m(j \mid \tau)} + \frac{1}{\hat{\sigma}_m(k \mid \tau)} \right), \tag{27}$$

The reciprocal of (27) is the natural weight for a feasible GLS version of (19). We report unweighted least squares as the baseline and weighted least squares as a robustness check, and note that the two disagree only if small cells carry the identification — itself worth knowing.

4.5. Computation

The alternative to a two-step estimator is to impose the equilibrium as a constraint and solve payoffs and shares together (Su and Judd 2012; Dubé, Fox, and Su 2012), a mathematical program with equilibrium constraints (Luo, Pang, and Ralph 1996) that needs an interior-point solver (Wächter and Biegler 2006). We do not use it, because the two-step estimator is already cheap. The likelihood (24) is concave in the parameters and its design is sparse. On the full panel — 1,240,526 rows over the 193,322 cells of all 11,136 markets, rather than the 193,178 cells of the 11,129 that Table 5 counts — with 104 parameters, a fit takes about a minute, and the clustered sandwich (25) costs more than the fit. The linear contrast of (19) is a single least-squares solve and runs in under a second. We keep it as a diagnostic because it displays the three identification contrasts directly and makes the role of zero cells, weights and the choice of reference visible. Neither step scales in the number of candidates once the cell counts are formed, which is the property the mean-field reduction was made for.

5. Monte Carlo

This section asks whether the estimator of Section 4 recovers parameters it is given, on data generated by the model of Section 2. The design follows the single-market study it replaces: fix the parameters, solve the mean-field equilibrium, draw a sample, estimate, and report bias, root mean squared error and confidence-interval coverage across replications.

One feature of the panel makes the exercise report something the single-market version could not. There, one market held 5,570 agents and the empirical share entering the prize B_j/s_j was precise enough that treating it as

known cost nothing. Here the sample is many small markets: the median municipality fields 55 candidates and the mean 83. The measurement error in the regressor is therefore $O(1/I)$ in the number of agents per market and does not fall in the number of markets at all, so we report both limits separately.

Two scales matter and they are not the same. The mean over all markets is 83 candidates across 5.9 lists, and the tables below use 86 and 6. The sample Section 6.4 actually estimates on is markets of sixty candidates or more, whose mean is 137.8 candidates across 8.69 lists, and the small markets that pull the pooled figure down are precisely the ones the estimation excludes. Section 5.6 reports the exercise at that second scale, and it is the one the application should be read against.

5.1. Data-generating process

Types are the $K = 32$ demographic profiles of the application and their weights $\mu(\tau)$ are the empirical ones, whose effective number $1/\sum_\tau \mu(\tau)^2$ is 13.0; a symmetric Dirichlet draw would give 24.3 and spread candidates across cells that the data concentrate. Council sizes are drawn from the empirical distribution, with median 9. Each market has $J = 6$ lists whose ex ante strength w_j is lognormal with scale 0.6 — a round number rather than a calibrated one, and the direction of the approximation is known: matching the simulated variance of the prize contrast to the panel's requires a scale of 0.865, so the exhibit below is *less* attenuated than the data and understates the problem rather than overstating it — normalised to sum to one, so that anticipated seats are $C_m w_j$ and the prize per candidate is B_j/s_j with $B_j = C_m w_j/I$.

The structural parameters are

$$\alpha = 1.5, \qquad \beta = (0.50,\ 0.30,\ -0.20,\ 0.80),$$

on the four components of $X(\tau)$, with ξ_p drawn normal with standard deviation 0.25, δ_p normal with mean -0.5 and standard deviation 0.60, and δ of the first party set to zero, which is the estimator's normalisation and the only normalisation either the design that generates these data or the design fitted to them imposes. One is the right count: the invariance group here is one-dimensional (Proposition C.1(ii), Remark C.1), the same design condition Section F.9 verifies on the fitted panel rather than assumes, and the single restriction meets every coset in exactly one point (Corollary C.1). No ξ convention is stated because none is needed, and every ξ_p is free in the fitted design. Had the two designs taken different reference parties, the recovered ξ_p and δ_p *levels* would differ from the generating ones by one common additive constant and nothing else would differ at all: by Corollary C.2 the group leaves α, every component of β, and every ξ and δ difference exactly where they are. Both $\{\xi_p\}$ and $\{\delta_p\}$ are drawn once and held fixed across markets, as Section 2 requires of a party constant; drawing them per market makes them an omitted variable correlated with the prize through the equilibrium.

The winner rule is the vote-share threshold of (A1). We solve Φ by damped iteration with a self-consistent regime to a sup-norm tolerance of 10^{-12}, draw I agents from μ, and let each choose from $\sigma^\star(\cdot \mid \tau)$. The simulated data are the tabulated counts $n_{m,\tau,j}$, from which every object the estimator uses is constructed.

5.2. Estimator

We run (24), the conditional logit of Section 4.3, whose indirect utility is (23), with the leave-one-cell-out prize (22) and standard errors clustered at the market, which is what the application clusters at. No part of the estimator

is reimplemented for the simulation: the simulated counts are assembled into the frame the vereador panel produces and passed to the same code. The coverage column is therefore the coverage the application claims.

5.3. Results at the empirical scale

Table 1: Monte Carlo at $M = 600$ markets of $I = 86$, $R = 40$

Parameter	True	Mean	Bias	RMSE	s.d.	95% coverage
α	1.500	0.826	−0.674	0.682	0.105	0.0%
β_{female}	0.500	0.490	−0.010	0.052	0.051	90.0%
$\beta_{\text{higher ed}}$	0.300	0.301	0.001	0.025	0.025	97.5%
β_{age}	−0.200	−0.204	−0.004	0.017	0.017	95.0%
$\beta_{\text{incumbent}}$	0.800	0.804	0.004	0.032	0.032	92.5%
Split-sample estimator of Section 5.7, same draws						
α	1.500	1.392	−0.108	0.209	0.179	95.0%
β_{female}	0.500	0.489	−0.011	0.053	0.051	90.0%
$\beta_{\text{higher ed}}$	0.300	0.301	0.001	0.025	0.025	97.5%
β_{age}	−0.200	−0.204	−0.004	0.017	0.017	95.0%
$\beta_{\text{incumbent}}$	0.800	0.803	0.003	0.032	0.032	92.5%

The first five rows of Table 1, the estimator of Section 4, split sharply. Every component of β is recovered: the largest bias is 0.010 and coverage runs from 90.0% to 97.5%. The prize coefficient is not. It comes back at 0.826 against a truth of 1.500 and its confidence interval covers the truth in none of the forty replications.

The asymmetry is the diagnosis. β multiplies $X(\tau)$, which is measured without error; α multiplies $B_j/\hat{s}_j$, and $\hat{s}_j$ is a share estimated from I draws.

5.4. Which limit the bias vanishes in

Table 2: More markets against larger markets, α only

M markets of $I = 86$					*$M = 400$ markets of I*				
M	Mean	Bias	s.d.	Cov.	I	Mean	Bias	s.d.	Cov.
150	0.850	−0.650	0.199	20.0%	55	0.471	−1.029	0.151	0.0%
300	0.842	−0.658	0.135	0.0%	86	0.832	−0.668	0.092	0.0%
600	0.842	−0.658	0.095	0.0%	200	1.138	−0.362	0.146	52.0%
1,200	0.829	−0.671	0.077	0.0%	500	1.411	−0.089	0.240	92.0%

The two halves of Table 2 behave differently, and the difference is the section's main point.

Adding markets does what adding observations usually does to the standard deviation, which falls from 0.199 to 0.077, and nothing at all to the bias, which sits between −0.650 and −0.671 throughout. Coverage therefore does not improve with the sample; it collapses, from 20% at 150 markets to zero at 300. **More markets buy precision without buying consistency, and the interval closes on the wrong number.**

Enlarging the markets is the limit that works. The bias falls from −1.029 at $I = 55$ to −0.089 at $I = 500$. A delta-method argument predicts $O(1/I)$ for a ratio with an estimated denominator, and the product $I \times |\text{bias}|$ —

57, 57, 72 and 45 across the four rows — is consistent with that rate without establishing it, since it rises over the third row before falling back.

5.5. Confirming the source

Re-running the $M = 600$, $I = 86$ design with s_j handed to the estimator rather than estimated returns $\hat{\alpha} = 1.565$, a bias of +0.065, and coverage of 88.0%. *Every other feature of the design is unchanged.* Nearly the whole failure in Table 1 is the measurement error in the share — a bias of −0.674 becomes one of +0.065 — and none of what it leaves is the mean-field approximation, the discrete regime, the clustering, or the two corrections of Section 4.2.

5.6. What this implies for the application

The tables above run at $I = 86$ over six lists, the mean across all markets. Section 6.4 estimates on markets of sixty candidates or more, which average 137.8 candidates over 8.69 lists. Repeating the exercise there:

Table 3: At the scale of the estimation sample, $M = 500$, $I = 138$, $J = 9$

Parameter	True	Mean	Bias	RMSE	s.d.	95% coverage
α	1.500	0.893	−0.607	0.634	0.183	13.3%
β_{female}	0.500	0.503	0.003	0.031	0.030	93.3%
$\beta_{\text{higher ed}}$	0.300	0.292	−0.008	0.021	0.019	90.0%
β_{age}	−0.200	−0.195	0.005	0.012	0.011	93.3%
$\beta_{\text{incumbent}}$	0.800	0.797	−0.003	0.020	0.019	90.0%
Split-sample estimator of Section 5.7, same draws						
α	1.500	1.365	−0.135	0.302	0.270	96.7%
β_{female}	0.500	0.502	0.002	0.031	0.030	93.3%
$\beta_{\text{higher ed}}$	0.300	0.292	−0.008	0.021	0.019	90.0%
β_{age}	−0.200	−0.195	0.005	0.012	0.011	93.3%
$\beta_{\text{incumbent}}$	0.800	0.797	−0.003	0.020	0.019	90.0%

The first five rows of Table 3 say the estimator of Section 4 recovers 60% of α there, against 55% at the pooled scale, and β is recovered throughout. Handing it the true share returns 1.485. The split-sample estimator recovers 91% on the same draws and leaves β where it was.

Three things carry over. The estimates of β in Section 6 can be read as they stand.

The clustered standard errors are correct for what they measure and misleading for what a reader wants. They describe the sampling variation of $\hat{\alpha}$ around a probability limit that is not α. A t-statistic against zero remains meaningful, since the bias is toward zero; a confidence interval for the magnitude does not, and we do not report one.

The correction is a matter of the regressor rather than the model, and Section 5.7 constructs one.

5.7. A split-sample correction

The regressor is B_j divided by a share counted from the cell's competitors: with x of the market's n other candidates on list j, the share is x/n and its reciprocal n/x. Two things are wrong with n/x as a measure of the

reciprocal of the population share. It is noisy, which attenuates. And it is biased upward, the more so the smaller the list, because the reciprocal is convex; an instrument does not remove a bias in the mean. The correction addresses each. In place of n/x it uses $(n+1)/(x+1)$, whose expectation is the reciprocal of the population share up to a term that vanishes geometrically in n when x is binomial. And it assigns every candidate to one of two halves by a fair coin, builds the regressor from each half's counts, and uses the regressor from one half as an instrument for the regressor from the other (Angrist and Krueger 1995): given the equilibrium, the two halves' counts are independent, so their errors are. The first stage is run within the market–type cell over clearing lists, the second stage is the likelihood (24) with the first-stage residual added as a control, and the two orientations of the halves are averaged. The outcome is the full counts throughout.

Table 4: The split-sample estimator in the two limits, α only

M markets of I = 86					*M* = 400 *markets of I*				
M	Mean	Bias	s.d.	Cov.	*I*	Mean	Bias	s.d.	Cov.
150	1.378	−0.122	0.361	100%	55	1.226	−0.274	0.331	72%
300	1.391	−0.109	0.229	96%	86	1.398	−0.102	0.190	100%
600	1.406	−0.094	0.161	96%	200	1.377	−0.123	0.189	100%
1,200	1.398	−0.102	0.114	100%	500	1.510	0.010	0.257	100%

The designs and the draws are those of Table 2, $R = 25$. Coverage is that of the nominal 95% interval formed from the second stage's clustered standard error.

The split-sample rows of Tables 1 and 3 and Table 4 are computed on the draws of the tables they sit beside; the split uses a separate random stream, so the estimator of Section 4 sees exactly the samples it saw before. At the empirical scale the split-sample estimator returns 1.392 where the estimator of Section 4 returns 0.826, and at the scale of the estimation sample 1.365 against 0.893: it recovers 91% to 93% of α where the other recovers 55% to 60%. Every component of β is unchanged to the third decimal. The standard deviation of $\hat{\alpha}$ rises by about half, from 0.183 to 0.270 at the scale of the estimation sample.

Both ingredients are needed. On a separate set of 100 draws at that scale, the instrument applied to n/x recovers 78% and $(n+1)/(x+1)$ without a split 69%, against 93% for the two together.

Table 4 shows what remains. A bias of about −0.10 does not fall as markets are added, so the corrected estimator is not consistent in M either; but it is a sixth of the bias it replaces. It is no smaller at $I = 200$ than at $I = 86$, and at $I = 500$ it is gone. At $I = 55$ the correction is weaker, 1.226 with 72% coverage, because half of fifty-five candidates leaves too few on a list for the first stage.

The coverage columns above use the second stage's clustered standard error, averaged over the two orientations. That standard error is wrong in two offsetting ways: it ignores the sampling error of the first stage, and it averages the standard errors of two estimates that share every observation, which overstates the standard error of their mean. The standard error the application reports, (26), corrects both. On 100 replications of three designs the standard deviation of $\hat{\alpha}$, the second-stage standard error and the corrected one are 0.214, 0.227 and 0.206 at $M = 600$, $I = 86$; 0.397, 0.443 and 0.411 at $M = 150$, $I = 86$; and 0.273, 0.293 and 0.272 at $M = 500$, $I = 138$, $J = 9$. With the corrected standard error the nominal 95% interval covers the truth in 90%, 96% and 94% of replications, and covers the estimator's own mean in 95%, 98% and 94%. The shortfall against the truth at the first design is the remaining bias and not the standard error. The standard errors of β track their standard

deviations in all three, the largest gap being 0.103 against 0.110.

The same estimator is used in Section 6.5 with a second regressor that shares the counted denominator, the control for local party strength of (30). Adding to the design an attraction $\gamma_0 z^\star$ that rises with local strength, with $\gamma_0 = 10$, and fitting (30) with each regressor instrumented by its counterpart from the other half, the split estimator returns 1.45 where $\alpha = 1.5$, 0.12 where $\alpha = 0$, and -0.41 where $\alpha = -0.5$ ($M = 500$, $I = 138$, $J = 9$, $R = 30$). It does not return a positive coefficient where there is none. The estimator of Section 4 with the same control returns 0.93, 0.06 and -0.29: it is attenuated and keeps the sign of the truth.

The closed-form factor. The attenuation factor can also be computed from the multinomial variance of the share, with no simulation. In the simulation it predicts the recovery ratio closely, 0.600 against 0.595 at the scale of the estimation sample. On the panel it does not behave as a correction: it is negative in the smallest market-size band, and above sixty candidates $\hat{\alpha}$ does not move in proportion to it. We therefore do not divide by it, and Section F.8 reports the comparison. The split-sample estimator is the correction we use instead. On the panel it raises the baseline coefficient from 1.20 to 1.92 (Section 6.4), a ratio of 0.63 against the 0.60 the simulation gives for the estimator of Section 4 at this scale (Table 3).

6. Application: Brazilian municipal councils

6.1. The institution

Brazil's 5,570 municipalities include two that elect no council — Brasília, which is not a municipality, and Fernando de Noronha, a state district — so 5,568 hold *vereador* elections, which is why the raw file below has 11,136 municipality-cycles. Councils are elected every four years. The 2016 election was held on the first Sunday of October; the 2020 election was not. Constitutional Amendment 107 of 2020 postponed it to 15 November on account of the pandemic and moved the close of registration from 15 August to 26 September. We return to what that does to the paper's out-of-sample exercise at the end of this subsection. Six features of the rules do the work in what follows, and none of them is incidental to the model.

The council's size is capped by law and chosen within the cap. Art. 29, IV of the Constitution, as amended in 2009, sets a *limite máximo* by population band — nine councillors up to fifteen thousand inhabitants, eleven to thirty thousand, thirteen to fifty thousand, and so on to fifty-five. The number actually seated is fixed by the municipality's *Lei Orgânica* within that ceiling. In our data councils run from nine seats to fifty-five with a median of nine and a ninetieth percentile of thirteen, and two-thirds of municipalities sit in the smallest band.

The distinction is not pedantic, because municipalities use the discretion. Matching IBGE population to the odd-sized councils in the frame, 86.3% seat exactly their ceiling and 13.3% seat fewer — and the second figure is concentrated in precisely the band this paper leans on. Where the ceiling is nine, 99.5% sit at it and none sits below, because nine is also the constitutional floor; the remaining half per cent is measured *above* its band, which the statute does not allow. We read that as error in the band assignment rather than in the council, and the likeliest source is the population vintage that fixes the band — the same difficulty Section 6.5 documents for its own design. Where the ceiling is eleven, only 75.7% do: **a quarter of municipalities entitled to eleven councillors**

choose nine or fewer. At ceilings of thirteen and fifteen the shares choosing down are 31.4% and 35.0%. Across all bands, 0.4% of markets sit above their ceiling.

Because the bands are steps, two towns of nearly the same size can still have councils differing by two seats, and that is the variation Sections 6.5 and 6.9 use. But the step is a ceiling rather than an assignment, so the design is fuzzy for a reason that sits in the statute and not only in our measurement. Section 6.5 reports what we can and cannot establish about the size of that step on population, the variable the law assigns on, and the answer is less than we would like.

Seats are won by a list and then by a person. A voter casts one vote, either for a named candidate or for a party label. A list's total is the sum of its candidates' personal votes and its label votes. The electoral quotient is the total valid vote divided by the number of seats, and a list takes one seat for each whole quotient it reaches; seats left over are allocated by highest average. *Within* a list the seats go to the candidates with the most personal votes, which is what makes the list open, and which is why a candidate cares how many others stand beside him and not only which party he joins.

One qualification matters for how we measure the prize. Since the reform of 2015, art. 108 of the Electoral Code seats a candidate on his party's quotient only if he has personally polled at least a tenth of the electoral quotient; places a list cannot fill under that floor pass to the highest-average round. So a list's seats are not distributed among its members by rank alone, and the model's per-candidate prize $B_{m,j}/s_{m,j}$ — the list's seats over the people who joined it — is the *average* of a distribution that the personal vote makes very unequal, not any individual's chance. Section 6.3 builds the prize as that average and Section 7 records the candidate-level version as the obvious next measurement.

The registry records the disposition of each candidacy: the category ELEITO POR QP — elected by quotient — identifies the lists that cleared, and the registry separately shows which lists took the remaining seats by highest average. Together the two categories are the published fact Section 4.1 conditions on, rather than a quantity we construct.

Who competes for the leftover seats, and the fact that it changed between our two cycles. The model pays a list nothing until it reaches the quotient, and that is the law in 2016 but for two lists, and not the law in 2020. Under art. 109 of the Electoral Code as it stood after Lei 13.165/2015, only parties and coalitions that reached the electoral quotient competed in the highest-average round. Lei 13.488/2017 replaced that with a rule admitting every party that contested the election, in force from 2018. The admission rule is all that changed. Art. 109, I still awards each leftover seat to the list with the largest average *and* an unelected candidate at a tenth of the quotient, so a list below the quotient whose candidates all sit beneath that floor gains nothing from the reform.

The registry shows the change with unusual clarity. Counting lists that took a seat by highest average without taking one by quotient — lists the model assigns δ_j and no prize at all:

	2016	2020
lists with a *média* seat and no quotient seat	2	4,691
as a share of lists	0.0%	11.7%
seats they take, as a share of all seats	0.0%	8.1%

The threshold rule is exact but for two lists in 2016 **and wrong for one list in eight in** 2020**.** Two consequences follow and we would rather state them than let a reader find them. Proposition 2 says a non-clearing list's log-odds depend on nothing but $\delta_j - \delta_k$; in 2020 that is false for the eighth of lists whose leftover prospects were real, and the dispersion of non-clearing shares which Section 2.3 reads as evidence for carrying δ_j has a second explanation in that cycle. And Section 6.8 fits on 2016 and predicts 2020 across this change as well as the others, which is one more thing the parameters are asked to survive.

So the model's winner rule is exact but for two lists in 2016 **and wrong for one list in eight in** 2020, and the estimator inherits that, because the clearing branch is read off ELEITO POR QP. What it costs is measurable, and Section 6.4 measures it: the change of law gives us a cycle in which the misclassification is two lists and a cycle in which it is one list in eight, which is as close to a placebo as an institution is likely to provide. We have not rebuilt the model on the post-2018 rule; the quotient branch remains the rule the model states, while the headline and the branch comparison in Section 6.4 are estimated on the any-seat branch. The cap, confound and concentration exhibits below are now estimated on the any-seat branch, with the headline.

The council is elected alongside a mayor. The two contests are held on the same day and on separate ballots: a voter casts one vote for mayor and one for the council, and the two are registered, counted and allocated separately. A party that fields a mayoral candidate campaigns its council list alongside that candidacy, and it is that association, not a joint ballot, that we record: a list one of whose parties also fielded a mayoral candidacy. The difference between such lists and the rest is large. Of the lists in our frame, 59.3% in 2016 and 41.7% in 2020 carry a mayoral candidacy by one of their parties. Those lists clear the quotient 74.0% of the time against 57.4% for the rest, and the lists whose mayoral candidate *won* clear 98.3% of the time and take 3.72 seats against 1.22.

That is a large fact, and the natural worry is that a candidate is choosing a mayoral ticket rather than a list, in which case the prize term would be standing in for the mayoral race. Section 6.5 adds the mayoral indicators to the within-market specification and $\hat{\alpha}$ moves by 0.020: the association is a determinant of list choice in its own right, and it is not confounded with the prize.

Coalitions existed and then did not. Until 2019 several parties could register a single joint list, pooling their votes against the quotient and dividing the resulting seats by personal vote across the whole coalition. Constitutional Amendment 97 of 2017 abolished proportional coalitions with effect from the 2020 election. The break is complete in the data: 80.0% of 2016 lists carry more than one party and 0.0% of 2020 lists do. A list is therefore not the same object in the two cycles, which is what makes the out-of-sample exercise of Section 6.8 demanding.

How many may stand is capped, and who may stand is constrained. Under art. 10 of Lei 9.504/1997 a party or coalition may register candidates up to 150% of the seats at stake, raised to 200% for a *coalition* in a municipality with at most a hundred thousand voters. Only 2.6% of our markets exceed that threshold, so in 2016 the operative ceiling was 200% for the four lists in five that were coalitions and 150% for the rest; after the ban, 150% for everyone. Both ceilings are visible and both bind (Section 6.6).

The same article requires each list to fill between 30% and 70% of its places with either sex. Women are 32.5% of 2016 candidacies and 34.7% of 2020 candidacies — just above the floor, which is the signature of a constraint that binds rather than a preference that happens to comply.

What a candidate knows when he chooses. Registration closes before the campaign — on 15 August in 2016 and, after the postponement, on 26 September in 2020. At that point the council's size is fixed and public, the parties' performance in the previous cycle is a matter of record, and this election's votes are not. That ordering is why Section 6.3 builds the prize from the first two and refuses the third, and it survives the change of date: what moved was the calendar, not the order of the calendar.

The second cycle is not the first cycle with the coalitions removed. Section 6.8 fits on 2016 and predicts 2020, and reads the exercise as a test of whether the parameters survive EC 97/2017. They had to survive more than that. The 2020 campaign was conducted under pandemic conditions, on a compressed calendar, with turnout and campaigning both disturbed. And two changes to campaign finance sit between the cycles: 2016 was the first election after corporate donations were struck down and spending limits imposed, and 2020 the second under the public campaign fund created in 2017, which changed how much money a party can put behind a candidate — that is, it changed ξ_p and δ_p, the objects the out-of-sample exercise holds fixed. The exercise asks the parameters to survive all of this at once. It does not isolate the coalition ban, and Section 6.8 does not claim that it does.

6.2. Data

Two families of file from the Tribunal Superior Eleitoral, for the municipal cycles of 2012, 2016 and 2020, restricted to the *vereador* (councillor) contest.

Candidacies, from `consulta_cand`: 926,900 candidate-cycles over the two analysis cycles, 437,368 in 2016 and 489,532 in 2020. Each record carries the municipality, the list the candidate stood on, party, gender, education, date of birth, and `DS_SIT_TOT_TURNO`, the disposition of the candidacy.

Votes, from `votacao_partido_munzona` for all three cycles and `votacao_candidato_munzona` for the two analysis cycles. The party files supply the lagged municipal vote shares that build the anticipated prize; 2012 enters only through that lag and through incumbency, and never as an estimation cycle. The vote schema changes before 2014 and the column names are handled separately for 2012.

Construction. A market is a municipality crossed with a cycle. Of 11,136 such markets, 11,129 have a council of positive size and form the frame; the seven dropped are municipalities whose results the registry does not record as filled. They hold 65,667 market-lists of the 65,742 in the raw files. Nothing else is excluded: every registered candidacy in a retained market enters, including candidacies later annulled, because the choice we model is the decision to register and it was made.

Incumbency is constructed by matching CPF across cycles, so it means *sat on this council*, not *ran before*. The type is the demographic profile — gender × higher education × four age bands × incumbency, thirty-two cells — and it is common across markets by construction, which is the separation Section 3 requires between what varies across markets and what does not.

Three rows of Table 5 carry the weight of the design. Almost every market has at least two clearing lists, 99.2% of them, so the contrast of Section 3.2 is available almost everywhere and the prize varies across alternatives within a market rather than only across markets. Three cells in five entered two or more lists, which is what a within-cell comparison needs. And the median cell holds two candidates while the mean holds five, so the panel is a great many small cells rather than a few large ones — the fact Section 5 turns on.

Table 5: The estimating frame

	mean	p10	median	p90	max
Markets, 11,129					
candidates	83.1	26	55	163	1,864
lists	5.9	2	5	11	33
council seats	10.4	9	9	13	55
Lists, 65,667 across 36 parties					
candidates	14.1	5	13	23	83
Type cells, 193,178					
candidates	4.79	1	2	11	221
Shares					
lists clearing the quotient					0.655
markets with two or more clearing					0.992
cells entering two or more lists					0.603
seats per clearing list, mean					2.59

Who stands. Candidates are 32.5% female in 2016 and 34.7% in 2020, 23.8% and 27.2% hold a degree, 9.5% and 8.4% sat on the council already,[3] and the mean age is 44.4 and 45.3. Thirteen per cent of 2016 candidacies and twelve per cent of 2020 candidacies ended in a seat.

List choice is not a demographic block decision. In a regression of a candidate's list choice on his demographics within municipality and cycle, the within-R^2 is 0.0008 in 2020 and 0.0007 in 2016; among type cells with ten or more candidates, essentially none is unanimous and the average cell spreads across about ten distinct lists. Section 2.5 draws on this fact to argue that list choice is an individual action rather than a block recruited by demographic factorial.

6.3. The prize, and what is anticipated

The model's B_j/s_j is a list-level index of seat opportunity per candidate, not an individual's probability of election. Its empirical counterpart is seats per candidate, and the whole difficulty of the application is that "seats" can mean several things.

Realised seats will not do. The obvious counterpart, $\text{seats}_{m,j}/n_{m,j}$, is an outcome of the same election in which the candidate chose. This is the simultaneity a demand system meets when the aggregate enters the payoff (Berry, Levinsohn, and Pakes 1995; Berry and Haile 2014), with the difference that the aggregate here is the divisor rather than a price. A list wins seats partly because this cell's members won them, so the prize is mechanically higher exactly where the cell is present. On this panel a cell's own members hold 5.6% of the seats that define their list's prize. Removing them moves the estimate by very much more than 5.6% — from +0.32 to −0.083 by the contrast estimator — because the leak is concentrated in precisely the variation that identifies α.

[3] Incumbency is coded from the candidate's tax identifier appearing on a winning row at $t-4$, in any municipality. Two things follow. About 0.06% of incumbent rows are flagged through a seat won in a different municipality from the one where the candidate now stands, so they mark someone who did not sit on *this* council. Larger, and running the other way, 2,559 candidates in 2016 were 2012 *suplentes* who took a seat mid-term: the electoral authority treats them as standing for re-election and this flag does not.

Removing them is not the repair either. What the correction subtracts is the cell's *realised* success, so a cell that placed many members on a list and won there has the most taken away, and a downward mechanical link replaces an upward one. Section 4.2 states the general point: no function of realised seats is clean.

Lagged seats have no leak and little signal. The natural alternative, the same party's seats in the same municipality last cycle, correlates with realised seats at 0.084 and is positive for only 29.0% of lists: for none in 2016, whose markets have no 2012 *seat* file behind them, and for 47.6% in 2020, so the specification collapses to a single cross-section. By the contrast estimator its coefficient is +0.004 with a standard error of 0.010. The likelihood, which reads the zero counts as well, gives it +0.398, less than half of what it gives the construction below, on a measure that tracks realised seats a seventh as well.

The construction we use. Two objects exist before anyone registers. The council's size is set ahead of the election, within the art. 29 ceiling, and does not depend on who registers. We measure it as C_m, the seats the market actually filled, because the statutory size is not in the registry — which means the measure does carry something from this election wherever a seat went unfilled, and the claim below that the numerator contains nothing from this election is true of the forecast and not of its scale. Even councils are the visible symptom: 1.4% of markets have one, and Brazilian councils are odd-sized. The party's share of the municipal vote last cycle, $v^{\text{lag}}_{m,p}$, is on disk for *both* cycles — 2012 for the 2016 markets, 2016 for the 2020 ones — which is why the vote share works where lagged seats do not. We forecast the seat share from those two and convert,

$$\frac{\text{seats}_{m,j}}{C_m} \;=\; \pi_0 + \pi_1 v^{\text{lag}}_{m,p(j)} + \pi_2 \big(v^{\text{lag}}_{m,p(j)}\big)^2 + \pi_3 \log C_m + \varepsilon_{m,j}, \tag{28}$$

and the prize is C_m times the fitted share over the leave-one-cell-out denominator, which is (22).

The first stage has $R^2 = 0.447$ against 0.112 for council size alone, so the lag adds 0.335, and $\hat{\pi}_1 = +0.688$. The resulting measure correlates with realised seats at 0.546, against 0.084 for lagged seats, and is built from a lagged vote share that is positive for 86.0% of lists against 29.0% for lagged seats. The forecast itself is clipped below at zero, which puts 1.0% of lists — 690 of them — on a prize of exactly zero. That violates Assumption 1's $B_j > 0$, and it is a live violation rather than a bounded one: 56.2% of the clipped lists took a seat, so 388 of them enter the clearing branch carrying a prize of zero. We note it and do not repair it here. What it costs is that the payoff of those lists is the constant $X(\tau)'\beta + \xi_j$ on a branch where the model expects the prize to vary, so they contribute to β and to ξ and nothing to α.

The prize is an average, and lists are not exchangeable. The quantity we build is the list's seats over its candidates, which is what every member would face if personal votes were spread evenly. They are not: the candidates who take a clearing list's seats hold 44.5% of its votes while being 16.7% of its candidates. Splitting the markets of sixty or more candidates into thirds by that concentration, $\hat{\alpha}$ is 2.27 in the least concentrated third and 1.83 in the most concentrated, a difference of 0.44 against a standard error of 0.18. The coefficient is lower where a list's seats are held by fewer of its candidates, which is the direction to expect if the list average overstates the opportunity most candidates on such a list face. The approximation therefore matters for the magnitude, by about a fifth across the variation Brazil supplies, and not for the sign; the data hold no exchangeable region to compare against (Section F.11).

(28) deliberately omits $\log n_{m,j}$. An earlier version included it, on the reasoning that a bigger list wins more seats, and that reintroduces the contamination through the fitted value: $n_{m,j}$ contains the cell whose choice is the outcome. The forecast must be built from objects that predate registration or it is not a forecast.

What each construction gives. The four, under the log-odds contrast and under the likelihood (24), on all markets and without the split-sample correction of Section 5.7, which is applied to the construction chosen here and not to the three it is chosen over:

prize	numerator leak	contrast	likelihood
realised seats	this cycle	+0.317	+1.965
realised, cell's seats removed	removed	−0.083	+0.163
lagged seats	none	+0.004	+0.398
ex ante (22)	**none**	**+0.098**	**+0.876**

Two-stage least squares, instrumenting the realised prize with the ex ante one at a first-stage correlation of +0.533, gives +0.441.

The first two rows bracket rather than sandwich, for the reason given above. The third has no leak and a measure that barely tracks seats. The fourth is the one with nothing from this election in its numerator, and the gap between its two columns is the subject of Section 4.3: the likelihood reads the zero counts that a log-odds cannot, and that difference is 91% of the gap. The contrast column is computed on every market and the likelihood column on the markets whose council filled a seat. The 0.876 is the uncorrected coefficient on all markets; the estimates of Section 6.4 are on markets of sixty candidates or more and carry the split-sample correction.

6.4. Estimates

The estimator is the split-sample estimator of Section 5.7. The prize (22) is built twice, from two random halves of a market's candidacies, with the reciprocal of the counted share replaced by its unbiased counterpart; one half's prize enters the likelihood (24) and the other half's instruments it, through a first-stage residual added as a control; the two orientations and two random splits are averaged. Party effects are at party by state. Standard errors are clustered at the municipality across both cycles and account for the first stage: each municipality's influence on the estimate is the second stage's score plus the first-stage coefficient's contribution, averaged over orientations and splits before the variance is formed. In the simulation of Section 5.7 that standard error matches the standard deviation of the estimates, 0.272 against 0.273 at the scale of this sample.

The sample is the markets fielding sixty candidates or more with a council that filled at least one seat: 5,089 markets, 100,970 cells and 903,542 triples of market, type and list, of which 56.6% are zero counts, with 883 party-by-state cells and 2,891 municipality clusters. 67.0% of lists take at least one council seat.[4] Three choices in this specification are not conventions.

The likelihood rather than log-odds contrasts, because a type that enters no member on a list supplies no log-odds and 56.6% of the sample is exactly that (Section 4.3).

The anticipated prize rather than realised seats, because a list wins seats partly because this cell's members won them, so realised seats per candidate is mechanically higher where the cell is present (Section 6.3).

[4]The any-seat branch; the quotient-clearing share is 0.655 on the full descriptive sample (Table 5).

Markets of sixty candidates or more, because the correction needs them. It instruments a prize counted from half of a market's candidates, and in simulation it recovers 91% to 93% of a known coefficient at 86 and 138 candidates but 82% at 55, with coverage of 72% (Table 4). Below sixty the first stage is too weak to carry the estimate.

Headline.

	all candidates	non-incumbent only
α	+1.9210 (0.0739)	+1.3305 (0.0778)
β, female	−0.0024 (0.0031)	−0.0028 (0.0031)
β, higher education	+0.3300 (0.0111)	+0.3271 (0.0112)
β, age (standardised)	+0.0570 (0.0042)	+0.0575 (0.0042)
β, incumbent	+1.9570 (0.0301)	

Notes: 5,089 markets; 903,542 rows in the first column and 666,481 in the second. β, incumbent is blank in the non-incumbent column because the covariate is identically zero on that sample. The four fits behind each $\hat{\alpha}$, two splits by two orientations, range from 1.897 to 1.958 in the first column and from 1.320 to 1.351 in the second.

The two columns differ in what they can be asked to mean. An incumbent is by construction someone whose party won here last cycle, and the anticipated prize is built from that party's lagged vote share, so an incumbent sits on a high-prize list for reasons of history as much as of choice. The second column drops them. It does not remove the tie altogether: 24.5% of the non-incumbents stood somewhere four years earlier, and standing four years earlier is not part of the type. **We read** $\alpha = 1.33$ **as the estimate and** 1.92 **as its upper end.**

The standard errors can be used. Section 5.7 finds that the corrected estimator still returns somewhat less than the truth, about 0.10 below a coefficient of 1.5 at these market sizes, and that nominal 95% intervals cover the truth between 90% and 96% of the time and the estimator's own centre between 94% and 98%. An interval around 1.33 is therefore a little too low in its location and close to right in its width. The errors on β cover at close to their nominal rate, and β is the same to the third decimal whether or not the prize is corrected.

The branch definition. The clearing branch can be read as the lists that took a seat by the party quotient or as the lists that took any seat. The two differ, in the registry, by 2 of 25,653 lists in 2016 and by 4,691 of 40,117 in 2020, the lists that took a seat by highest average without reaching the quotient (Section 6.1). Redefining the branch moves those lists and leaves the prize, the covariates and the rows as they were.

markets of 60+ candidates		by quotient	by any seat
2016, all candidates	*(2 of* 25,653 *lists differ)*	1.8516 (0.0868)	1.8515 (0.0868)
2020, all candidates	*(*4,691 *of* 40,117 *differ)*	2.0401 (0.0970)	2.0555 (0.1058)
both cycles, all candidates		1.9768 (0.0693)	1.9210 (0.0739)
both cycles, non-incumbent only		1.4123 (0.0732)	1.3305 (0.0778)

Notes: the 2016 fit pools 421 of 806 ξ levels and 214 of 756 δ levels onto donors; the 2020 fit pools none. The two cycle rows share the estimator and the specification but not that restriction.

In 2016 the statute made the two definitions agree, and $\hat{\alpha}$ moves by 0.0001: re-cutting the branch has no mechanical effect. In 2020, where an eighth of the lists change branch, it moves by 0.015, well inside one standard error, and in the pooled rows by 0.06 and 0.08, about one standard error. The definition of the branch is therefore not a material choice for the prize coefficient under this estimator. We report the any-seat branch throughout, because it is the branch each cycle's law implemented; the quotient branch is the rule the model is written on (Section 2.3), and the table gives it beside the headline.

The coefficient is larger in 2020 than in 2016, 2.06 against 1.85 on the any-seat branch, a difference of 0.20 against standard errors of about 0.10 on each. Four things changed between the cycles (Section 6.1), and we do not attribute the difference to any one of them.

The value of a party's name, in two regimes. δ disperses about five times as widely across parties as ξ does: within a state, the standard deviation of the 809 free δ levels is 0.727 and that of the 689 free ξ levels is 0.144. The comparison is made within states because the level of the party effects is identified only there (Section 3.4); taken across states, where an arbitrary shift per state enters, the two standard deviations are 0.733 and 0.171. What a party's label is worth to a prospective candidate varies much more among lists that will not clear than among lists that will. A label is worth something to a candidate because it is worth something to voters (Snyder and Ting 2002), and Brazilian labels differ enough in what they signal for that to bite (Power and Zucco Jr. 2009; Zucco Jr. and Power 2024).

Responsiveness by type. Letting a taste coefficient vary across observed groups is the discrete counterpart of the random coefficients a demand system carries (Nevo 2000). Figure 1 plots $\alpha(\tau)$ for the 32 types with their 95% intervals, arranged as the factorial the type is: panels by gender and education, age on the horizontal axis, incumbency as the two series.

All 32 estimates are positive, and 31 are more than two standard errors from zero; the exception is women without a degree under thirty-five who are not incumbents, at +0.13 with a standard error of 0.11. The median is +4.77 and the range runs from +0.13 to +8.27. The common α of 1.92 comes from a different model, one that restricts the thirty-two coefficients to be equal, and it is not an average of them: in a logit the restricted coefficient depends on the distribution of the prize and of the choice probabilities within each type as well as on the types' sizes. The large types are non-incumbents, whose coefficients are the smaller: the mean over the sixteen non-incumbent types is 1.91 and over the sixteen incumbent types 6.89.

The one large regularity is incumbency, at +4.99 between the incumbent and non-incumbent means. We do not report it as a preference parameter, for the reason the figure's note gives and the one the next paragraph adds.

Education is the margin on which the coefficient and β point the same way with precision. Types with a degree average +4.80 against +4.00, a gap of +0.80 with a standard error of 0.13, and β on higher education is +0.33 in the headline: the prize moves the choice among lists more for the educated, and a clearing list is more attractive to them at any prize. The gap belongs to non-incumbent candidates. Among the sixteen non-incumbent types it is +1.18 (0.06), 2.50 against 1.32; among the sixteen incumbent types it is +0.41 (0.25). Women average +4.61 against men's +4.19, a gap of +0.43 (0.14). Age rises across all four bands, from +3.58 under thirty-five to +4.32, +4.71 and +4.99.

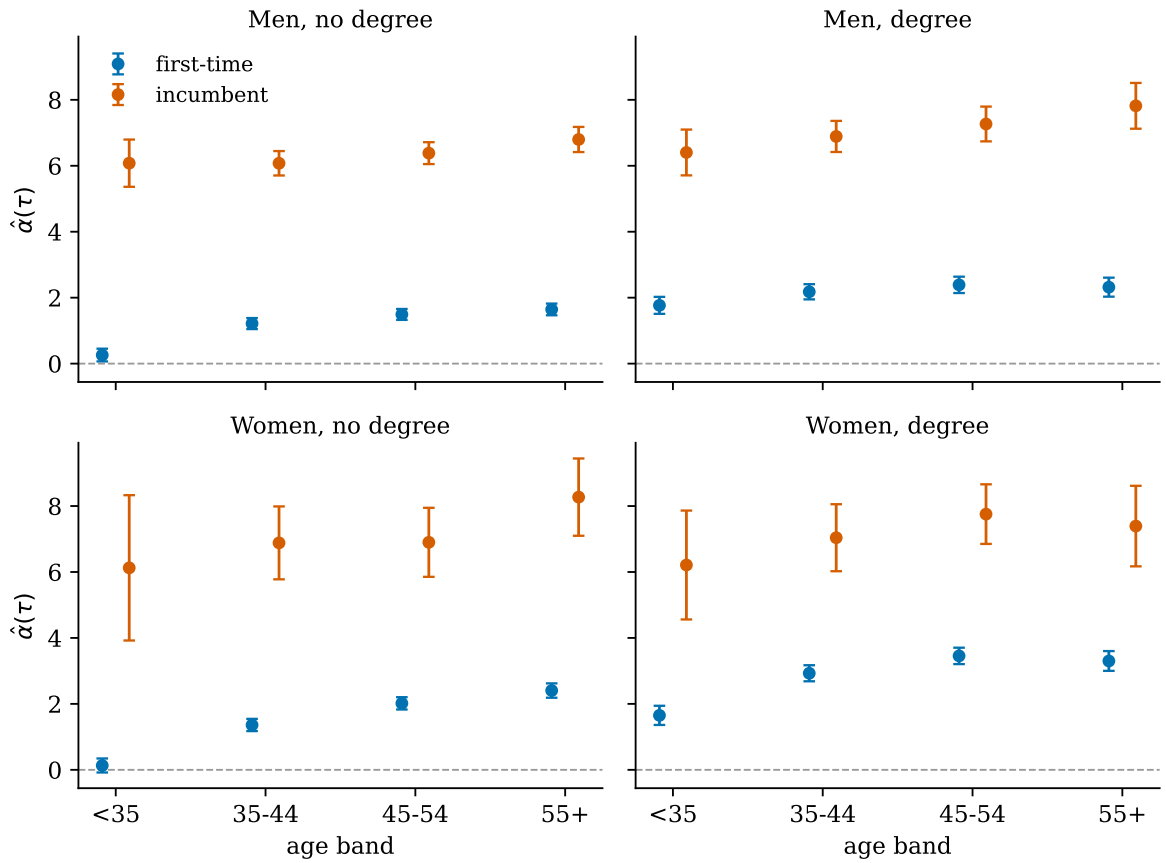


Figure 1: Marginal utility of the anticipated per-candidate chance of a seat, by type.

Notes: $\hat{\alpha}(\tau)$ from the split-sample estimator on markets of sixty candidates or more, with 95% intervals from municipality-clustered standard errors that account for the first stage. Types are gender × higher education × age band × incumbency. The dashed line marks zero. The incumbent series is inflated by the tie between incumbency and the party's lagged vote share, from which the prize is built, and should be read as an upper bound; the non-incumbent series carries a weaker form of the same tie, since 24.5% of non-incumbents stood four years earlier and the type does not record it.

The education gap with an effect for every list in every market. The by-type fit absorbs list appeal with party-by-state effects, so a list's local appeal stays in the error. Proposition 3 says that differences in α across types survive any list-level term common to the types, and the estimator that uses it puts a free effect on every list in every market. We fit it on the by-type fit's own frame as a Poisson regression of the cell counts with market-by-type and market-by-list effects, whose first-order conditions for the remaining parameters are the conditional logit's. Levels of α are absorbed by the list effects, and the coefficient of the reference type is a normalisation.

The split-sample correction is applied here as it is elsewhere. Seat opportunity is one number per list, anticipated seats times the unbiased reciprocal of the list's share among a random half of the market's candidacies; the other half's instruments it, in a first stage across the clearing lists of a market, and the first-stage residual enters beside the regressor, each interacted with type. Standard errors account for the first stage and are formed from influences averaged over two splits and both orientations.

The gap between the mean coefficients of the degree and the non-degree types is +2.60 with a standard error of 0.17 among the sixteen non-incumbent types, and the four fits behind it range from 2.52 to 2.65. Among the

sixteen incumbent types it is +0.60 (0.61), with fits from −0.22 to +1.44, and over all types +1.60 (0.32). Local list appeal that is common to the types therefore does not produce the education gap; removing it leaves a larger one among non-incumbents and none that can be distinguished from zero among incumbents.

A placebo checks that the estimator does not produce a gap of its own, which it could, since a list's own candidates are in the denominator of its seat opportunity. Redrawing the counts 30 times from the model fitted with every difference in α restricted to zero, rebuilding the halves and the regressor from the redrawn counts and re-estimating, the mean gap among non-incumbent types is +0.01, its standard deviation across draws 0.094 against a mean standard error of 0.095, and a nominal five-per-cent test rejects in 2 of the 30 draws. Over all types the mean is −0.01 and the standard deviation 0.21 against a mean standard error of 0.24.

The two estimates of the gap among non-incumbents, +1.18 with party-by-state effects and +2.60 with an effect for every list, agree in sign and differ in size by more than their standard errors. They rest on different assumptions about list appeal (Section 3.3), and we report both and do not average them.

The prize is not the same quantity for every type on a list. The model hands every type on list j the same $B_{m,j}/s_{m,j}$. The institution does not: on a clearing list an incumbent takes 0.596 seats per candidate and a non-incumbent 0.108. The largest single feature of Figure 1, the gap between incumbent and non-incumbent types, lies along exactly the dimension on which the prize is most mismeasured, and we do not read that gap as responsiveness. Section F.7 reports the diagnostic and what it does and does not explain.

Patterns we do not interpret. Figure 1 shows a gender gap, within-sex incumbency gaps that are nearly equal at +5.06 for men and +4.92 for women, and an age profile that rises and flattens. We do not interpret them, because under the likelihood of Section 4 they were sensitive to the construction of the prize: computed on realised seats instead of the anticipated prize, the gender ordering reversed, the incumbency gap took opposite signs for the sexes, and age traced an inverted U. We have not repeated that comparison under the split-sample estimator.

What is settled and what is not. The prize coefficient is positive in every specification of this subsection and the next: for all candidates and for non-incumbents, on either definition of the branch, in each cycle, for every one of the thirty-two types, and with and without the control for local party strength (Section 6.5). Its magnitude without that control is 1.33 among non-incumbents and 1.92 among all candidates.

The correction for the measurement error in the prize accounts for most of what earlier made the magnitude unreportable. Section 5 shows the likelihood of Section 4 recovering three-fifths of a known coefficient at these market sizes, and on this panel that likelihood gives 0.72 and 1.20 where the split-sample estimator gives 1.33 and 1.92: ratios of 0.54 and 0.63 against a simulated 0.60. Two results of the uncorrected likelihood that had no explanation do not appear under the correction: two significantly negative type coefficients, and a reversal of the sign among non-incumbents when local strength was controlled.

Three things are not settled. The corrected estimator is still biased toward zero by about 0.10 in simulation, a bias that does not fall as markets are added, so 1.33 is more likely low than high. The level rests on the restriction that list appeal beyond a party-by-state effect is unrelated to the prize (Section 3.5), and Section 6.5 shows that the one design we have for relaxing it does not give a stable answer. And the coefficient is not the same in the two cycles, nor, on the evidence of that design, across council sizes.

6.5. Local party strength, and what the council-size jump identifies

One threat to Section 6.4 is not addressed by any of the choices made there. A party that is strong in a particular municipality has a larger anticipated prize, because the prize is built from its lagged vote share, and plausibly also a larger unobserved attraction: a machine, a sitting mayor, a claim on local patronage. Party effects at party by state absorb the national and the regional components of that attraction. They cannot touch the municipal one, which is exactly the component that moves with the regressor.

A separate simulation prices the residue. Adding a market-specific attraction correlated with party strength to the design of Section 5 takes the estimate from 0.01 to 4.85 times the truth, depending on how the prize is built — the confound is the difference between an α near a half and one near three and a half — against 0.78 to 12.16 in the configurations with no measurement error, where the confound alone does the damage. The run is not reported in Section 5, which carries the attenuation exercise only; it is in the verification log of 2026-09-05. This is the last thing in the paper that nothing bounds.

The mayoral ticket is a confound and it is measurable. The council is elected beside a mayor (Section 6.1), and a list carrying the mayoral candidacy is a different object: it clears 74.0% of the time against 57.4%, and 98.3% where its mayoral candidate won. If candidates are choosing a ticket, $\xi_{m,j}$ is largely the mayoral race and α inherits whatever that correlates with.

Unlike the local attraction of the previous paragraph, this one is observed. Building from the *prefeito* rows of the same registry — which party fielded a mayoral candidate in each municipality, and which won — and crediting a coalition with all of its members' candidacies, the indicators enter the likelihood as list-level terms, free to shift a list's appeal differently in the clearing and non-clearing branches.

markets of 60+ candidates	$\hat{\alpha}$	(s.e.)
baseline	1.9210	(0.0739)
+ carries a mayoral candidate	1.8770	(0.0735)
+ and that mayoral candidate won	1.8764	(0.0734)

$\hat{\alpha}$ moves by 0.044, about 0.6 of a standard error, which is the number that matters for whether the prize term was standing in for the ticket. It was not.

The mayoral terms themselves are not negligible: on a clearing list a winning mayoral candidacy is worth about a fifth of what the prize moves across the median cell, and a mayoral candidacy attracts on a list that clears and repels on one that does not. Section F.1 reports the coefficients by branch and their size in the prize's own units.

So the mayoral association is a real determinant of list choice. What the table shows is the narrower thing: it is not *confounded* with the prize. How much of the association runs through the seats a list wins, which the prize already measures, the table cannot say, because a coefficient that holds still under an added control does not decompose paths.

The instrument is in the Constitution. Population thresholds that assign local institutions are a standard source of quasi-experimental variation, and the design here is the one Folke (2014) uses to recover party effects under

proportional rules. Art. 29 caps the size of the council by population band, so the ceiling on C_m steps from nine seats to eleven at fifteen thousand inhabitants.

Two things make the step fuzzy and only one of them is the statute's. Art. 29 sets a maximum and the *Lei Orgânica* chooses within it, and Section 6.1 reports that a quarter of the municipalities entitled to eleven councillors seat nine or fewer. On top of that we measure C_m as seats *filled*, which understates the statutory size wherever one went unfilled.

The first stage is weaker than the statute suggests, and Section F.2 reports it in full. On IBGE population, the variable the law assigns on, the estimated jump in the probability of seating eleven or more falls from 0.319 at a bandwidth of eight thousand inhabitants to 0.044 at one thousand, where it cannot be told from zero. We attribute that to the vintage of the population files we hold and not to the law being ignored, but we cannot confirm a local jump on population. On the municipal vote total, the variable the estimation frame is built on, the share of markets with eleven seats or more rises from 0.3% below 7,000 votes to 80.2% between 11,500 and 14,000. The results that lean on the design, here and in the counterfactual check of Section 6.9, should be read as suggestive.

The exclusion restriction. Write the anticipated prize (22) as

$$P_{m,j} \;=\; C_m\, z_{m,j}, \qquad z_{m,j} \;=\; \frac{\widehat{\text{seatshare}}\big(v^{\text{lag}}_{m,p(j)},\, C_m\big)}{n_{m,j} - n_{m,\tau,j}}, \tag{29}$$

so the council enters multiplicatively and z carries everything about local party strength. *The confounding attraction is a function of that strength. It is not a function of the council's size.* Entering both columns,

$$V_{m,\tau,j} \;=\; \mathbf{1}\{j \in W_m\}\big[X(\tau)'\beta + \alpha\, C_m z_{m,j} + \gamma\, z_{m,j} + \xi_{p(j)}\big] + \mathbf{1}\{j \notin W_m\}\,\delta_{p(j)}, \tag{30}$$

γ absorbs any attraction that is a smooth function of local strength and α is identified only by how much more strongly shares respond to z where the council is larger. Within a market the two columns are proportional and nothing separates them; across markets their ratio is C_m, and near the threshold C_m is as good as randomly assigned.

Result. Estimating (30) with the split (29), on non-incumbent and on all candidates, by the split-sample estimator of Section 5.7. Both regressors, $C_m z_{m,j}$ and $z_{m,j}$, carry the counted share in their denominator, so each is built from one random half of the market's candidacies and instrumented by its counterpart from the other half, with its own first stage:

window	markets	α, no γ (s.e.)	α, with γ (s.e.)	γ (s.e.)	$t(\gamma)$
all markets, non-incumbent	11,129	1.4587 (0.0568)	0.5651 (0.2370)	8.4014 (2.1403)	3.93
all markets, all candidates	11,129	1.9427 (0.0555)	2.2924 (0.2298)	−3.4773 (2.0710)	−1.68
60+ candidates, non-incumbent	5,089	**1.3305** (0.0778)	0.9953 (0.2331)	3.1505 (2.2425)	1.40
60+ candidates, all candidates	5,089	1.9210 (0.0739)	2.4800 (0.2301)	−6.4159 (2.2067)	−2.91
votes 4k–20k, non-incumbent	6,331	1.4256 (0.0750)	2.4678 (0.3814)	−10.1746 (3.5605)	−2.86
votes 4k–20k, all candidates	6,331	2.1071 (0.0782)	3.6592 (0.3833)	−15.1577 (3.5340)	−4.29
votes 6k–14k, non-incumbent	3,444	1.4797 (0.1056)	3.1042 (0.5678)	−16.0276 (5.3819)	−2.98
votes 6k–14k, all candidates	3,444	2.2166 (0.1113)	4.0845 (0.5800)	−18.4815 (5.4302)	−3.40
votes 7.5k–11.5k, non-incumbent	1,697		withheld[a]		
votes 7.5k–11.5k, all candidates	1,697		withheld[a]		
votes 8.25k–10.5k, non-incumbent	993	1.9597 (0.2033)	3.8859 (0.9866)	−19.3456 (9.2348)	−2.09
votes 8.25k–10.5k, all candidates	993	2.7177 (0.2263)	4.4427 (1.0285)	−17.3732 (9.4686)	−1.83

[a] The specification stops on a registered pre-fit gate on this window; the paper prints no estimate from it.

The table reports non-incumbent and all-candidate fits of (30), with and without the control γ, on the paid branch, the sample restricted to markets whose council filled at least one seat, and party-by-state pooled maps built on each design's own frame. Standard errors are clustered by municipality and account for both first stages. The non-incumbent, no-γ cell on 60+ candidates is the headline estimate of Section 6.4. The all-markets rows include markets below sixty candidates, where Section 5.7 finds the correction weaker.

What the table shows. Three things can be read off it.

The prize coefficient is positive in every cell. Without the control it is between 1.33 and 2.72, with t-ratios from 10 to 35. With the control it is between 0.57 and 4.44, and the smallest t-ratio is 2.4. No sample and no specification gives a negative coefficient.

The control does not move the coefficient in one direction. The confound this design was built to detect is an attraction to locally strong parties that inflates $\hat{\alpha}$; it predicts $\hat{\gamma} > 0$ and a smaller $\hat{\alpha}$ once γ is in. That is what the two non-incumbent rows of the wide samples show: on all markets $\hat{\gamma} = 8.40$ with a t-ratio of 3.9 and $\hat{\alpha}$ falls from 1.46 to 0.57; on the markets of sixty or more $\hat{\gamma} = 3.15$ is not distinguishable from zero and $\hat{\alpha}$ falls from 1.33 to 1.00. In the other eight cells the control raises $\hat{\alpha}$ and $\hat{\gamma}$ is negative, significantly so in most of them.

A negative γ is not the confound, and it is not uninterpretable either. The two columns enter as $z\,(\alpha C_m + \gamma)$. On the markets of sixty or more with all candidates, $\hat{\alpha} = 2.48$ and $\hat{\gamma} = -6.42$ give a response to local strength of 15.9 at nine seats and 45.7 at twenty-one, positive at every council size the panel contains; a negative γ says that the response is not proportional to the council's size and has a negative intercept. At least three things would produce that. The value of a unit of seat opportunity may rise with the size of the council. A list that is strong per candidate is one whose votes are held by a few of its candidates (Section 6.3), so that for most candidates its seats are less attainable than the list average the model assigns them; the coefficient is indeed lower, 1.83 against 2.27, in the third of markets where a list's seats are most concentrated among its candidates (Section 6.3). Or utility may be convex in seat opportunity. The design cannot tell these apart.

What the design does not deliver. In the narrow windows C_m takes little more than two values, and the data determine the combination $\alpha C_m + \gamma$ much better than its two parts. On the narrowest window the controlled non-incumbent estimates, 3.89 and −19.3, give 19.5 at ten seats, and the uncontrolled estimate, 1.96, gives 19.6: the control has redistributed one response between two coefficients. The window estimates of γ should not be read at face value, and the growth of $\hat{\alpha}$ as the window narrows is the same redistribution. Identification is not confined to the threshold either, since in the full sample and the wide windows part of the variation in C_m is large councils in large municipalities. Density tests for manipulation of the population threshold are inconclusive, for the reason the population first stage is. Section F.3 reports all three.

What follows for the level. The design gives no stable estimate of how much of the uncontrolled coefficient is local strength. Read with its least favourable cell it puts the non-incumbent coefficient at 0.57 in place of 1.46 on all markets, and at 1.00 in place of 1.33 on the markets of sixty or more; read with its other cells it puts the coefficient above the uncontrolled one. We therefore keep 1.33 as the estimate, report 1.00 and 0.57 as what a positive local attraction of the estimated size would leave of it, and do not claim that the design has either removed the confound or shown it to be absent. What it does establish is the sign: the coefficient is positive with the control and without it.

The simulation of Section 5.7 bears on how far that can be trusted. With an attraction to local strength in the design and the same two-regressor estimator, the estimate is 1.45 where the coefficient is 1.5, 0.12 where it is zero, and −0.41 where it is −0.5: the estimator does not return a positive coefficient where there is none. The simulation holds market size fixed across council sizes, which the data do not, and it has no counterpart of the negative γ.

6.6. The registration cap

Section 6.1 states the rule: a list may register up to 150% of the seats at stake, raised to 200% for a coalition in a municipality with at most a hundred thousand voters. In 2016 four lists in five were coalitions and only 2.6% of markets clear the voter threshold, so the operative ceiling was 200% for most lists and 150% for the rest; the abolition of coalitions left 150% for everyone in 2020. Whether the cap binds is not a detail. If it does, list size is not a free choice, and the relation between a list's size and its seats — which is what the prize B_j/s_j is built on — cannot be read off the cross section.

It binds, hard. Figure 2 is the whole argument. In 2016 the ratio of list size to council size, among the coalition lists the 200% ceiling governs, has its ninetieth and ninety-ninth percentiles both at exactly 2.00; among the solo lists governed by 150% the ninetieth percentile is 1.52. Taking all 2016 lists together the ninetieth, ninety-fifth and ninety-ninth percentiles are 2.00. Grouping lists into bins of width 0.1, the bin just below the ceiling holds 1,094 lists, the bin containing it holds 4,546, and the bin just above holds 13. In 2020 the same shape appears at the new ceiling: 4,783 lists in $[1.5, 1.6)$ and five in $[1.6, 1.7)$. Seventeen and three-quarter per cent of 2016 lists sit within 0.02 of two times the council.

The excess mass is measurable by the usual bunching arithmetic (Saez 2010; Chetty et al. 2011; Kleven 2016). Against a counterfactual density fitted away from the ceiling, the window $[1.88, 2.03]$ holds 5,636 lists where 1,531 are expected, an excess of 4,105 and a normalised bunching of 8.04 bins. Under the usual bunching

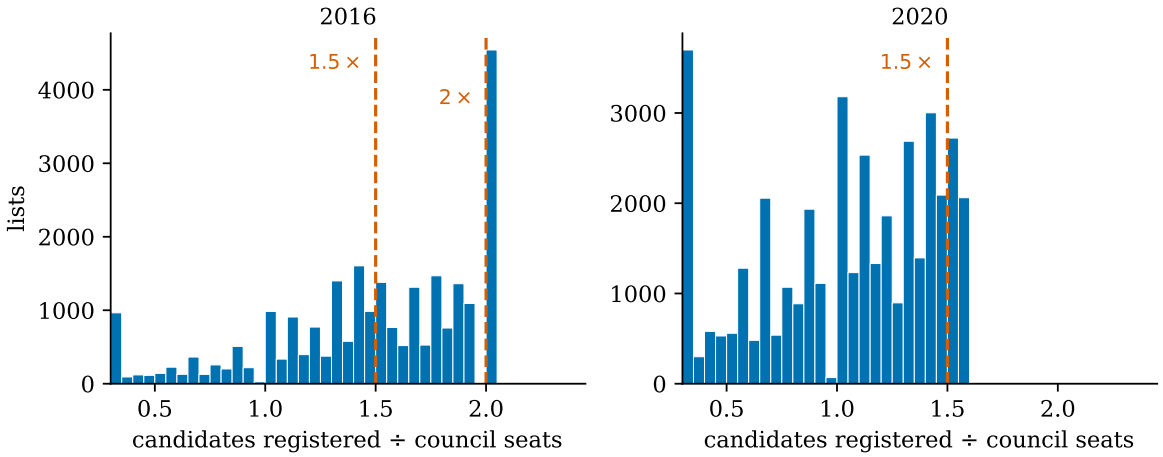


Figure 2: Registered candidates per council seat, by cycle.

Notes: one bar per 0.05 of the ratio of a list's registered candidates to the municipality's council seats, over all lists in markets with a council of positive size. Dashed lines mark the statutory multiples of art. 10: 150% always, and 200% for a coalition in a municipality of at most a hundred thousand voters, which in 2016 covers four lists in five. Coalitions were abolished for 2020, leaving one ceiling. The council is measured as seats filled, which understates the statutory size wherever one went unfilled, so some mass sits just beyond the lower line; the cliff rather than the line is what locates the ceiling.

arithmetic the marginal buncher wanted a ratio of 2.40: **the constrained lists wanted about a fifth more candidates than the law allowed them**.

What the cap change identifies. The move from 2.0 to 1.5 between the cycles is a shock to list size that is not a shock to a party's prospects. Differencing party by municipality across the two cycles and taking treatment to be whether the party sat at the 2016 ceiling:

	first stage	reduced form	implied
	Δ candidates	Δ seats	seats per candidate
A. no partner control	+0.637 (0.065)	−0.038 (0.019)	−0.060
B. + partner count	+0.587 (0.069)	−0.113 (0.019)	−0.192
C. + partner count squared	+0.580 (0.069)	−0.113 (0.019)	−0.195
D. coalition members only	+0.762 (0.067)	−0.110 (0.019)	−0.144

The first stage is strong in every specification. The reduced form is negative and, once the number of coalition partners is controlled, sharply so. The implied causal relation runs from −0.06 to −0.19 seats per additional candidate.

What moves list size here is the coalition ban, not the cap. The first stage is positive, and it has to be read carefully, because the cap *tightened* between the cycles — from twice the council to one and a half times for the parties in the treatment group, who were at the coalition ceiling in 2016. A tightening cap cannot raise anyone's list. What raises it is that a 2016 coalition dissolves into its member parties, each of which then fields a list of its own instead of a share of a joint one, so the party's *own* candidate count goes up by 0.637 against the control. The instrument is therefore EC 97/2017 operating through list size, and the cap enters as the reason the treated

parties were at a ceiling to begin with. This design does not separate the two, and the clean version — 2016 solo lists, whose ceiling of 150% did not move and which have no exposure to the ban — is not one we have run.

The comparison that matters is with the cross section, which gives +1.27. A party that fields more candidates in a municipality wins more seats there, by a wide margin, and essentially all of it is selection: parties field more where they expect to do well. Made to field more by a change in the law and not by their own prospects, they win slightly *fewer*. The sign reverses.

Why this bears on the estimator. Section 4.2 insists on removing a cell's own members from the prize and Section 6.3 on building the numerator from objects that predate the election. Both look fussy until this table. The uncorrected relation between how many candidates a list carries and how many seats it takes is off by more than the whole magnitude of the effect and has the wrong sign, so a prize built from realised seats over realised list size inherits that. This is the same defect, measured by a different design, that made Section 6.4 discard its first headline.

Does the binding cap censor the estimating sample? A list at the ceiling holds the number of candidates the law allowed and not the number free choice would produce, so its $\hat{\sigma}_m(j \mid \tau)$ is censored, and in the estimating sample of markets with sixty candidates or more 18.2% of lists are at the ceiling. The censoring bites little. Dropping every list at the ceiling, and every candidate who stood on one, moves $\hat{\alpha}$ from 1.92 to 1.83, a fall of about a twentieth and a little more than one standard error of either estimate (Section F.5). The cap constrains *how many* a list may register; parties also choose *whom*, and that we do not test.

Two things the design does not deliver. Treatment is binary, whether a party sat at the ceiling. Continuous measures of cap exposure are available and are not used, because exposure to the cap and exposure to the coalition ban move together: the correlation of sitting at the cap with being in a coalition is +0.163, and coalitions are what the ban abolished. Specifications B through D control the partner count directly, and the off-diagonal mass — 21,827 cells, 68% of the panel, that are treated on one margin and not the other — is what separates them at all.

And the estimand is the prize *function*, not α. It tells us how B_j responds to s_j, which the model takes as given. It does not identify how candidates value B_j/s_j, which is what Section 6.4 estimates and what the rest of Section 6 is about.

6.7. Selecting among equilibria

Multiplicity is the standard obstacle in the econometrics of games, and the standard responses are to set-identify (Tamer 2003; Ciliberto and Tamer 2009; de Paula 2013) or to estimate the selection rule alongside the payoffs (Bajari et al. 2010). Neither is needed for estimation here. Following Menzel (2016), we estimate conditional on the realised equilibrium (see also Moro 2003; Bisin, Moro, and Topa 2011), and the conditioning object — the observed seat-winning set, which the application uses for the clearing set of Assumption 2 — is recorded in the electoral data (Proposition 7). Selection matters instead for counterfactuals. A change in the quotient, in the prize, or in the registration cap may move a market from its observed equilibrium to another element of the equilibrium correspondence, and the model does not say which.

We measure how much room is at stake by enumerating all 2^J regimes at the estimated parameters, over the 9,125 markets fielding between two and eight lists — 82.0% of markets and 61.4% of lists — and checking (A8) at each solved fixed point. A regime is a set A of lists that clear, and the prize is the anticipated $B_{m,j}$ of Section 6.3, which (4) takes as a primitive that does not vary with the regime. The enumerated object is therefore the one Proposition 1 is stated for: a regime is self-fulfilling exactly when its fixed point is a mean-field equilibrium in the sense of Definition 1, and a market has an equilibrium if and only if some regime is self-fulfilling.

Proposition 1 does not make the fixed point of a regime unique, so a count of self-fulfilling regimes is in general a lower bound on the number of equilibria. We check the gap numerically. Re-solving every regime in every market from three further starting points — the observed shares and two random draws from the simplex — none of the 1,294,266 converged solves reaches a fixed point further than 10^{-10} from the one first found. That diagnostic does not establish uniqueness, but it finds no additional fixed points. The counts below are counts of self-fulfilling regimes, each with the one fixed point these searches find. Figure 3 plots the count against the number of lists.

lists	2	3	4	5	6	7	8
markets	1,622	1,673	1,757	1,459	1,158	860	596
regimes enumerated	4	8	16	32	64	128	256
mean self-fulfilling	1.01	1.20	1.54	2.31	3.32	4.70	6.42
share with a unique one	0.988	0.815	0.596	0.352	0.229	0.166	0.141
share with none	0	0	0	0	0	0	0
observed one is self-fulfilling	0.994	0.946	0.898	0.858	0.735	0.620	0.542

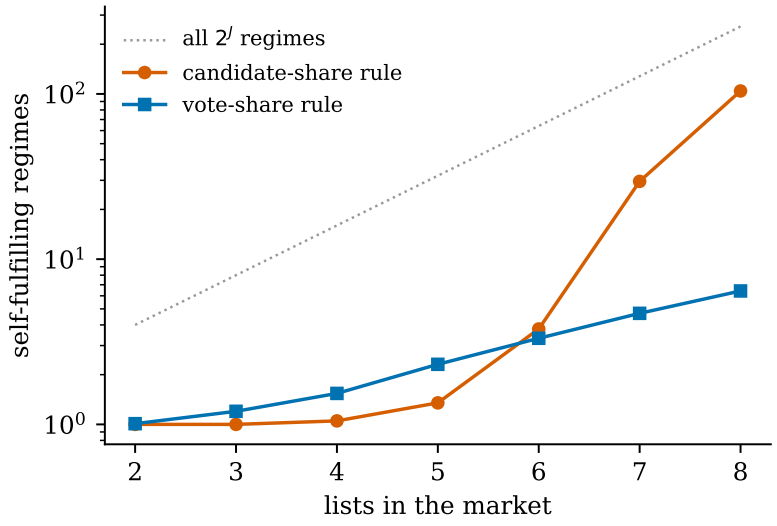


Figure 3: Self-fulfilling regimes against the number of lists, under the two winner rules.

Notes: mean number of regimes satisfying (A8) at the estimated parameters, over the 9,125 markets fielding between two and eight lists. The dotted line is the number of regimes enumerated, 2^J. The vertical scale is logarithmic. The candidate-share series applies a candidate-share threshold in place of the vote-share threshold of (A1).

The enumeration has three results. A self-fulfilling regime is found in every one of the 9,125 markets.

Proposition 1 leaves that step open, and Section A.4 shows that it can fail at admissible primitives, so this is an empirical finding about these markets. Multiplicity is present and modest, rising from one self-fulfilling regime at two lists to 6.4 at eight: a counterfactual in a small market is well defined, and one in a crowded market depends on selection. And the regime that occurred is self-fulfilling in 84.7% of markets overall, and in just over half at eight lists.[5]

The closed form of Proposition A.4 and a direct evaluation of (A8) agree in every one of the 431,440 regime-market pairs, which is the proposition checked on real primitives. The branch the vote rule adds — the case in which no non-member can reach the quotient at any member share — binds in none of the pairs enumerated.

Three candidate selection rules, none of which narrows the set. Section F.4 asks whether anything outside the model narrows the set of self-fulfilling regimes, and examines three candidates. Stability under best-response dynamics does not refine the correspondence: on the ten equilibria of the illustration in Section A.5 every one fails the undamped test and every one is locally attracting once the dynamic is damped, because the instability is overshooting. Coordination on the previous cycle finds no support: at the quotient cutoff of 2012 we detect no discontinuity in a list's 2016 candidate share or candidate count at any bandwidth. An extremal rule, the self-fulfilling regime whose members hold the largest total candidate share, picks the observed regime in 52.4% of the 4,106 markets with more than one self-fulfilling regime, against 29.8% for a uniform draw, which is too inaccurate to serve as a selection rule.

What follows. We therefore impose no selection rule. Estimation conditions on the observed regime, and Section 6.9 computes its counterfactual by solving both worlds from the observed share vector. Where a single clearing set is self-fulfilling that is the full response; where several are, it is the response of the fixed point reached from that start.

6.8. Does the estimated model predict what it was not fitted to?

A simulation drawn from the model cannot answer that. Its data-generating process *is* the model, so no misspecification is in the sample, and Section 5 is accordingly a statement about the estimator rather than about the model. Only the data can answer it, and only through predictions the estimation did not use.

Two are available and neither is delicate. The likelihood (24) compares lists *within* a cell, so it works from log-odds and never from the level of the share vector; and it takes the clearing set as given, reading it off the published record of which lists took a seat. The model says more than the likelihood uses. It says the realised shares must be a fixed point of the choice probabilities they induce,

$$s = \mu'\sigma(s;\hat{\theta}), \qquad P_{m,j} = \frac{B_{m,j}}{s_j}, \qquad W_m = \left\{ j : \frac{\rho_j s_j}{\sum_k \rho_k s_k} \geq q_m \right\}. \tag{31}$$

Solving (31) at $\hat{\theta}$ and comparing to the data is therefore over-identifying on both counts.

[5] Under a candidate-share threshold in place of the vote-share threshold, the same enumeration gives 104.2 self-fulfilling regimes of 256 at eight lists and finds the observed regime self-fulfilling in 60.0% of markets. That is what imposing homogeneous vote productivity, $\rho_{m,j} \equiv \rho$, costs in self-fulfilment.

Every comparison below is against the model *re-estimated* with the prize term dropped, not against the fitted model with α switched off. ξ and δ are fitted alongside α and partly complement it, so zeroing α after the fact handicaps the benchmark and flatters the prize. The full model's parameters are estimated with the split-sample estimator of Section 5.7, as everywhere in the application; the restricted model has no prize term and is the same under either estimator.

What the restricted model keeps is worth naming, because it bounds what the comparison can show. Dropping α drops the prize term and nothing else: ξ and δ are still fitted, at the party level, so the benchmark retains a national brand for every party and loses only the channel through which a party's *local* standing enters a candidate's index. The anticipated prize is built from the lagged municipal vote share, so that channel is α's alone. The restricted model is therefore not a model without local party strength; it is a model with national brand and without local strength, and the gain over it is a gain over that benchmark rather than over the confound the literature names.

Why δ is not a common constant. Section 2.3 carries a list-specific $\delta_{m,j}$ on the non-clearing branch rather than a common constant, because a common constant would imply equal choice probabilities for all non-clearing lists in a market. On the vereador panel the within-market coefficient of variation of non-clearing shares is 0.48 in 2016 and 0.43 in 2020, against 0.18 and 0.20 among clearing lists, which carry genuinely different prizes and are supposed to differ. This is a description and not a test: realised shares differ by sampling variation even under equal probabilities, and we have not compared the dispersion with what that variation alone would produce.

What is scored. Each market's fixed point is found by damped iteration from a stated starting vector, and the statistics below measure the distance between the observed shares and the fixed point the iteration reaches. That is not the residual of the observed shares in (31), and the iteration need not reach the fixed point nearest the data; where a market has several, the starting vector takes part in choosing among them, which is why two starts are reported.

In sample. The statistic is one minus the sum of squared errors over the same sum for a uniform share vector within each market.

	share vector			clearing set	
start	restricted	full	gain	restricted	full
observed shares	0.2726	0.3283	+0.0557	0.9108	0.9066
uniform	0.1665	0.2386	+0.0722	0.8852	0.8860

The two rows are different questions. From the observed shares the test asks whether the data sit *at* a fixed point; from uniform, whether the model *predicts* them from primitives. The second does worse, and the gap is the multiplicity of Section 6.7 showing up in a fit statistic.

The prize response adds about 0.056 from the observed shares and 0.072 from uniform, which is about a fifth of the level. It adds nothing to predicting the clearing set — 0.9066 against 0.9108 without it — because under the vote rule that is already near the ceiling.

Out of sample, across a change in what a list is. Fitting on 2016 and predicting 2020 asks the parameters to survive EC 97/2017, which abolished proportional coalitions: 2016's lists are alliances, 2020's are single parties. Ninety-nine point nine per cent of 2020 lists have a party with a 2016 estimate.

It asks them to survive more than that, and we should not pretend otherwise. The 2020 cycle was postponed to November and fought under pandemic conditions; two campaign-finance changes sit between the cycles (Section 6.1), the later of which moved exactly the party-level objects the exercise holds fixed; and the rule deciding which lists take the leftover seats changed as well, which is the one that moves the model's own winner rule. Estimated cycle by cycle on markets of sixty or more, $\hat{\alpha}$ is 1.85 in 2016 and 2.06 in 2020 (Section 6.4). *The parameter is not the same number in the two cycles*, so what the exercise below shows is that the fitted model transfers, not that α is stable. What follows is therefore a test of whether the parameters transfer across a four-year gap containing the coalition ban, not a measurement of the coalition ban's effect on them. It is the harder test of the two, which is why we report it; it is not the cleaner one.

	$1 - \text{SSE/SST}$	corr	mean abs. error	clearing
restricted, fitted on 2016	0.2665	0.8812	0.0328	0.8681
full, fitted on 2016	**0.2896**	0.8856	0.0327	0.8613
gain	**+0.0230**	+0.0044	−0.0001	−0.0067

The gain survives transfer: +0.023 out of sample, on a cycle the parameters never saw and under rules that redefined the choice set. It is less than half the in-sample gain of +0.056, so what the exercise shows is the survival of a gain and a small one. A gain that survives transfer is harder to produce by over-fitting one cycle than a gain in sample is, and that is what the exercise establishes. It does not establish that α is behaviour rather than local party strength, because the benchmark it beats is the one described above: party brands held fixed, local standing absent. The prize term is the only route by which the lagged local vote reaches the index, so a gain over that benchmark is consistent with α carrying behaviour and consistent with it carrying local strength that persists across the ban.

The gain is also smaller than it was before the measurement error in the prize was corrected. With parameters from the likelihood of Section 4 fitted without the split, the same exercise gives 0.3030 out of sample and 0.3339 in sample, gains of +0.036 and +0.061. The corrected prize coefficients are larger, the fixed point uses the model's own share and not a counted one, and the larger coefficients fit the 2020 shares slightly worse. We report the corrected figures because they are the paper's estimates, and note that the correction does not improve this statistic.

One threshold definition throughout. The fixed point (31) applies the quotient rule, while the parameters above are fitted with the paid lists defined as those that took a seat, and the solved clearing set is scored against that definition. Refitting on the quotient definition and scoring against it puts the estimates and the fixed point on one rule. Out of sample nothing moves, because the parameters are fitted on 2016, where the two definitions differ in 2 lists: the full model reaches 0.2895 against 0.2665, a gain of +0.023, and the solved clearing set agrees with the quotient-clearing set at 0.956, where it agrees with the seat-taking set at 0.861. In sample, where the 2020 lists enter the fit, the gain from the observed shares is +0.065 against +0.056. The comparisons in this subsection

are therefore not produced by mixing the two definitions. They remain exercises on the quotient threshold and not on the 2020 allocation rule, under which a list below the quotient can be paid.

What the vote rule costs: ρ is not forecastable. The winner rule (3) needs ρ, the votes a list draws per candidate, and the fixed-point exercises above set it to what the election delivered. Replacing it with a ρ built from the previous cycle's party vote shares fails. At the fixed point the lagged ρ agrees with the published clearing set at 0.650, against 0.655 for the rule that every list clears, and the shares it solves out are worse than uniform, with $1 - \text{SSE}/\text{SST}$ of -0.186 (Section F.6).

The estimates of Section 6.4 are untouched, because the likelihood conditions on the published clearing set and never evaluates (3): no ρ enters them. What is touched is the reach of the fixed point. The agreement between the solved clearing set and the published one, the multiplicity counts of Section 6.7 and the counterfactual of Section 6.9 are conditional on the realised ρ. They are statements about what the model does *given* the vote conversion, not about what could have been predicted before any vote was cast.

What this does not establish. The model explains 33% of the within-market variation in shares in sample and 29% out of it, against a uniform benchmark. That is a real improvement over a benchmark that is not a straw man — the restricted model with party effects, demographics and the equilibrium condition already reaches 0.27 — and it leaves most of the variation unexplained. The claim the paper makes is that α carries information the estimating equation did not put there, which the table supports. It is not the claim that the model is an adequate description of how candidates choose lists, which it does not.

6.9. A counterfactual, and a check on it

An estimated equilibrium model earns its machinery by answering questions the data do not contain. This one does so once, and the exercise is chosen for what can be checked rather than for what would be most striking.

What is not available. The registration cap is the obvious candidate. It moved from twice the council to one and a half times between the cycles, which is a policy change of the right kind. The model cannot answer it. There is no cap in the model: candidates choose freely and nothing constrains a list's size, so imposing one requires a rationing rule saying who is turned away and where they go. That is an extension, not a counterfactual. The cycle change is in any case confounded with the coalition ban, so even a rationing rule could not be validated against 2020.

What is. The council's size is a primitive and a policy variable. Art. 29 fixes it by population band, the bands have been amended, and C_m enters the model twice: the quotient is $1/C_m$ and the anticipated prize is C_m times a forecast seat share. Raising it lowers the bar and raises the purse at once. It is also the margin this paper identifies best, since Section 6.5 uses the jump in C_m to bound the confound.

We raise every council by two seats, the size of one step in the bands, and re-solve at $\hat{\theta}$, the split-sample estimate of Section 6.4. Seats scale with the council and the forecast seat *share* is held: the $\log C_m$ term in (28) is a fitting device, and proportional representation is the claim that a larger council gives each party proportionally more.

	baseline	+2 seats	change
lists clearing	3.829	4.131	**+0.302**
effective lists, 1/HHI	5.597	5.590	−0.007
largest list's share	0.2889	0.2904	+0.0015
concentration, HHI	0.2528	0.2532	+0.0004

Both worlds solve in all 11,129 markets. The effect is almost entirely on the extensive margin: three-tenths of a list more clears the quotient, and how candidates divide themselves barely moves. Two more seats change who is paid, not who stands where.

The response grows with the number of lists, as Figure 4 shows, from nothing in a two-list market to two-thirds of a list where there are nine, because a crowded market has more lists sitting near the quotient for the lower bar to catch. The market counts here are the counterfactual frame's, and at five lists they differ by one from the count Section 6.7 reports for the enumeration frame:

lists in the market	3	4	5	6	7	8	9
markets	1,673	1,757	1,458	1,158	860	596	449
extra lists clearing	+0.052	+0.119	+0.212	+0.340	+0.531	+0.612	+0.659

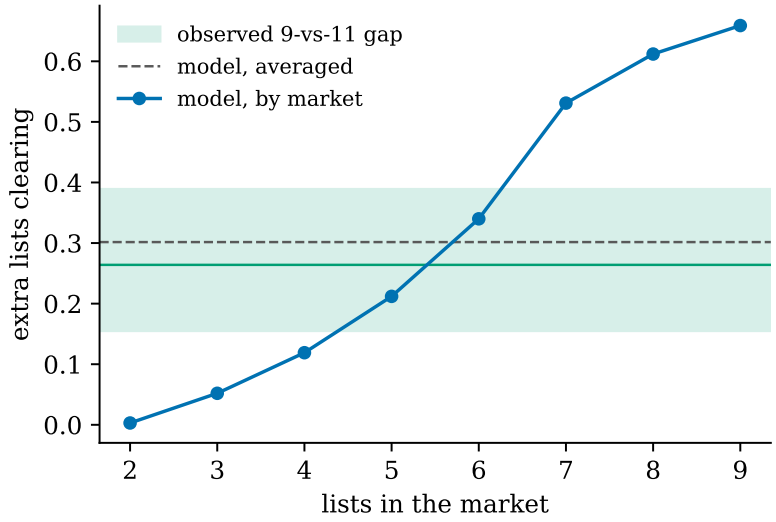


Figure 4: Lists brought over the quotient by two more council seats.

Notes: the marked series is the model's prediction, solved at $\hat{\theta}$ from the observed configuration, by the number of lists the market fields; the dashed line is its average over all 11,129 markets. The band is the range of the observed gap between nine- and eleven-seat municipalities across the three windows of municipal votes reported below, and the solid line inside it is the middle window. The observed gap is not resolved by market size, which is why it is drawn as a level.

The check. Municipalities on either side of the nine-to-eleven population band differ by exactly the two seats the counterfactual adds. Restricting to a window of municipal votes around the band makes them otherwise comparable, and the model was not fitted to the gap between them.

window, municipal votes	markets with 9	with 11	extra lists clearing
6,000–14,000	2,135	1,284	+0.389
7,500–11,500	1,017	672	+0.264
8,250–10,500	595	393	+0.155
the model's prediction	7,230 nine-seat markets		**+0.244**

The prediction sits inside the observed range and next to the middle window. That is the agreement this can establish: the observed gap also carries whatever else differs between towns of nine and eleven seats, and the three windows trade that contamination against precision.

The concentration margin does not check out, and we say so. The model predicts the largest list's share rising by 0.002; the same three windows give −0.012, +0.008 and +0.024, changing sign as the window narrows. The data are too noisy on that margin to test the prediction, and we do not select the window that agrees.

What kind of counterfactual this is. Both worlds are fixed points reached by the same damped iteration started at the observed share vector. The baseline is therefore the model's fixed point reached from the data and not the data, which need not be a fixed point of the estimated model, and the reported change is the difference between two solved fixed points. The exercise does not search the equilibrium set. Where a model admits several equilibria a counterfactual is a correspondence and not a number (de Paula 2013), and reporting a number requires saying which selection is being held fixed. Where a single clearing set is self-fulfilling that distinction is empty, which covers 99% of two-list markets and 60% of four-list ones; where it is not, the number is the response of the fixed point reached from the observed shares and not of the correspondence. Section 6.7 gives the counts. This is the sense in which counterfactuals here are local, and it is a weaker claim than the model would support if selection were settled.

7. Conclusion

Half a million people a cycle choose a party to stand for, knowing that the seats their list wins will be divided among everyone who made the same choice. That is a game of incomplete information among tens of thousands of players per country and fifty-five per town, and the estimators built for such games enumerate a space that has 10^{38} elements in the median Brazilian municipality. Replacing the equilibrium concept rather than the algorithm removes the obstacle: in the anonymous formulation the fixed point has one dimension per list and none per candidate, and what a candidate cares about — how many others crowd onto the list he is considering — survives the aggregation intact.

What the threshold rule buys, and what it costs. A list is paid only where it clears the quotient, so the prize B_j/s_j is bounded exactly where it is paid, and the singularity a winner-take-all formulation must handle at the boundary cannot arise. Fixing a clearing set then makes the best-response operator continuous, and each regime has a fixed point unconditionally (Proposition A.1).

Whether any of those fixed points is self-fulfilling is a different question and we cannot settle it. Section A.4 gives primitives satisfying every assumption at which none of them is. The mechanism that does it — a list made

less attractive by the promise of a prize, for every type in the market and not merely for one — is available at no list in our own sample, which is close to a restatement of the model's fit rather than a finding in its own right. What it bears on is how the search below should be read: the estimated configuration sits far from the region in which the known route to failure opens. It does not buy existence, the condition being sufficient and never necessary. On existence, then, the comparison with the winner-take-all rule favours that rule. Its operator is repaired by a participation bound and delivers existence outright, one equilibrium per list (Section A.6). It is the threshold that turns existence into a regime-consistency problem.

Uniqueness fails too, and that failure has a closed form on half of it. Non-members of the clearing set are indistinguishable up to a constant, so the $J - |A|$ conditions on them collapse to one whatever their number (Proposition A.4); the members' conditions do not collapse. On the vereador data a self-fulfilling regime exists in all 9,125 markets enumerated, and the collapsed condition agrees with a direct evaluation in every one of 431,440 regime-market pairs. The branch the vote rule adds to that condition binds in none of them.

What the three contrasts buy, and where they stop. The threshold pays several lists at once, so two of them can be compared inside one town and the prize difference stands alone — which a winner-take-all rule, paying one alternative per market, cannot arrange. That no probability of not standing appears anywhere follows from it, and is not on its own the contribution: a conditional logit tolerates a subset of the choice set regardless (McFadden 1978). One more difference — across types on the same pair of lists — removes the list-level unobservable exactly, so differences in responsiveness are identified without assuming anything about it. The level is identified only under a restriction on the local component of that unobservable, and the council-size design that targets the component does not give a stable estimate of it. That division is what the design delivers. Two clearing lists give the marginal utility of the prize, two non-clearing lists the relative value of a party's name, and a mixed pair the taste parameters — which survive there precisely because they do not survive the first contrast. The quantity a single-market design is least able to observe is absent from the estimator rather than better measured.

Two implementation facts turned out to matter more than the algebra. *The likelihood is not a convenience.* A type that puts nobody on a list supplies no log-odds, and 56.6% of the estimating sample is exactly that; the discarded zeros account for 87% of the difference between the two estimators, and a candidate's decision to avoid a list entirely is where most of the information about α lives. *And the prize cannot be built from realised seats.* A list wins seats partly because this cell's members won them, and correcting for that mechanically over-corrects, so the numerator has to be assembled from objects that predate registration.

What the data say. The marginal utility of the anticipated per-candidate chance of a seat is positive. With the measurement error in the prize corrected by the split-sample estimator of Section 5.7, it is 1.33 among candidates who did not win a seat here last cycle, and so carry no mechanical tie to their party's record here, with a standard error of 0.08, and 1.92 among all candidates. It is positive on either definition of a paid list, in each cycle, for each of the thirty-two types, and with and without a control for local party strength. Candidates with a university degree respond more than candidates without one: the difference between the type means among non-incumbents is 1.18 with party-by-state effects and 2.60 with an effect for every list in every market, and among incumbents it is not distinguishable from zero under either.

Fitting on 2016 and predicting 2020, the prize term improves the fit of a share vector the estimator never targeted, by 0.023 against 0.056 in sample. That crossing contains the amendment abolishing proportional

coalitions and also a postponed, pandemic-era cycle and two changes to campaign finance, so it is a demanding test of transfer rather than a measurement of the amendment, and the gain that survives it is small.

What is not settled. The magnitude, for three reasons. The correction leaves a bias toward zero of about 0.10 in simulation that does not fall as markets are added, so 1.33 is more likely low than high. The level rests on list appeal beyond a party-by-state effect being unrelated to the prize, and the one design we have for relaxing that, the constitutional step in council size, does not move the coefficient in one direction: it lowers the non-incumbent coefficient to 1.00 on the markets of sixty candidates or more and to 0.57 on all markets, and raises it in the windows around the step, where its control takes a sign the confound does not predict (Section 6.5). And the coefficient differs between the two cycles, 1.85 against 2.06 for all candidates, and on the evidence of that design with the size of the council. A single number for the magnitude is therefore an average over a parameter that is not constant.

What is bounded. Counterfactuals are local. Solving both worlds from the observed configuration, two more council seats bring 0.244 more lists over the quotient on nine-seat markets and leave the division of candidates among lists almost unchanged; municipalities that really do differ by two seats differ by 0.16 to 0.39 on the same margin, and the model was not fitted to that gap. Where a single clearing set is self-fulfilling the exercise is unconditional, which covers most small markets; where several are, it is the response of the fixed point reached from the observed shares. Neither standard way of narrowing the set works here. Stability under damped best-response dynamics does not refine the equilibria we examine: their unstable roots are real and negative, so undamped tâtonnement fails by overshooting and a sufficiently damped update restores local attraction, which it could not do for a real root above one. History gives the coordination account no support, and an extremal rule beats chance without approaching selection.

What a second rule would test. The estimate is read off choices made under one payment rule, and nothing inside a single rule regime separates a property of the candidates from a property of the rule that identifies it. Section G sets out a test that does: it states in advance what would count as passing, what would count as a failure worth reporting and what would count as no result, and it says why this paper does not report the answer.

What is open. Entry, and one half of it more than the other. The registration cap binds hard enough that constrained lists wanted about a fifth more candidates than the law allowed, so a list at the ceiling carries the number the law permitted rather than the number free choice would produce. That half we test: removing every capped list and every candidate on one moves the estimate by 0.014 (Section 6.6). The other half we do not. Parties choose *whom* to register as well as how many, so the menu a candidate faces is chosen by someone with an interest in it, and the model has candidates choosing among fixed alternatives. That is the assumption to push on.

Second, the winner rule is an approximation. The model supports the realised configuration in 85% of the markets enumerated, and multiplicity rises with the number of lists while remaining limited, at 6.4 self-fulfilling regimes at eight lists. A list's conversion of candidates into votes is nonetheless treated as a primitive when it is plainly a choice.

Third, the prize is an average. Art. 108 bars a candidate polling less than a tenth of the quotient from taking one of his list's seats, so within a list the chance of a seat is concentrated at the top of the personal-vote ranking

and is zero as a matter of law for everyone beneath the floor, however many seats the list wins (Section 6.1). Every member of a cell is given the list average $B_{m,j}/s_{m,j}$. The candidate-level prize — the anticipated chance that this candidate takes one of those seats — is the obvious next measurement, and it requires a model of the personal vote that this paper does not have.

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Appendix A. Existence of Mean-Field Equilibrium under a Threshold Winner Rule

A.1. Setup

Types $\tau \in \mathcal{T}$ form a finite set with $|\mathcal{T}| = K$ and weights $\mu(\tau) \geq 0$ summing to one. Each agent chooses one alternative from a set $\mathcal{J}$, and the share vector $s = (s_j)_{j \in \mathcal{J}}$ lies in the unit simplex $\Delta(\mathcal{J})$. Two choice sets are in play and it is worth separating them before anything is proved. The paper's model is $\mathcal{J} = \{1, \ldots, J\}$, with *no outside option*, matching Section 2.1 and the estimator's choice set: there $\Delta(\mathcal{J}) = \Delta^{J-1}$ and $\sum_{k \geq 1} s_k = 1$. This appendix proves the general case $\mathcal{J} = \{0, 1, \ldots, J\}$, which carries an outside option $j = 0$ with payoff normalised to zero: there $\Delta(\mathcal{J}) = \Delta^{J}$ and $\sum_{k \geq 1} s_k = 1 - s_0$, and the paper's model is the face $s_0 = 0$. Every result below states which of the two choice sets it carries, because the specialisation is immediate for some of them and not for others; Corollary A.1 and Proposition A.4 are the forms the application uses. This appendix drops the market subscript throughout, so its $\mathcal{J}$ is Section 2.1's $\mathcal{J}_m$ in the paper's model and that set together with the outside option in the general case, and "list" always means a $j \geq 1$.

A list earns a prize when its share of the vote reaches the threshold. Writing $\rho_j > 0$ for the votes list j draws per candidate,

$$W(s) \;=\; \left\{ j \in \{1, \ldots, J\} : \frac{\rho_j s_j}{\sum_{k \geq 1} \rho_k s_k} \geq q \right\}, \qquad q \in (0, 1), \tag{A1}$$

with $q = 1/C$ the electoral quotient expressed as a share of the vote and C the size of the council. When $\rho_j = \rho$ for every j the weights cancel and this reduces to $s_j \geq q$, a threshold on the share of *candidates*. The cancellation uses the absence of an outside option: were one present the shares would be normalised by $1 - s_0$ and the reduction would read $s_j \geq q(1 - s_0)$.

That reduction is not innocuous. It is tempting to argue that the two rules agree whenever the vote function is increasing in a list's size, on the ground that $\{j : V(s_j) \geq \bar{q}\} = \{j : s_j \geq V^{-1}(\bar{q})\}$. The step treats V as one function of s_j alone. Votes are converted from candidates at a rate that differs across lists, so V_j is list-specific and the quotient is crossed by a *share* of the total, which depends on the whole vector; monotonicity in a list's own size does not deliver the equivalence, and equality of the ρ_j does.

The distinction is quantitatively first-order rather than a technicality. In simulation at the empirical market size, a model whose agents anticipate the candidate rule while the election applies the vote rule misclassifies the regime of a third of lists, and the misclassification alone moves the likelihood's estimate of α from 0.98 of its true value to −0.09. Replacing the candidate rule by (A1) cuts the misclassification to 9.4%, where the same exercise recovers α at 0.20 of its true value, and roughly doubles what the estimator recovers. The three misclassification rates on record — 0% (recovering 0.98), 9.4% (recovering 0.20), and roughly 33% (recovering −0.09) — trace a convex relation between misclassification and recovery rather than a linear one: the decline is steep at first and flattens well before reaching zero. The design is Section 5's; the winner-rule arm is a separate run, reported here rather than there (`scripts/python/structural/pr_floor_mc_voterule.py`).

On the data the correction is larger still. Solving the fixed point at the estimated parameters and scoring the resulting share vector against a uniform benchmark within each market, the candidate rule reaches 0.151 and the vote rule 0.282; agreement with the published clearing set rises from 0.782 to 0.966, and the candidate rule's systematic over-prediction — it calls 79.4% of lists clearing where 65.5% do — disappears, the vote rule

predicting 65.8%. Section 6.7 reports what the same correction does to the multiplicity.

Payoffs are, with ξ_j the list-level unobservable that (4) carries on the clearing branch,

$$\pi(j, s, \tau) = \begin{cases} X(\tau)'\beta + \alpha(\tau)\,\dfrac{B_j}{s_j} + \xi_j & j \in W(s), \\ \delta_j & j \notin W(s),\ j \geq 1, \\ 0 & j = 0, \end{cases} \tag{A2}$$

whose third line is the general case alone: in the paper's model the alternative $j = 0$ is absent and (A2) is the two cases of (4). Choice probabilities are logit, $\sigma(j \mid \tau; s)$ summing to one over $\mathcal{J}$, with aggregate operator

$$\Phi(s)_j = \sum_{\tau} \mu(\tau)\,\sigma(j \mid \tau; s). \tag{A3}$$

A mean-field equilibrium is $s^\star$ with $\Phi(s^\star) = s^\star$.

Assumption 1 is what every result below assumes. It is stated in Section 2.4, where it is first invoked, and reads in this appendix's market-free notation: $\mathcal{T}$ is finite and $\mu(\tau) \geq 0$ sums to one; $q \in (0, 1)$; $0 < \underline{\rho} \leq \rho_j < \infty$ for every j; $|\alpha(\tau)| < \infty$ for every τ and $0 < B_j < \infty$ for every j; β finite and $X(\tau)$ bounded on $\mathcal{T}$; each ξ_j and each δ_j finite.

Two remarks on what this does and does not ask. It bounds $\alpha(\tau)$ but does not sign it, and it is as well that it does not, since two of the thirty-two type-specific estimates in Section 6.4 are negative. The sign matters in two places below, and they are not on a par. The bound in Lemma A.1 carries $\max_\tau |\alpha(\tau)|$ rather than α, which would not be a bound when some $\alpha(\tau) < 0$; Proposition A.1 uses only the lemma's conclusion, so existence does not depend on the sign. The ceiling (A9) behind (A10) is the substantive case: the plain formula $\alpha(\tau)B_j/\underline{s}_j$ is the largest clearing payoff only where $\alpha(\tau) > 0$, and the sign-split form below removes the need for the restriction rather than assuming it away. And it is stated for the heterogeneous $\alpha(\tau)$ of (4) rather than a scalar, since that is the model the application estimates and at which Section 6.7 solves the fixed point.

The term δ_j is the payoff to standing on a list that does not clear, measured against not standing at all. The natural first pass sets it to zero: a candidate on a list that wins no seat wins no seat. That normalisation is convenient and, on the vereador data, false — standing on a small party that will not clear buys ballot exposure and a claim on the next cycle, and the implication the normalisation carries is rejected sharply. Section A.5 gives the test and its consequence. We therefore carry δ_j throughout; setting $\delta \equiv 0$ returns every statement below to its simpler form.

A.2. Regime cells and the truncated operator

The winner rule (A1) makes the payoff (A2) discontinuous at the quotient: crossing it switches the prize on. Whether Φ itself jumps at a given tie point is a sharper question, settled by the primitives in the way the next paragraph records, and we do not settle it; what the argument below needs and uses is only that Φ is not *known* to be continuous there, so a single Brouwer argument on $\Delta(\mathcal{J})$ is not available. Existence in discontinuous games is available under conditions on the payoff correspondence (Dasgupta and Maskin 1986; Reny 1999); we do not need them, because the discontinuity here partitions the simplex into finitely many pieces on each of

which the operator is continuous, and truncating the prize at the floor established below extends the operator to a continuous self-map of the whole simplex, to which Brouwer applies directly.

The discontinuity is not removable, and no tie-breaking convention repairs it. At a share vector s at which list j sits exactly at the quotient, the two branches of (A2) differ by

$$D_j(s,\tau) \;=\; X(\tau)'\beta + \alpha(\tau)\,\frac{B_j}{s_j} + \xi_j \;-\; \delta_j, \tag{A4}$$

which depends on the type, is evaluated at the share at which the crossing happens, and has no common sign — Assumption 1 bounds $\alpha(\tau)$ without signing it, and (A10) below is exactly the configuration in which D_j is negative at every type in the market. What rules out the repair is therefore not the magnitude of D_j but where a convention is able to act. A tie-breaking rule assigns a winner set on the tie locus $\{s : \rho_j s_j = q \sum_{k\geq 1} \rho_k s_k\}$ and nowhere else, so it leaves both one-sided limits of the choice probabilities at such an s where they were. Continuity of $\sigma(\cdot \mid \tau; \cdot)$ at s is equivalent to the payoff jump being constant *across the whole choice set*, since that and only that is what the logit absorbs. When j is the only list at the quotient the remaining alternatives do not jump, so the condition is $D_j(s,\tau) = 0$; and $D_j(s,\tau) = 0$ at every tied j remains the condition whenever the tied lists are a proper subset of $\mathcal{J}$, because the untied ones pin the constant at zero. One case escapes: with *every* list tied at once, a common non-zero value $D_j(s,\tau) \equiv c$ shifts all alternatives alike and σ is continuous after all. In this paper's choice set that needs $\sum_j v_j(s) = Jq = 1$ exactly, and in the general case it cannot happen at all, the outside option's payoff being fixed at zero and so never shifting. In each case Φ inherits continuity when the condition holds at every $\tau \in \operatorname{supp}\mu$ — which is a coincidence of the primitives and not something a convention can produce. The gap is finite at every such s, since $\rho_j s_j = q \sum_{k\geq 1} \rho_k s_k$ forces $s_j \geq \underline{s}_j$ by (A6), so $|D_j(s,\tau)| \leq 2\bar{\pi}$ with $\bar{\pi}$ the bound of Lemma A.1. Two observations make the discontinuity harmless.

First, the winner set takes at most 2^J values, and each value is realised on a convex set. The vote-share condition $\rho_j s_j / \sum_{k\geq 1} \rho_k s_k \geq q$ is $\rho_j s_j \geq q \sum_{k\geq 1} \rho_k s_k$, which is linear in s, so replacing the candidate rule by the vote rule costs nothing here. For $A \subseteq \{1,\dots,J\}$ define the *regime cell*

$$C_A \;=\; \Big\{ s \in \Delta(\mathcal{J}) : \rho_j s_j \geq q \textstyle\sum_{k\geq 1} \rho_k s_k \;\forall j \in A, \quad \rho_j s_j \leq q \sum_{k\geq 1} \rho_k s_k \;\forall j \in \{1,\dots,J\} \setminus A \Big\}, \tag{A5}$$

an intersection of $\Delta(\mathcal{J})$ with J half-spaces, hence a compact convex polytope. In the paper's model it is non-empty if and only if $|A|\, q \leq 1$ when $A \neq \emptyset$, and if and only if $Jq \geq 1$ when $A = \emptyset$, since no list can sit below the quotient unless there are enough lists to divide the vote between them. In the general case every cell also contains the degenerate point $s_0 = 1$, at which (A1)'s ratio is undefined and (A5)'s cleared form holds vacuously. Nothing below uses either count: the cell is used only to locate a given s.

Second — and this is what the threshold rule buys — on C_A the prize of a clearing list is bounded away from its singularity. Write $\underline{\rho} = \min_{k\geq 1} \rho_k$ and

$$\underline{s}_j \;=\; q\,\underline{\rho}/\rho_j \;>\; 0. \tag{A6}$$

In the paper's model $\sum_{k\geq 1} s_k = 1$, so $\sum_{k\geq 1} \rho_k s_k \geq \underline{\rho}$, and $j \in A$ therefore implies $s_j \geq q \sum_{k\geq 1} \rho_k s_k / \rho_j \geq \underline{s}_j$ and $B_j/s_j \leq B_j/\underline{s}_j$. The singularity of the prize term at $s_j \to 0$ cannot occur where the prize is paid. In the general case the same holds on the subset of C_A where the participation share $1 - s_0$ is bounded below, and $\underline{s}_j$

is scaled by that bound; nothing in the application needs it, since Corollary 1 identifies the model without an outside option.

The floor is used below in two ways, and only one of them is sensitive to the choice set. As the *constant* inside the truncation (A7) it is a function of primitives alone, positive by Assumption 1, and that is all Lemma A.1 and Proposition A.1 ever ask of it: both hold verbatim on $\Delta(\mathcal{J})$ for either choice set. As a *lower bound on the share of a clearing list* — which is what makes the truncation inactive, and is what Proposition A.2 uses — it is the display above, and that display rests on $\sum_{k\geq 1} s_k = 1$. So the outside option does not supply the floor; it dilutes it, and the paper's model is the case in which the floor cannot be raised — $\sum_{k\geq 1} \rho_k s_k$ descends to $\underline{\rho}$ as the mass concentrates on an arg min-ρ list, so no larger constant serves for every s, though the bound is approached and not attained.

Fix A and define the *truncated* payoff

$$\pi^A(j,s,\tau) \;=\; \begin{cases} X(\tau)'\beta + \alpha(\tau)\,\dfrac{B_j}{\max\{s_j,\ \underline{s}_j\}} + \xi_j & j \in A, \\ \delta_j & j \notin A,\ j \geq 1, \\ 0 & j = 0, \end{cases} \tag{A7}$$

whose last line is again the general case alone, and with induced operator $\Phi^A : \Delta(\mathcal{J}) \to \Delta(\mathcal{J})$ defined as in (A3). By the previous paragraph the truncation is inactive throughout C_A in the paper's model, so $\pi^A = \pi$ there. Under the candidate rule ρ is constant, $\underline{s}_j = q$, and (A7) is the earlier truncation unchanged.

LEMMA A.1 (Continuity and a uniform payoff bound). *Under Assumption 1, for either choice set $\mathcal{J}$ and for every A, the map Φ^A is continuous on all of $\Delta(\mathcal{J})$, and*

$$\sup_{s,\tau,j} |\pi^A(j,s,\tau)| \;\leq\; \bar{\pi} \;\equiv\; \max_{\tau} |X(\tau)'\beta| + \max_{\tau} |\alpha(\tau)| \max_j \left(B_j/\underline{s}_j\right) + \max_j |\xi_j| + \max_j |\delta_j| \;<\; \infty.$$

PROOF. $s \mapsto \max\{s_j, \underline{s}_j\}$ is continuous and bounded below by $\underline{s}_j > 0$, so $s \mapsto B_j/\max\{s_j, \underline{s}_j\}$ is continuous and bounded above by $B_j/\underline{s}_j$; multiplying by $\alpha(\tau)$ of either sign bounds the term in absolute value by $|\alpha(\tau)|B_j/\underline{s}_j$, which is where the absolute value in the display is used. The winner set is fixed at A inside π^A, so no indicator varies, and ξ_j and δ_j do not depend on s at all. Hence each $\pi^A(j,\cdot,\tau)$ is continuous and uniformly bounded, the logit map is smooth on bounded arguments, and Φ^A is a finite convex combination of continuous functions. No step uses the choice set: the floor (A6) enters only as the positive constant $q\underline{\rho}/\rho_j$, so the argument is the same word for word on Δ^{J-1} and on Δ^J, and the extra alternative of the general case contributes the constant payoff 0. □

The bound carries $\max_\tau |\alpha(\tau)|$ and $\max_j |\xi_j|$ because the inequality with α alone would be false whenever some $\alpha(\tau) < 0$, which two of the thirty-two estimates are. What the lemma is used for is its conclusion — that Φ^A is continuous on $\Delta(\mathcal{J})$ — and that follows from Assumption 1 alone, through the floor (A6), without the printed constant. Proposition A.1 invokes the conclusion and never the inequality.

A.3. Existence

PROPOSITION A.1 (Regime fixed points exist unconditionally). *Under Assumption 1, for every $A \subseteq \{1, \dots, J\}$ the operator Φ^A admits a fixed point $s^A \in \Delta(\mathcal{J})$. This holds for either choice set: a fixed point in Δ^{J-1} in the paper's model, and one in Δ^J in the general case.*

PROOF. By Lemma A.1, whose bound rests on the floor (A6) as a positive constant and not as a bound on any share, Φ^A is a continuous self-map of $\Delta(\mathcal{J})$, which is compact and convex for either choice set. Brouwer's theorem applies. No restriction on (α, B, β) is used, and the choice set enters only through which of the two simplices Brouwer is applied to. □

PROPOSITION A.2 (When a regime fixed point is an equilibrium). *Take the paper's model, $\mathcal{J} = \{1, \dots, J\}$, and let $s^A = \Phi^A(s^A)$. If*

$$\text{(A8)} \qquad \frac{\rho_j s_j^A}{\sum_{k\geq 1} \rho_k s_k^A} \geq q \quad \forall j \in A \qquad \text{and} \qquad \frac{\rho_j s_j^A}{\sum_{k\geq 1} \rho_k s_k^A} < q \quad \forall j \notin A,$$

so that s^A lies in the cell (A5), then $W(s^A) = A$, $\pi^A(\cdot, s^A, \cdot) = \pi(\cdot, s^A, \cdot)$, and s^A is a mean-field equilibrium. Conversely, every mean-field equilibrium $s^\star$ is a fixed point of Φ^A for $A = W(s^\star)$ and satisfies (A8), both halves, by the definition of W.

PROOF. Under (A8) the realised winner set is exactly A, so the indicator in (A2) agrees with the fixed set used in (A7). For the truncation, take $j \in A$: the member half of (A8) gives $\rho_j s_j^A \geq q \sum_k \rho_k s_k^A \geq q\underline{\rho} \sum_k s_k^A = q\underline{\rho}$, the last step because the shares sum to one — which is where the absence of an outside option is used, an outside option leaving $\sum_{k\geq 1} s_k^A = 1 - s_0^A < 1$ and the bound short by that factor. Hence $s_j^A \geq q\underline{\rho}/\rho_j = \underline{s}_j$ and $\max\{s_j^A, \underline{s}_j\} = s_j^A$, so the truncation is inactive. The two payoffs coincide at s^A, hence so do the operators, and $\Phi(s^A) = \Phi^A(s^A) = s^A$. The converse is immediate on setting $A = W(s^\star)$. □

Proposition A.2 is the one result in the appendix that the paper's model makes easier rather than harder, and the restriction sits in the truncation step rather than in the regime bookkeeping. In the general case, under (A1), the member half of (A8) delivers only $s_j^A \geq \underline{s}_j\,(1 - s_0^A)$, which permits the truncation to be active at s^A; the proposition then holds as stated at any fixed point with $1 - s_0^A \geq \underline{p} > 0$, provided the truncation in (A7) is set at $q\underline{\rho}\,\underline{p}/\rho_j$ rather than at (A6). Proposition A.1 is unaffected either way, since it asks of the floor only that it be positive.

The strict inequality for $j \notin A$ cannot be weakened: with the convention $W(s) = \{j : s_j \geq q\}$, a fixed point with $s_j^A = q$ for some $j \notin A$ would place j in the realised winner set and contradict the regime. Existence of an equilibrium therefore reduces to a finite collection of regime-specific fixed-point problems: 2^J of them, each unconditionally solvable by Proposition A.1, with self-fulfilment (A8) to be checked at every fixed point of each, since Proposition A.1 does not make that fixed point unique.

A.4. The search can come back empty

Proposition A.2 is an equivalence, and we cannot discharge its right-hand side. This subsection shows why not: the map $A \mapsto W(s^A)$ can cycle, and where it has no fixed point the game has no equilibrium.

EXAMPLE 1 (No mean-field equilibrium). *Let $J = 2$, $K = 1$, $\rho = (1, 1)$, $q = 2/5$, $\alpha = 1$, $X(\tau)'\beta = 0$, $\xi \equiv 0$, $B = (1/2,\ 1/20)$ and $\delta = (1/10,\ 1)$, in the paper's choice set $\mathcal{J} = \{1, 2\}$. Every clause of Assumption 1 holds, and the floor is $\underline{s}_j = q\underline{\rho}/\rho_j = 2/5$. The four regime fixed points are*

A	s^A		*consistency*
$\emptyset$	0.2891	0.7109	$s_2 \geq q$: *list 2 clears, contradiction*
$\{1\}$	0.5000	0.5000	$s_2 \geq q$: *list 2 clears, contradiction*
$\{2\}$	0.5000	0.5000	$s_1 \geq q$: *list 1 clears, contradiction*
$\{1, 2\}$	0.6545	0.3455	$s_2 < q$: *list 2 fails, contradiction*

and no regime is self-fulfilling, so no mean-field equilibrium exists.

Each line is immediate. Under $A = \{1\}$ the payoffs at $s_1 = 1/2$ are $\alpha B_1/s_1 = 1$ and $\delta_2 = 1$, so the logit returns $(\frac{1}{2}, \frac{1}{2})$ and list 2, which the regime denies a prize, holds two fifths of the market. Under $A = \{1, 2\}$ list 2's share falls below its floor, its truncated prize is $\alpha B_2/\underline{s}_2 = 1/8$, and it cannot hold the two fifths the regime requires.

The mechanism is one comparison. List 2 is *less* attractive when promised the prize than when denied it, because its capped prize $1/8$ is below its losing payoff $\delta_2 = 1$. Both of the two regimes that single out one list land on $(1/2, 1/2)$, where both lists clear, so $W(s^{\{1\}}) = W(s^{\{2\}}) = \{1, 2\}$; the regime map in full is $\emptyset \mapsto \{2\} \mapsto \{1, 2\} \mapsto \{1\} \mapsto \{1, 2\}$, and the two-cycle is $\{1\} \leftrightarrow \{1, 2\}$.

The comparison generalises, and it is made type by type. Write

$$\bar{\pi}_j(\tau) \;=\; X(\tau)'\beta + \xi_j \;+\; \alpha(\tau)^+ \frac{B_j}{\underline{s}_j} \;-\; \alpha(\tau)^- B_j, \qquad \alpha^+ = \max\{\alpha, 0\}, \quad \alpha^- = \max\{-\alpha, 0\}, \tag{A9}$$

for the largest payoff list j can offer type τ when it clears, at any share the regime admits: where $\alpha(\tau) > 0$ the prize is largest at the floor and the α^+ term applies; where $\alpha(\tau) < 0$ it is largest at a unit share and the α^- term does. Whenever

$$\delta_j \;>\; \max_{\tau \in \operatorname{supp} \mu} \bar{\pi}_j(\tau) \qquad \text{for some } j, \tag{A10}$$

the promise of a prize is a demotion for list j — not for a type that happens to dislike prizes, but for every type in the population, so that j's aggregate share falls whatever the type weights — and the map can cycle at any J and any q. The quantifier is not decoration. If (A10) holds for some types and fails for others, the sign of the change in j's aggregate share depends on the type weights and on the rest of the choice set, and nothing follows. Where (A10) fails for every j — that is, where some type in the support is not demoted — the mechanism above is unavailable in its weight-free form; we have not shown that it is the only one, so this is not a sufficient condition for existence.

The failure is not a knife edge. Perturbing B and α by up to a tenth, δ by up to 0.05 and q by up to 0.02, all four regimes remain inconsistent in every one of 4,000 draws. Nor does it depend on the liberty of choosing q freely: at $C = 2$, so $q = 1/2$, with $B = (1, 1/2)$ and $\delta = (3/2,\ 5/2)$, the four fixed points are $(0.2689, 0.7311)$, $(0.3775, 0.6225)$, $(0.6225, 0.3775)$ and $(0.6381, 0.3619)$, and again none is self-fulfilling.

Is the mechanism live in the application? Not through this route, at $\hat{\theta}$ — the fitted parameters this appendix has carried since drafting and has not re-evaluated against the current structural estimates; we flag the numbers below by vintage rather than restate them. Evaluating (A10) over the 65,667 market-lists, the condition holds at no list and in no market: the smallest margin between δ_j and $\max_\tau \bar{\pi}_j(\tau)$ is +0.424 and the median is +3.417. Nor does the aggregate version fire. At the fixed point of the observed regime, at that fixed point with j's own share pushed to its floor, at the observed shares, and at the observed shares with j's own share at the floor, no list's aggregate share falls when the prize is promised; the median list's rises by about eleven points. What does hold widely is the weaker comparison that asks only for *some* demoted type, at 61.7% of lists and 96.4% of markets — and that comparison bounds the share from above while implying nothing about the aggregate, since it is consistent with the prize raising every one of those lists' shares, which is what in fact happens. The statistic restricts the maximum to the types the market's own distribution puts weight on, and it does not license the assertion that the prize *would be* a demotion.

Two cautions on how to read the zeros. They are one margin distribution seen twice rather than two pieces of evidence: the largest of the 35 party δ's of the panel-wide fit this appendix evaluates is +0.060, against a median ceiling above +3, so the two incidences are collinear in this sample rather than independent draws. (That fit is not the one Section 6.4 reports, whose δ's are 847 party-by-state effects on the 60+ markets with a median of +0.061; the two are different parameters and the figures are not comparable.) And at $\hat{\theta}$ the zero is close to being implied by fit — over the thirty types with $\hat{\alpha}(\tau) > 0$, even the *minimum* clearing payoff, which is (A9) evaluated at a unit share rather than at the floor and so equals $X(\tau)'\beta + \xi_j + \alpha(\tau)B_j$ for these types, exceeds δ_j at every one of the 65,667 lists. This is a correction of what the draft claimed, not a discovery.

What the data do support, with both quantifiers in place, is a uniform statement. On at least 38.3% of lists the promise of the prize cannot lower list j's aggregate share at *any* share vector the regime admits — the simplex under the truncation convention of (A7) — and under *any* type distribution supported on the market's types; that holds for every list in the market simultaneously in 3.6% of markets, 399 of 11,129. Where the uniform bound fails, no demotion has been shown to occur; the set on which it fails carries no claim.

At $\hat{\theta}$ the 38.3% happens to be exactly the complement of the 61.7% above. The two sets are disjoint by construction but need not exhaust, and on a constructed control they do not, so the coincidence is a fact about these estimates rather than a second measurement. It does place the 47 single-list markets, which are the "at least" of the uniform statement: there is no other list for a candidate to move to, so the aggregate share cannot change at all, and those lists are not counted in the 38.3%; under the exhaustion they therefore sit in the 61.7%. That is not a contradiction, because the two statistics quantify over different objects. The 61.7% counts lists holding a type whose payoff ceiling the prize lowers, a comparison made type by type, and it fires at $J = 1$ as readily as anywhere; the aggregate share of a market's only list is pinned at one whatever its types do.

None of this establishes existence. Condition (A10) is sufficient for the mechanism and never necessary, so its failure everywhere leaves the possibility open. What it bears on is how the search Section 6.7 reports should be read: the estimated configuration sits far from the region in which the known route opens, so what that subsection finds is a property of $\hat{\theta}$ rather than of the parameter region.

There are three ways to buy existence back. A sufficient condition ruling out cycles in $A \mapsto W(s^A)$ would be the cleanest, and (A10) is where to start, but it is not enough on its own and we have not found the rest. Allowing public randomisation over regimes convexifies the regime map and restores existence by Kakutani, at the cost of

an equilibrium object the data cannot speak to. The third is to soften the rule, and it is the one we have looked at.

A soft edge. Pay list j its prize with probability $w_j(s) = F\big((v_j(s) - q)/h\big)$ for a smooth increasing F, so that

$$\pi_j \;=\; w_j(s)\Big[X(\tau)'\beta + \alpha(\tau)\frac{B_j}{\max\{s_j, \underline{s}_j\}}\Big] \;+\; \big(1 - w_j(s)\big)\,\delta_j. \tag{A11}$$

Φ is then continuous on the whole simplex, Brouwer applies directly with no consistency condition to discharge, the continuity hypothesis behind Section 2.6 holds, and $h \to 0$ recovers (A1).

Two things are worth reporting about what that buys. The first is that the guarantee is about existence and not about reachability. At the primitives of Example 1, damped iteration on (A11) settles cleanly for $h \geq 0.01$ — list 2 is paid with probability about 0.6, neither clearing nor failing — and then stops settling: at $h = 0.005$ the iteration enters a cycle of amplitude 0.031, at $h = 0.001$ one of amplitude 0.057, with w_2 collapsing toward zero without ever resting there. A fixed point exists at every h; below $h \approx 0.01$ this dynamic does not find it. That is the non-existence of Example 1 reappearing in the smoothed model as a limit that will not settle, which is the honest form of the same fact.

The second is that on the vereador panel the fit statistics settle even where the iteration does not. The iteration reaches a residual below 10^{-11} at $h = 0.010$, $h = 0.030$ and $h = 0.100$; at $h = 0.003$ and $h = 0.001$ it does not, stopping at about 8×10^{-2}, so those two columns are non-converged iterates rather than fixed points, and we do not know what fails to settle in them. What fails to settle is confined to markets that do not move the panel: at $h = 10^{-5}$ and $h = 10^{-6}$ the three statistics the table reports land on the sharp rule's values to every digit we print, while the residual stays at that same 8×10^{-2}. So the statistics are not in doubt and the iteration's settling at small h is. Solving (A11) at $\hat{\theta}$ over all 11,129 markets:

h	0.001	0.003	0.010	0.030	0.100
corr with observed shares	0.918	0.915	0.909	0.912	0.910
1 − SSE/SST	0.316	0.289	0.243	0.262	0.245
clearing-set agreement	0.899	0.885	0.863	0.864	0.878

The sharp rule, from the same start and the same $\hat{\theta}$, gives a correlation of 0.921, a 1 − SSE/SST of 0.334 and clearing-set agreement of 0.910. From the narrowest bandwidth to the widest the fit degrades gently: over two orders of magnitude clearing-set agreement falls 2.1 points end to end, from 0.899 at $h = 0.001$ to 0.878 at $h = 0.100$, and 3.6 at its trough (0.863 at $h = 0.010$); 1 − SSE/SST falls 7.1, from 0.316 to 0.245. Between the ends the ordering breaks: clearing-set agreement is higher at $h = 0.100$ than at either $h = 0.030$ or $h = 0.010$, and the approach to the sharp values is monotone only over the three narrowest bandwidths.

What h should be, and the price of finding out. h is not a nuisance parameter. It says how much of a list's clearing is genuinely uncertain when the candidate registers, and that is measurable: project the realised vote share on everything a candidate could know in August and the standard deviation of what is left is the uncertainty he faces.

v_j forecast from	resid. s.d. = h	R^2
nothing (the raw spread of v)	0.1497	—
council size and the number of lists	0.1070	0.490
+ the party's lagged vote share	0.1008	0.546
+ market fixed effects	0.0926	—

Even the last row, which gives the candidate a forecast no one inside the market could beat, leaves $h \approx 0.093$. The median list faces a quotient of 0.100. **The uncertainty about whether a list clears is about as large as the bar it has to clear.** At that h, 63.6% of lists sit within one bandwidth of the quotient and 84.1% have a clearing probability between 0.1 and 0.9: for most lists, at the moment the registration decision is made, clearing is genuinely open.

That is where the exercise turns uncomfortable. The calibrated h sits at the poorly-fitting end of the table above — $1 - \text{SSE/SST}$ of 0.245 against 0.316 at $h = 0.001$, and clearing agreement of 0.878 against 0.899 — and the fit improves steadily only once h falls below 0.010. The model fits best when it is told candidates knew what would happen, and worse when it is told only what they could have known. This is the same finding as Section 6.8's about ρ, arriving by a second route: the model's agreement with the data leans on hindsight, and both places where we have tried to remove the hindsight it has cost us.

The contrast between Example 1 and the vereador panel is the point. Where the sharp rule has no equilibrium the smoothed dynamic will not settle; where it has one — and Section 6.7 finds one in every market enumerated — the smoothed dynamic settles at every h, to a residual below 10^{-11}. The soft edge is therefore a repair rather than a different model, and adopting it in place of (A1) is what we would do in a version of this paper that rebuilt Section 6.7 and what follows around it. We have not rebuilt them, and the numbers those sections report are the sharp rule's.

Example 1 and the check at $\hat{\theta}$ are in `scripts/python/structural/pr_existence_counterexample.py` and `pr_existence_condition.py`.

A.5. Which regimes are self-fulfilling

Through Corollary A.2 this subsection sets $\rho_j = \rho$, so that the quotient is crossed by a share of candidates and the conditions read on s directly. In the paper's model that reduction is (A1) exactly, because $\sum_{k \geq 1} s_k = 1$. In the general case it is not, and the difference is the one Section A.1 warns of: (A1) normalises by the participating mass and reduces to $s_j \geq q(1 - s_0)$, not to $s_j \geq q$. The general case of this subsection therefore carries the *population* rule

$$W(s) \;=\; \big\{\, j \geq 1 : s_j \geq q \,\big\}, \tag{A12}$$

under which the quotient is a share of the whole population rather than of the participants. This is the rule Proposition A.3 is stated for and the rule the illustration below is computed under, and declaring it buys two things. The conditions of (A15) are then the rule's own conditions, so the biconditional holds in both directions. And the truncation is inactive at any such s^A for free: with $\rho_j = \rho$ the floor (A6) is $\underline{s}_j = q$, so $j \in A$ gives $s^A_j \geq q = \underline{s}_j$ with no participation factor to leak, and the restriction Proposition A.2 needs in the general case is

not needed here. The two rules coincide on the face $s_0 = 0$, so nothing that reaches the application is affected either way.

Keeping (A1) in the general case instead is possible and we do not use it. The non-member condition $e^{\bar\delta_A} s_0^A < q(1 - s_0^A)$ would give $s_0^A < q/(e^{\bar\delta_A} + q)$ in place of $s_0^A < q\, e^{-\bar\delta_A}$, hence $S_A > 1 - q(1 + D_A)/(e^{\bar\delta_A} + q)$ in place of (A15), and the member half would read $\min_{j \in A} s_j^A \geq q(D_A + S_A)/(1 + D_A)$. Both conditions specialise to (A16) unchanged when the outside option is deleted, which is why the application is indifferent; what they would change is the general case's printed floor and the counts of the illustration, which are computed under (A12).

Proposition A.4 states the vote rule in full, and the reduction is exact. "General case" names the choice set $\mathcal{J} = \{0, 1, \ldots, J\}$ of Section A.1, not the winner rule; within this subsection the general choice set is paired with (A12).

Proposition A.2 reduces existence to a finite collection of regime-specific fixed-point problems, and solving them shows the threshold game to be a coordination game with several self-fulfilling regimes — on the vereador data, one at two lists rising to 6.4 at eight. Enumeration is not a characterisation, however, and the search can be replaced by a closed-form condition. The step that makes this possible is that the non-member lists are indistinguishable from one another up to a constant — and, where there is one, from the outside option too. Throughout this subsection a *non-member* is a list $j \in \{1, \ldots, J\} \setminus A$; the outside option, where present, is counted separately, so that D_A and $\bar\delta_A$ below never range over $j = 0$. And one convention governs the degenerate regime $A = \{1, \ldots, J\}$, at which there is no non-member, $D_A = 0$ and $\bar\delta_A$ is a maximum over the empty set: in (A15), (A16) and Corollary A.2 the *second* condition is then vacuous and read as satisfied, the product $q\, e^{-\bar\delta_A} D_A$ being an indeterminate form rather than zero. Such a regime is self-fulfilling exactly when $\min_j s_j^A \geq q$, which by (A5) needs $Jq \leq 1$; it is not a curiosity, since a median municipality has four lists clearing.

LEMMA A.2 (Non-member shares). *Let s^A be a fixed point of Φ^A, and write $S_A = \sum_{j \in A} s_j^A$ and $D_A = \sum_{j \notin A} e^{\delta_j}$. For either choice set, for all non-members j, k and every τ,*

$$\text{(A13)} \qquad \frac{\sigma(j \mid \tau; s^A)}{\sigma(k \mid \tau; s^A)} \;=\; e^{\delta_j - \delta_k}, \qquad \textit{and hence} \qquad s_j^A \;=\; e^{\delta_j}\, c_A$$

for a constant c_A common to every non-member. In the general case the outside option, though not a non-member by the convention above, satisfies (A13) with $\delta_0 = 0$, so $c_A = s_0^A$ and $s_j^A = e^{\delta_j} s_0^A$. In the paper's model, provided $D_A > 0$, $c_A = (1 - S_A)/D_A$.

PROOF. Under (A7) the payoffs $\pi^A(j, \cdot, \tau) = \delta_j$ and $\pi^A(k, \cdot, \tau) = \delta_k$ of two non-members depend neither on τ nor on s. The logit denominator is common to the two, so the ratio of probabilities is $\exp(\delta_j - \delta_k)$ for every type. Aggregating with the type weights, $s_j^A = \sum_\tau \mu(\tau)\sigma(j \mid \tau) = e^{\delta_j - \delta_k} \sum_\tau \mu(\tau)\sigma(k \mid \tau) = e^{\delta_j - \delta_k} s_k^A$, so $e^{-\delta_j} s_j^A$ takes a value c_A that does not depend on which non-member it is computed from. In the general case the outside option pays the constant 0 as well, so it may be taken as k with $\delta_0 = 0$, giving $c_A = s_0^A$. In the paper's model the non-members hold $1 - S_A$ between them and $\sum_{j \notin A} e^{\delta_j} c_A = c_A D_A$, so $c_A = (1 - S_A)/D_A$ whenever $D_A > 0$; where $D_A = 0$ there is no non-member and the statement is vacuous. □

PROPOSITION A.3 (Characterisation, with an outside option). *Take the general case $\mathcal{J} = \{0, 1, \ldots, J\}$. With S_A*

and D_A as in Lemma A.2 and $\bar{\delta}_A = \max_{j \notin A} \delta_j$, then

$$s_0^A \;=\; \frac{1 - S_A}{1 + D_A}, \tag{A14}$$

and s^A is a mean-field equilibrium if and only if

$$\min_{j \in A} s_j^A \;\geq\; q \qquad \text{and} \qquad S_A \;>\; 1 - q\, e^{-\bar{\delta}_A}\,(1 + D_A). \tag{A15}$$

Proof. Shares sum to one, so by Lemma A.2, $1 = s_0^A + \sum_{j \notin A} e^{\delta_j} s_0^A + S_A = s_0^A(1 + D_A) + S_A$, which is (A14). The first half of (A15) is the member half of (A12), and (A8) is not invoked here: it is (A1)'s condition and in the general case the two differ by the factor $1 - s_0^A$. For the second, the non-member half of (A12) requires $s_j^A < q$ for every $j \notin A$; by Lemma A.2 this is $e^{\delta_j} s_0^A < q$ for every such j, which binds only at the largest δ_j and is therefore equivalent to $s_0^A < q\, e^{-\bar{\delta}_A}$. Substituting (A14) and rearranging gives the stated floor. □

Two features deserve comment. The $J - |A|$ separate conditions on non-members collapse to *one*, because the non-members differ only by a constant and so only the most attractive of them can threaten the regime. And what survives is a floor on the total share the members must capture, stated in primitives: given q and δ, (A15) says how much of the population a coalition of clearing lists must hold to keep the rest below the quotient.

Corollary A.1 (No outside option: the form the application uses). *Take the paper's model, $\mathcal{J} = \{1, \ldots, J\}$, so that the choice set is the J lists. Then $s_j^A = e^{\delta_j}(1 - S_A)/D_A$ for every non-member j, and s^A is a mean-field equilibrium if and only if*

$$\min_{j \in A} s_j^A \;\geq\; q \qquad \text{and} \qquad S_A \;>\; 1 - q\, e^{-\bar{\delta}_A}\, D_A. \tag{A16}$$

(At the degenerate regime $A = \{1, \ldots, J\}$, where $D_A = 0$ and $\bar{\delta}_A$ is a maximum over the empty set, the second condition is read by the convention of Section A.5 and not by $\infty \cdot 0 = 0$, under which it is vacuous and satisfied.)

Proof. Every ingredient of Proposition A.3 survives the deletion, and Proposition A.2 is strengthened by it. Lemma A.2 holds for either choice set and already gives $c_A = (1 - S_A)/D_A$, which is the first display. Proposition A.1 is untouched, the floor (A6) entering Lemma A.1 only as a positive constant. Proposition A.2 is stated in the paper's model to begin with, the absence of the outside option being exactly what makes (A6) a bound on the share of a clearing list rather than the weaker $\underline{s}_j(1 - s_0^A)$. What is left is the accounting in (A14): the outside option is the alternative with $\delta_0 = 0$, contributing $e^{\delta_0} = 1$ to that sum, so deleting it replaces $1 + D_A$ by D_A and leaves the non-members holding $1 - S_A$ between them. The member half of (A8) is unchanged; the non-member half still binds at the largest δ_j, and $c_A < q\, e^{-\bar{\delta}_A}$ rearranges to the stated floor. □

Corollary A.1 is the version the application uses, since Corollary 1 identifies the model without an outside option and the estimator's choice set is the lists. Enumerating regimes at the estimated parameters, over the 9,125 markets fielding between two and eight lists, the closed form and a direct evaluation of (A8) at the solved fixed point agree in all 431,440 regime-market pairs, and a self-fulfilling regime exists in every market. The regime that actually occurred is one of them in 84.7% of markets under (A1), against 60.0% under the candidate-share rule.

The characterisation extends to the vote rule of (A1) without losing its closed form, and the same collapse happens: however many non-members there are, only the one that converts candidates into votes most effectively can threaten the regime.

PROPOSITION A.4 (Characterisation under the vote rule). *Let W be given by* (A1) *and take the paper's model, $\mathcal{J} = \{1, \dots, J\}$. Write*

$$R_A = \sum_{j \in A} \rho_j s_j^A, \qquad E_A = \sum_{j \notin A} \rho_j e^{\delta_j}, \qquad M_A = \max_{j \notin A} \rho_j e^{\delta_j},$$

so that R_A is the vote weight of the members and E_A, M_A are the total and the largest vote weight per unit of non-member mass. Then s^A is a mean-field equilibrium if and only if

$$\min_{j \in A} \rho_j s_j^A \;\geq\; q\Big(R_A + \frac{1 - S_A}{D_A} E_A\Big) \tag{A17}$$

and either $M_A \leq q E_A$, in which case no non-member can reach the quotient at any S_A, or

$$S_A \;>\; 1 - \frac{q\, R_A\, D_A}{M_A - q\, E_A}. \tag{A18}$$

PROOF. Lemma A.2 is unaffected by the winner rule, which enters nowhere in it, so $s_j^A = e^{\delta_j} c$ for every non-member j with $c = c_A = (1 - S_A)/D_A$. Total vote weight is therefore $R = \sum_k \rho_k s_k^A = R_A + c E_A$.

The member half of (A8) is $\rho_j s_j^A / R \geq q$ for every $j \in A$, which is (A17). The non-member half is $\rho_j e^{\delta_j} c < qR$ for every $j \notin A$, binding at the largest $\rho_j e^{\delta_j}$, hence $c\, M_A < q(R_A + c E_A)$, that is $c\,(M_A - q E_A) < q R_A$. If $M_A \leq q E_A$ the left side is not positive and the inequality holds for every $c > 0$. Otherwise it is $c < q R_A / (M_A - q E_A)$, and substituting $c = (1 - S_A)/D_A$ and rearranging gives (A18). □

Setting $\rho_j = \rho$ recovers Corollary A.1: then $R_A = \rho S_A$, $E_A = \rho D_A$, $M_A = \rho e^{\bar{\delta}_A}$, and (A18) reads $S_A > 1 - q S_A D_A / (e^{\bar{\delta}_A} - q D_A)$, which rearranges to $(1 - S_A)/D_A < q e^{-\bar{\delta}_A}$, the earlier floor. The new content is the case $M_A \leq q E_A$, which has no analogue under the candidate rule: when the non-members are numerous and none of them converts votes much better than the others, the regime is safe from below whatever share its members hold, because a non-member can only cross the quotient by out-drawing the field and not merely by being large.

Evaluating Proposition A.4's closed form against a direct evaluation of (A8) at the same solved fixed points, over the same 431,440 regime-market pairs, the two agree in every one, and Lemma A.2 holds there to 4.1×10^{-11}. Two things should be said about what that check does and does not show. It is the characterisation tested on estimated primitives rather than invented ones. But the branch $M_A \leq q E_A$ — the proposition's advertised new content — binds in *none* of the 431,440: on this panel the vote rule buys a statement of theory that the data never call on, and every pair is decided by (A18). The edge case $M_A = q E_A$ with $R_A = 0$ should be read as excluded, the non-member condition there being strict; it never arises in the enumeration.

COROLLARY A.2 (Symmetric non-members). *Suppose $\delta_j = \delta$ for every non-member. In the general case the floor in* (A15) *is $1 - q\,(J - |A| + e^{-\delta})$, which is increasing in δ and decreasing in q. In the paper's model the floor in* (A16) *is $1 - q\,(J - |A|)$, which is decreasing in q and does not depend on δ at all.*

PROOF. $D_A = (J - |A|)e^{\delta}$ and $\bar{\delta}_A = \delta$, so $e^{-\bar{\delta}_A}(1 + D_A) = e^{-\delta} + J - |A|$ and $e^{-\bar{\delta}_A} D_A = J - |A|$. □

Corollary A.2 carries the economics, and it carries different economics in the two cases. In the general case, raising δ makes lists that will not clear more attractive to stand on, which pushes them toward the quotient and makes any regime harder to sustain; as $\delta \to -\infty$ the floor falls without bound and the floor condition is satisfied for every regime, because nobody is left to push a non-member over the threshold by accident. That channel runs against the outside option, which is what the δ's are measured against, and the paper's model has none of it: with every non-member equally attractive and nothing else to stand on, the level of δ cancels in the normalisation among the non-members, and only the *dispersion* of δ across them — which is what the panel estimates, and what (A16) reads through $e^{-\bar{\delta}_A} D_A$ — moves the floor. That effect is signed. Since $D_A \le (J - |A|)e^{\bar{\delta}_A}$ with equality only when the non-member δ's are equal, $e^{-\bar{\delta}_A} D_A \le J - |A|$, so the floor in (A16) is lowest at symmetry and *dispersion makes every regime harder to sustain*: spreading the δ's concentrates the non-member mass on the most attractive of them, which is the only one that can threaten the regime.

The q comparative static does not transfer to the equilibrium count. What Corollary A.2 signs is the *non-member* floor, which does fall in q. The member condition $\min_{j \in A} s_j^A \ge q$ rises in q, and the two conditions bear on different objects — a minimum and a sum — so they collapse into a single requirement only at $|A| = 1$, where the lone member's share must satisfy

$$\max\{\, q,\ 1 - q\,(J - 1)\,\},$$

V-shaped in q and minimised at $q = 1/J$: in the paper's model at $J = 2$ it is 0.90 at $q = 0.10$, 0.50 at $q = 1/2$ and 0.90 again at $q = 0.90$. For larger A the member condition still forces $S_A \ge |A|\, q$, which rises in q while the floor falls, so the opposition of the two conditions is general. A higher quotient also raises the truncation floor $q\underline{\rho}/\rho_j$, thinning the members further, and destroys large regimes outright, since (A5) needs $|A|\, q \le 1$ — at $J = 2$ the regime $A = \{1, 2\}$ is unavailable for every $q > 1/2$. The count is therefore non-monotone in q, low at both ends, and we have not computed it: the sweep below is in δ, and Section 6.7 tabulates by J at the estimated q. No claim about the direction is made.

Numerical illustration. This illustration is the general case, its regimes scored by (A15). At $J = 4$, $q = 1/9$, $K = 30$ and $\alpha B = (0.65, 0.50, 0.40, 0.30)$, ten of the sixteen regimes satisfy (A15) under $\delta \equiv 0$, and eleven under $\delta = (-0.8, -0.3, -1.5, -0.1)$. Lemma A.2 and (A14) hold at the computed fixed points to 3.3×10^{-12} and 2.3×10^{-12}. Every failure is the same failure: the members are too thin, s_0 does not fall below $q\, e^{-\bar{\delta}_A}$, and a non-member clears when the regime says it should not. No singleton survives under either δ specification, and at $\delta \equiv 0$ the reason is immediate: (A15) asks a lone member for $1 - q(J - 1 + 1) = 5/9$ of the population, which no prize delivers. Sweeping δ with the other primitives fixed takes the count from sixteen at $\delta = -3$ to five at $\delta = +0.5$, monotonically, as Corollary A.2 requires in the general case.

What multiplicity does and does not break. It does not break estimation. Which lists cleared the quotient is observed, so the estimator conditions on the realised regime and the inversion of Section 3.2 runs inside it unchanged. What it breaks is counterfactual simulation: the response of shares to a change in q, in B, or in the registration cap is a set rather than a point, and reporting it as a point would need a selection rule the model does

not supply. We report bounds and record in Section 6.7 the three attempts to narrow them that failed.

Why δ_j is carried rather than normalised. Lemma A.2 is refutable without the outside option, which is what makes the specification testable at all: under $\delta \equiv 0$ every non-clearing list in a market must hold the same share. On the vereador panel the within-market coefficient of variation of non-clearing shares is 0.48 in 2016 and 0.43 in 2020, against 0.18 and 0.20 among clearing lists, which carry genuinely different prizes and are supposed to differ. Non-clearing lists are the *more* heterogeneous group where the prediction is homogeneity. The zero normalisation is rejected, and δ_j is what replaces it.

The weaker implication that survives — that $\delta_j - \delta_k$ does not vary with τ, so the log ratio of two non-clearing lists' type-conditional shares is one number per market — we do not claim to have tested. A parametric bootstrap conditioned on the sample's own selection returns $p = 0.205$ for non-clearing pairs, but the positive control fails: clearing pairs, which must violate the same null through $\alpha(\tau)$, return $p = 0.770$, their observed dispersion sitting *below* the null mean, 0.1039 against 0.1078. A test that cannot detect the violation built into it is uninformative about one it did not build in. The obstacle is cell size, and a version of the test on coarser types or larger municipalities is left open.

A.6. Remark: the argmax proof, and how to repair it

An earlier existence appendix for this project, written for the argmax rule and now retired, restricts Φ to $\mathcal{D}_{\bar{s}} = \{s : s_j \geq \bar{s}\ \forall j, \ldots\}$ and asserts a self-map by choosing $\bar{s}$ with $\bar{s} \leq \sigma_{\min}(\bar{s})$, on the grounds that $\sigma_{\min}(\bar{s})$ "approaches a positive constant as $\bar{s} \to 0^{+}$". It does not. As $\bar{s} \to 0$ the winner's prize $\alpha B_{j^\star}/\bar{s}$ diverges, the winner's choice probability tends to one, and every other probability is crushed at rate $\exp(-\alpha B_{j^\star}/\bar{s})$. So $\sigma_{\min}(\bar{s})$ vanishes exponentially in $1/\bar{s}$ while $\bar{s}$ vanishes linearly, and the required inequality fails for *every* $\bar{s}$; numerically, at $\alpha B = 0.647$ and $J = 3$, $\sigma_{\min}$ runs from 8.7×10^{-2} at $\bar{s} = 0.30$ to 1.1×10^{-281} at $\bar{s} = 0.001$, never once exceeding $\bar{s}$. Three further gaps: $\mathcal{D}_{\bar{s}}$ as written is not convex, so Brouwer does not apply to it; it presumes the identity of $j^\star$, which the equilibrium is supposed to determine; and the claim that randomising uniformly across ties "preserves continuity" is false, since the payoff jumps by a finite amount across the tie set.

The argmax case is nonetheless repairable, by a shorter argument than the one attempted, and the repair is the argument that follows. It is written with an outside option, so everything in this paragraph is the general case of Section A.1. The winner's share is bounded below for free: since $\sum_{j \geq 1} s_j = 1 - s_0$, the largest inside share satisfies $\max_{j \geq 1} s_j \geq (1 - s_0)/J$. So the prize is bounded wherever s_0 is bounded away from one, and s_0 is self-limiting — a small winner share means a large prize, which drives the outside option down. Since the winner's share is at most one, its payoff is at least $\underline{\pi} = \min_\tau X(\tau)'\beta + \min_\tau \alpha(\tau) \min_j B_j$, so $\Phi(s)_0 \leq 1/(J + e^{\underline{\pi}})$ for *every* s: the operator maps the simplex into the compact convex set $\{s : s_0 \leq 1 - \eta\}$ with $\eta = 1 - 1/(J + e^{\underline{\pi}})$, and no $\bar{s}$ is chosen at all. Fixing the regime as above then makes the operator continuous, and the resulting fixed point is its own arg max whenever $X(\tau)'\beta + \alpha(\tau)B_j \geq 0$, which delivers one equilibrium per candidate rather than the one the earlier argument sought. That last condition belongs to the argmax model and does not transfer: it is derived there from losing lists that all pay zero and an $\alpha(\tau)$ signed positive, and with the δ_j of (A2) and Assumption 1's unsigned $\alpha(\tau)$ the comparison would be against $\bar{\delta}_A$ rather than against zero, and would have to be reproved.

Appendix B. Finite Markets: Separation from the Quotient and the ϵ-Equilibrium Bound

Definition 1 is a limit object. The question this appendix answers is what it licenses at the market sizes Section 2.5 reports, and the answer is conditional on a quantity that can be computed market by market.

B.1. Why the cited hypothesis is not available

The static large-game approximation of Bodoh-Creed (2013) asks that payoffs be continuous in the aggregate. Equation (4) is not: crossing the quotient switches the prize on, which is the content of Section A.2 and the reason Proposition A.1 is proved regime by regime rather than by a single Brouwer argument. Restoring continuity on a regime cell does not restore the hypothesis the conclusion needs. In a market of I_m agents one agent's relocation moves every share by $1/I_m$, so a market whose lists sit within one agent's influence of the quotient admits a unilateral deviation that leaves the cell; at such a deviation the payoff moves by the difference between the two branches of (4), which Section A.4 shows is first order and of either sign. A bound asserted on a cell bounds no deviation that leaves it, and those are the deviations an ϵ-equilibrium claim is about.

What follows therefore does not invoke that theorem. It proves the ϵ-equilibrium bound directly, in this game's own notation, under a separation condition on the distance between a market's vote shares and its quotient. The condition is a comparison of two numbers per market, both functions of primitives and observables, and Section 2.6 states it where the claim is made.

B.2. Notation

This section carries the market subscript. Market m holds I_m agents with types $\tau_1, \dots, \tau_{I_m}$ whose empirical distribution is μ_m, so that (1) reads $s_{m,j} = I_m^{-1} \sum_i \sigma_m(j \mid \tau_i)$. Write

$$v_{m,l}(s) \;=\; \frac{\rho_{m,l}\, s_{m,l}}{R_m(s)}, \qquad R_m(s) \;=\; \sum_{k \in \mathcal{J}_m} \rho_{m,k}\, s_{m,k}, \tag{A19}$$

for the vote share (3) thresholds, and $\bar{\rho}_m = \max_k \rho_{m,k}$ beside Assumption 1's $\underline{\rho}_m = \min_k \rho_{m,k}$. Set

$$\Lambda_m \;=\; \frac{2\bar{\rho}_m - \underline{\rho}_m}{\underline{\rho}_m}, \qquad \Gamma_m \;=\; \frac{2\bar{\rho}_m}{\underline{\rho}_m}, \tag{A20}$$

both finite and at least one by Assumption 1, with $\Lambda_m \le \Gamma_m$ and $\Lambda_m = \Gamma_m - 1$. For a profile $a \in \mathcal{J}_m^{I_m}$ let $\hat{s}(a)_j = I_m^{-1} \#\{i : a_i = j\}$, and write $(z)_+ = \max\{z, 0\}$ and ι_n for the vector of n ones.

In the finite-I_m Bayesian game agent i observes τ_i and $\boldsymbol{\varepsilon}_i$, chooses $a_i \in \mathcal{J}_m$, and receives $\pi(a_i, \hat{s}(a), \tau_i) + \varepsilon_i(a_i)$ with π given by (4). A type-conditional strategy σ is an *ϵ-equilibrium* if no agent can raise his interim expected payoff by more than ϵ by departing from it when every other agent plays it.

B.3. How far one agent can move a vote share

LEMMA B.1 (Vote-share sensitivity and unilateral influence). *Let $s, s' \in \Delta^{J_m - 1}$. Under Assumption 1,*

$$\max_{l \in \mathcal{J}_m} \left| v_{m,l}(s') - v_{m,l}(s) \right| \;\le\; \frac{\bar{\rho}_m}{\underline{\rho}_m} \left(\|s' - s\|_\infty + \|s' - s\|_1 \right) \;\le\; \Gamma_m \, \|s' - s\|_1 . \tag{A21}$$

If in addition $s' = s + (e_j - e_k)/I_m$ *for some* $j \neq k$ *with* $s_{m,k} \geq 1/I_m$ *— one agent relocating from list* k *to list* j *— then*

$$\max_{l \in \mathcal{J}_m} \left| v_{m,l}(s') - v_{m,l}(s) \right| \;\leq\; \frac{\Lambda_m}{I_m}, \tag{A22}$$

and (A22) *holds with equality in the homogeneous case* $\rho_{m,j} \equiv \rho_m$*, where* $\Lambda_m = 1$.

Proof. *Step 1: an exact difference.* Write $\Delta_l = \rho_{m,l}(s'_{m,l} - s_{m,l})$ and $\Delta R = R_m(s') - R_m(s) = \sum_k \Delta_k$. Then

$$v_{m,l}(s') - v_{m,l}(s) \;=\; \frac{\rho_{m,l} s'_{m,l}}{R_m(s')} - \frac{\rho_{m,l} s_{m,l}}{R_m(s)} \;=\; \frac{R_m(s)\Delta_l - \rho_{m,l} s_{m,l}\, \Delta R}{R_m(s)\, R_m(s')} \;=\; \frac{\Delta_l - v_{m,l}(s)\, \Delta R}{R_m(s')},$$

the middle equality by putting the two fractions over a common denominator and the last by dividing numerator and denominator by $R_m(s)$.

Step 2: the general bound. Both s and s' lie in $\Delta^{J_m - 1}$, so $R_m(s') \geq \underline{\rho}_m \sum_k s'_{m,k} = \underline{\rho}_m > 0$ by Assumption 1, and $0 \leq v_{m,l}(s) \leq 1$. Also $|\Delta_l| \leq \bar{\rho}_m \|s' - s\|_\infty$ and $|\Delta R| \leq \bar{\rho}_m \|s' - s\|_1$. Substituting into Step 1 gives the left inequality of (A21); the right inequality follows from $\|\cdot\|_\infty \leq \|\cdot\|_1$.

Step 3: one agent. For $s' = s + (e_j - e_k)/I_m$ we have $\Delta_l \in \{\rho_{m,j}/I_m,\, -\rho_{m,k}/I_m,\, 0\}$, so $|\Delta_l| \leq \bar{\rho}_m/I_m$, and $\Delta R = (\rho_{m,j} - \rho_{m,k})/I_m$, so $|\Delta R| \leq (\bar{\rho}_m - \underline{\rho}_m)/I_m$. Step 1 and $R_m(s') \geq \underline{\rho}_m$ then give $\big(\bar{\rho}_m + \bar{\rho}_m - \underline{\rho}_m\big)/(\underline{\rho}_m I_m) = \Lambda_m/I_m$, which is (A22).

Step 4: the homogeneous case. If $\rho_{m,j} \equiv \rho_m$ then $R_m(s) = \rho_m$ for every $s \in \Delta^{J_m - 1}$ and $v_{m,l}(s) = s_{m,l}$, so the left side of (A22) is exactly $1/I_m$; and $\Lambda_m = (2\rho_m - \rho_m)/\rho_m = 1$. □

Lemma B.1 is the influence bound the separation condition is written against. It is the quantity the model section means by "one agent's influence": a single relocation cannot move any list's vote share by more than Λ_m/I_m, and Λ_m is a ratio of the largest to the smallest vote weight in the market, equal to one when the weights are equal and growing linearly in the spread Section 6.8 measures.

B.4. The separation condition

definition B.1 (Quotient margin and separation). *For* $s \in \Delta^{J_m - 1}$ *the* quotient margin *of market* m *at* s *is*

$$\eta_m(s) \;=\; \min_{l \in \mathcal{J}_m} \left| v_{m,l}(s) - q_m \right|, \tag{A23}$$

with $v_{m,l}$ *the vote share* (A19)*, and at a mean-field equilibrium* $s^\star_m$ *we write* $\eta^\star_m = \eta_m(s^\star_m)$*. Market* m *is*

$$\text{weakly separated} \quad \textit{if} \quad \eta^\star_m \;>\; \frac{3\,\Gamma_m J_m}{I_m}, \tag{A24}$$

$$\text{strongly separated} \quad \textit{if} \quad \eta^\star_m \;\geq\; \Gamma_m J_m \sqrt{\frac{\log I_m}{I_m}}. \tag{A25}$$

Four things about Definition B.1, because the condition is the whole content of the claim it supports.

Both sides of (A24) and (A25) are numbers. The left side is read off the realised vote shares and the published quotient; the right side is J_m, I_m and the ratio $\bar{\rho}_m/\underline{\rho}_m$ of (A20), all of which Section 6.8 measures. Nothing in

either is a qualitative property.

Equation (A24) is (A22) with a cushion. The sharp statement that no unilateral relocation crosses the quotient is $\eta^\star_m > \Lambda_m/I_m$, by Lemma B.2 below; (A24) asks for $3\Gamma_m J_m/I_m$ instead, which is larger by the factor $3\Gamma_m J_m/\Lambda_m$, because the proof of Proposition B.1 must absorb three displacements at once — the deviator's own move, at most $1/I_m$; the displacement of the others' leave-one-out mean from the equilibrium share vector, at most $1/(I_m - 1)$; and the sampling fluctuation of the others' realised shares about that mean — and bounds all three in one norm, paying J_m to pass from $\|\cdot\|_\infty$ to $\|\cdot\|_1$ and Γ_m to apply (A21). The cushion is a cost of the proof and not of the game.

Equation (A25) implies (A24) whenever $I_m \geq 6$, since then $I_m \log I_m > 9$ and hence $\sqrt{\log I_m/I_m} > 3/I_m$.

The orders of the two conditions differ, and the difference is the point. Equation (A24) is of order J_m/I_m and is what makes the bound of Proposition B.1 non-trivial. Equation (A25) is of order $\sqrt{\log I_m/I_m}$ and is what makes the regime-crossing term of that bound negligible against the rate, so it is (A25) and not (A24) that returns the $O(I_m^{-1/2})$ reading.

LEMMA B.2 (No unilateral crossing). *Let $s \in \Delta^{J_m-1}$ with $\eta_m(s) > \Lambda_m/I_m$. Then $W(s') = W(s)$ for every $s' = s + (e_j - e_k)/I_m$ in Δ^{J_m-1}.*

PROOF. Fix $l \in \mathcal{J}_m$. By (A22), $|v_{m,l}(s') - v_{m,l}(s)| \leq \Lambda_m/I_m < \eta_m(s)$. If $l \in W(s)$ then $v_{m,l}(s) \geq q_m$ by (3) and $|v_{m,l}(s) - q_m| \geq \eta_m(s)$ by (A23), so $v_{m,l}(s) \geq q_m + \eta_m(s)$ and therefore $v_{m,l}(s') > q_m$, whence $l \in W(s')$. If $l \notin W(s)$ then $v_{m,l}(s) < q_m$ and $v_{m,l}(s) \leq q_m - \eta_m(s)$, so $v_{m,l}(s') < q_m$ and $l \notin W(s')$. □

LEMMA B.3 (Lipschitz payoffs within a regime). *Let $s, s' \in \Delta^{J_m-1}$ with $W(s) = W(s') = A$. Under Assumption 1, for every $j \in \mathcal{J}_m$ and every $\tau \in \mathcal{T}$,*

$$\left|\pi(j, s, \tau) - \pi(j, s', \tau)\right| \;\leq\; L_m \|s - s'\|_\infty, \qquad L_m \;=\; \max_\tau |\alpha(\tau)| \max_{j \in \mathcal{J}_m} \frac{B_{m,j}\, \rho^2_{m,j}}{q^2_m\, \underline{\rho}^2_m}.$$

PROOF. If $j \notin A$ then $\pi(j, s, \tau) = \pi(j, s', \tau) = \delta_{m,j}$ by (4) and the difference is zero. If $j \in A$ then both s and s' lie in the cell C_A, so $s_{m,j} \geq \underline{s}_{m,j}$ and $s'_{m,j} \geq \underline{s}_{m,j}$ with $\underline{s}_{m,j} = q_m \underline{\rho}_m/\rho_{m,j}$ by (A6). The terms $X(\tau)'\beta$ and $\xi_{m,j}$ of (4) do not depend on s, so

$$\left|\pi(j, s, \tau) - \pi(j, s', \tau)\right| = |\alpha(\tau)|\, B_{m,j} \left|\frac{1}{s_{m,j}} - \frac{1}{s'_{m,j}}\right| = |\alpha(\tau)|\, B_{m,j} \frac{|s'_{m,j} - s_{m,j}|}{s_{m,j} s'_{m,j}} \leq \frac{|\alpha(\tau)| B_{m,j}}{\underline{s}^2_{m,j}} \|s - s'\|_\infty,$$

and $\underline{s}^{-2}_{m,j} = \rho^2_{m,j}/(q_m \underline{\rho}_m)^2$. Taking the maximum over the finitely many types and lists, each factor bounded by Assumption 1, gives L_m. □

B.5. The bound

PROPOSITION B.1 (The mean-field equilibrium is an ϵ-equilibrium of the finite market, with ϵ a function of the margin). *Let $(\sigma^\star_m, s^\star_m)$ be a mean-field equilibrium of market m in the sense of Definition 1, let $I_m \geq 3$, and let $\bar{\pi}_m$ denote the uniform payoff bound of Lemma A.1. Set*

$$t_m \;=\; \left(\frac{\eta^\star_m}{\Gamma_m J_m} - \frac{3}{I_m}\right)_+. \tag{A26}$$

Then, under Assumption 1, $\sigma_m^\star$ is an ϵ_m-equilibrium of the finite-I_m Bayesian game with

$$\epsilon_m \;\le\; \underbrace{2L_m\left(\sqrt{\frac{\log(2J_m)}{2(I_m-1)}}+\frac{3}{I_m}\right)}_{\text{within-regime term}} \;+\; \underbrace{8\bar\pi_m J_m \exp\!\big(-2(I_m-1)\,t_m^2\big)}_{\text{regime-crossing term}}. \tag{A27}$$

Proof. Fix an agent i and write $\tau=\tau_i$. Every other agent plays $\sigma_m^\star(\cdot\mid\tau_{i'})$, independently across i'.

Step 1: the others' share vector and its mean. Let $\bar s_{-i,j}=(I_m-1)^{-1}\#\{i'\neq i: a_{i'}=j\}$ and $s^\dagger_{-i,j}=\mathbb{E}\,\bar s_{-i,j}=(I_m-1)^{-1}\sum_{i'\neq i}\sigma_m^\star(j\mid\tau_{i'})$. With $S_j=\sum_{i'}\sigma_m^\star(j\mid\tau_{i'})$ we have $s^\star_{m,j}=S_j/I_m$ by (1), so

$$\big|s^\dagger_{-i,j}-s^\star_{m,j}\big|=\left|\frac{S_j-\sigma_m^\star(j\mid\tau_i)}{I_m-1}-\frac{S_j}{I_m}\right|=\frac{\big|S_j-I_m\sigma_m^\star(j\mid\tau_i)\big|}{I_m(I_m-1)}\;\le\;\frac{1}{I_m-1},$$

the inequality because $0\le S_j\le I_m$ and $0\le\sigma_m^\star(j\mid\tau_i)\le 1$.

Step 2: the realised share vector when i plays j. Writing $\hat s(j,a_{-i})$ for the share vector of all I_m agents when i chooses j,

$$\hat s(j,a_{-i})=\frac{I_m-1}{I_m}\,\bar s_{-i}+\frac{1}{I_m}\,e_j,\qquad\text{so}\qquad \|\hat s(j,a_{-i})-\bar s_{-i}\|_\infty=\frac{1}{I_m}\|e_j-\bar s_{-i}\|_\infty\le\frac{1}{I_m}.$$

With $Z_i=\|\bar s_{-i}-s^\dagger_{-i}\|_\infty$ and Step 1, and using $1/(I_m-1)+1/I_m\le 3/I_m$ for $I_m\ge 2$,

$$\|\hat s(j,a_{-i})-s^\star_m\|_\infty\;\le\;Z_i+\frac{3}{I_m}\qquad\text{for every } j\in\mathcal{J}_m. \tag{A28}$$

Step 3: concentration. For each j, $(I_m-1)\bar s_{-i,j}$ is a sum of I_m-1 independent indicators, so Hoeffding's inequality gives $\Pr(|\bar s_{-i,j}-s^\dagger_{-i,j}|\ge t)\le 2\exp(-2(I_m-1)t^2)$, and a union bound over $\mathcal{J}_m$ gives

$$\Pr(Z_i\ge t)\;\le\;2J_m\exp\!\big(-2(I_m-1)t^2\big). \tag{A29}$$

Each coordinate is therefore sub-Gaussian with variance proxy $1/(4(I_m-1))$, and the maximal inequality for J_m such coordinates gives

$$\mathbb{E}\,Z_i\;\le\;\sqrt{\frac{\log(2J_m)}{2(I_m-1)}}. \tag{A30}$$

Step 4: the good event, on which the regime does not move. Let $G_i=\{Z_i<t_m\}$ with t_m as in (A26), and suppose $t_m>0$. On G_i, by (A28), $\|\hat s(j,a_{-i})-s^\star_m\|_\infty<\eta^\star_m/(\Gamma_m J_m)$ for every j, so by (A21) and $\|\cdot\|_1\le J_m\|\cdot\|_\infty$,

$$\max_l\big|v_{m,l}(\hat s(j,a_{-i}))-v_{m,l}(s^\star_m)\big|\;\le\;\Gamma_m J_m\|\hat s(j,a_{-i})-s^\star_m\|_\infty\;<\;\eta^\star_m.$$

The argument of Lemma B.2, applied with $s=s^\star_m$ and this displacement in place of Λ_m/I_m, gives $W(\hat s(j,a_{-i}))=W(s^\star_m)$ for every $j\in\mathcal{J}_m$ on G_i. By (A29), $\Pr(G_i^c)\le 2J_m\exp(-2(I_m-1)t_m^2)$.

Step 5: the payoff error, branch by branch. Fix j and set $\Xi_j=|\pi(j,\hat s(j,a_{-i}),\tau)-\pi(j,s^\star_m,\tau)|$. On G_i the

two share vectors lie in the same regime cell by Step 4, so Lemma B.3 and (A28), (A30) give

$$\mathbb{E}\big[\Xi_j \mathbf{1}_{G_i}\big] \;\le\; L_m\,\mathbb{E}\Big[Z_i + \tfrac{3}{I_m}\Big] \;\le\; L_m\left(\sqrt{\frac{\log(2J_m)}{2(I_m-1)}} + \frac{3}{I_m}\right).$$

On G_i^c the branches may differ. Every share vector realised in the finite game satisfies $s_{m,l} \ge \underline{s}_{m,l}$ for each $l \in W(s)$ by (A6), so π agrees there with the truncated payoff (A7) and Lemma A.1 bounds it by $\bar{\pi}_m$; hence $\Xi_j \le 2\bar{\pi}_m$ and $\mathbb{E}[\Xi_j \mathbf{1}_{G_i^c}] \le 2\bar{\pi}_m \Pr(G_i^c) \le 4\bar{\pi}_m J_m \exp(-2(I_m-1)t_m^2)$. Writing E_m for the sum of the G_i bound and the G_i^c bound just displayed, $\mathbb{E}\,\Xi_j \le E_m$ for every $j \in \mathcal{J}_m$, so that E_m is exactly half the right side of (A27), term by term: $L_m(\cdot)$ against $2L_m(\cdot)$ and $4\bar{\pi}_m J_m e^{(\cdot)}$ against $8\bar{\pi}_m J_m e^{(\cdot)}$.

Step 6: from payoff error to ϵ. Let $U_i(j) = \mathbb{E}_{a_{-i}}\pi(j, \hat{s}(j, a_{-i}), \tau) + \varepsilon_i(j)$ be agent i's interim payoff from j in the finite game and $\hat{U}_i(j) = \pi(j, s_m^\star, \tau) + \varepsilon_i(j)$ the payoff he computes at the mean-field belief. By Step 5, $|U_i(j) - \hat{U}_i(j)| \le E_m$ for every j and every realisation of $\boldsymbol{\varepsilon}_i$. By (6), $\sigma_m^\star(\cdot \mid \tau)$ is the law of $\arg\max_j \hat{U}_i(j)$, the arg max being unique almost surely under the Type I extreme value assumption. For any rule d measurable in $(\tau, \boldsymbol{\varepsilon}_i)$,

$$\mathbb{E}_{\boldsymbol{\varepsilon}} U_i(d) \;\le\; \mathbb{E}_{\boldsymbol{\varepsilon}} \hat{U}_i(d) + E_m \;\le\; \mathbb{E}_{\boldsymbol{\varepsilon}} \hat{U}_i\big(\arg\max_j \hat{U}_i(j)\big) + E_m \;\le\; \mathbb{E}_{\boldsymbol{\varepsilon}} U_i\big(\arg\max_j \hat{U}_i(j)\big) + 2E_m,$$

the middle inequality because the arg max maximises $\hat{U}_i$ pointwise. So no deviation gains more than $2E_m$, which is (A27).

Step 7: the case $t_m = 0$. If $t_m = 0$ the regime-crossing term of (A27) is $8\bar{\pi}_m J_m$, which exceeds the trivial bound $2\bar{\pi}_m$ available from Lemma A.1 alone, so (A27) holds and asserts nothing. □

COROLLARY B.1 (The rate, and its constant). *Fix J_m and let market m be strongly separated,* (A25). *Then the regime-crossing term of* (A27) *is $o(I_m^{-1/2})$ and*

$$\epsilon_m \;\le\; L_m\sqrt{\frac{2\log(2J_m)}{I_m - 1}}\,\big(1 + o(1)\big),$$

so $\epsilon_m = O(I_m^{-1/2})$ with constant $L_m\sqrt{2\log(2J_m)}$, where L_m is the Lipschitz constant of Lemma B.3.

PROOF. Under (A25), $t_m \ge \sqrt{\log I_m/I_m} - 3/I_m$ and $\sqrt{\log I_m/I_m}\,(I_m/3) = \sqrt{I_m \log I_m}/3 \to \infty$, so $t_m = (1 + o(1))\sqrt{\log I_m/I_m}$ and $2(I_m - 1)t_m^2 = (2 + o(1))\log I_m$. Hence the regime-crossing term is $O(J_m I_m^{-2+o(1)})$, which for fixed J_m is $o(I_m^{-1/2})$. The within-regime term is $2L_m\sqrt{\log(2J_m)/(2(I_m-1))} + 6L_m/I_m$, whose first summand is the display and whose second is $o(I_m^{-1/2})$. □

B.6. What the condition does and does not settle

Proposition B.1 is conditional and the condition is not discharged here. Three statements fix its reach.

The bound is a bound on unilateral deviations at a given mean-field equilibrium, so it inherits the multiplicity of Section A.5: $\eta_m^\star$ is a property of the equilibrium selected, and a market can be strongly separated at one self-fulfilling regime and weakly separated at another. Section 3.6 conditions on the realised regime, so the margin to compute is the realised one.

The constant in Corollary B.1 is not one. It is $L_m\sqrt{2\log(2J_m)}$ with L_m of Lemma B.3, which rises with $\max_\tau |\alpha(\tau)|$, with the prize, and with the spread of the vote weights, and falls with the quotient. A figure obtained by evaluating $I_m^{-1/2}$ at a representative I_m reports the rate factor and not ϵ_m.

Where a market fails (A24) the proposition says nothing, and the reason is not slack in the proof. A list within one agent's influence of the quotient is a list at which one agent's relocation changes the regime, and at such a deviation the payoff moves by the branch difference (A4) — a first-order quantity bounded by $2\bar{\pi}_m$ and, by (A10), of either sign. No $O(I_m^{-1/2})$ statement survives there, and the honest report is the fraction of markets on each side of (A24) and (A25).

Which threshold a reported fraction was computed against has to be said, because three thresholds are in play and they are far apart. The sharp no-crossing threshold is Lemma B.2's $\eta_m^\star > \Lambda_m/I_m$, computed from the market's own ρ vector and from nothing else; (A24) is $3\Gamma_m J_m/I_m$, larger by the factor $3\Gamma_m J_m/\Lambda_m$, which at $J_m \approx 6$ and homogeneous vote weights is about 36; and (A25) is larger again by $\sqrt{I_m \log I_m}/3$. A fraction computed against Λ_m/I_m therefore measures Lemma B.2 and is an *upper* bound on the fraction satisfying (A24); it is not the fraction on which Proposition B.1 is non-vacuous, which is the smaller number obtained by comparing $\eta_m^\star$ with $3\Gamma_m J_m/I_m$ itself. The three are reported separately or not at all.

Appendix C. The Invariance Group of the Payoff and the Logit

Section 2.3 fixes one location by convention and Corollary 1 says what the convention costs. This section derives the group those statements are about: the set of transformations of $(\beta, \{\alpha(\tau)\}, \{\xi_p\}, \{\delta_p\})$ that leave every choice probability in every market unchanged. The count is two or one according to whether $X(\tau)$ spans a constant. The design estimated here does not span one — it carries no intercept and enters age as a single standardised score, Remark C.1 — so the count is one, one convention fixes the location, and $X(\tau)'\beta$ is identified outright. In either case the reason is *not* that only within-branch differences of ξ and of δ appear — they do not, and (12) is the proof.

C.1. The parameter, the observable, and the equivalence

This section restores the market subscript and works at Corollary 1's condition (ii): the local components vanish, so $\xi_{m,j} = \xi_{p(j)}$ and $\delta_{m,j} = \delta_{p(j)}$ with $p(j)$ the party of list j and $\mathcal{P}$ the set of parties, of which there are $n_\mathcal{P}$. The prize is written $P_{m,j}$ throughout, so the party count is never abbreviated to P. The parameter is

$$\theta \;=\; \big(\beta,\; \{\alpha(\tau)\}_{\tau\in\mathcal{T}},\; \{\xi_p\}_{p\in\mathcal{P}},\; \{\delta_p\}_{p\in\mathcal{P}}\big) \;\in\; \Theta \;=\; \mathbb{R}^{n_X} \times \mathbb{R}^{K} \times \mathbb{R}^{n_\mathcal{P}} \times \mathbb{R}^{n_\mathcal{P}}.$$

The observable is, for every market m, the realised winner set W_m of Assumption 2, the prizes $P_{m,j}$ of (9), and the choice probabilities $\sigma_m(j \mid \tau)$ for every $j \in \mathcal{J}_m$ and every τ in market m's support. Two parameters θ, θ' are *observationally equivalent*, $\theta \sim \theta'$, if they generate the same choice probabilities in every market for every type in its support.

Write $\mathcal{X}$ for the $K \times n_X$ matrix whose rows are the $X(\tau)'$, and say that $X(\tau)$ *spans a constant* if $\iota_K \in \mathrm{col}(\mathcal{X})$, that is if some $e \in \mathbb{R}^{n_X}$ has $X(\tau)'e = 1$ for every $\tau \in \mathcal{T}$. A design containing an explicit intercept spans a constant, and so does one built from the exhaustive levels of any single factor.

LEMMA C.1 (Logit location). *Fix a market m and a type τ. Then $\sigma_m(\cdot \mid \tau; \theta') = \sigma_m(\cdot \mid \tau; \theta)$ at the realised (s_m, W_m) if and only if there is a scalar $c_{m\tau}$ with $\pi_{\theta'}(j, s_m, \tau) - \pi_\theta(j, s_m, \tau) = c_{m\tau}$ for every $j \in \mathcal{J}_m$.*

PROOF. If the difference is the constant $c_{m\tau}$, then $\exp(c_{m\tau})$ factors out of the numerator and the denominator of (6) and cancels. Conversely, equality of the probabilities gives, for every pair (j, k), $\log\{\sigma_m(j \mid \tau)/\sigma_m(k \mid \tau)\} = \pi_{\theta'}(j) - \pi_{\theta'}(k) = \pi_\theta(j) - \pi_\theta(k)$, so $\pi_{\theta'}(j) - \pi_\theta(j)$ does not depend on j; call it $c_{m\tau}$. □

Because (9) is affine in θ, the payoff difference produced by $\theta' = \theta + h$ with $h = (b, a, x, d) \in \Theta$ does not depend on θ:

$$\Delta_h(j, m, \tau) \;=\; \mathbf{1}\{j \in W_m\}\big[X(\tau)'b + a(\tau)P_{m,j} + x_{p(j)}\big] \;+\; \mathbf{1}\{j \notin W_m\}\, d_{p(j)}. \tag{A31}$$

By Lemma C.1, $\theta \sim \theta + h$ for one θ if and only if it holds for all of them, and the condition is that $j \mapsto \Delta_h(j, m, \tau)$ be constant on $\mathcal{J}_m$ for every market m and every type τ in its support. The set of such h is a linear subspace of Θ; call it the *invariance space* $\mathcal{H}$. The observationally equivalent classes are its cosets, which is why the questions to ask of it are its dimension and a basis.

ASSUMPTION C.1 (Design conditions for the level). ***(D1) Prize variation across markets.*** *For every $\tau \in \mathcal{T}$ there are parties $p \neq p'$ and markets $m \neq m'$ such that a list of p and a list of p' clear in m, a list of p and a list of p' clear in m', and τ has positive probability on all four lists. Writing $j_m(p)$ for the clearing list of party p in market m, the two prize gaps differ:*

$$P_{m,j_m(p)} - P_{m,j_m(p')} \;\neq\; P_{m',j_{m'}(p)} - P_{m',j_{m'}(p')}.$$

(D2) Connectedness. *The bipartite graph joining parties to markets is connected through clearing appearances and, separately, through non-clearing appearances — condition (iii) of Corollary 1.* ***(D3) Mixed reach.*** *Every $\tau \in \mathcal{T}$ has, in some market, positive probability on a clearing list and on a non-clearing list.* ***(D4) Type design rank.*** $\operatorname{rank}(\mathcal{X}) = n_X$.

(D2) is condition (iii) of Corollary 1 and Section F.9 reports that it holds on this panel. (D1) *is* condition (i) of that corollary, and it is stronger in two ways than the within-market condition that asks only that *some* market hold a clearing pair with $P_{m,j} \neq P_{m,k}$ and two types with positive probability on both. First, (D1) asks for two markets whose prize gaps differ at a common party pair, where the within-market condition asks for one market whose two lists differ. The difference is not cosmetic: variation within a market does not separate a slope from a party intercept, because a party pair that clears together in only one market consumes its own free ξ difference there and leaves nothing over to identify the slope against. Second, (D1) asks for such a configuration at *every* $\tau \in \mathcal{T}$, where the within-market condition asks for two types; that is what carries $a(\tau) = 0$ at all types rather than at two in Step 2 of Proposition C.1. Both strengthenings are used below and neither implies the other. (D3) is what makes the mixed block of Section F.9 bind on every type. (D4) is the condition under which β is a parameter rather than a direction, and it fails exactly when the type design is collinear.

C.2. The group

PROPOSITION C.1 (The invariance space). *Under Assumption C.1,*

$$\text{(A32)} \qquad \mathcal{H} \;=\; \big\{\,(b,a,x,d)\in\Theta \;:\; a=0,\quad x=\eta\,\iota_{n_{\mathcal{P}}},\quad d=\zeta\,\iota_{n_{\mathcal{P}}},\quad \mathcal{X}b=(\zeta-\eta)\,\iota_K,\quad \eta,\zeta\in\mathbb{R}\,\big\}.$$

Consequently:

(i) if $X(\tau)$ spans a constant, $\dim\mathcal{H}=2$*, and a basis is*

$$h_1=(0,\,0,\,\iota_{n_{\mathcal{P}}},\,\iota_{n_{\mathcal{P}}}),\qquad h_2=(e,\,0,\,0,\,\iota_{n_{\mathcal{P}}}),$$

with e the unique vector satisfying $\mathcal{X}e=\iota_K$;

(ii) if $X(\tau)$ does not span a constant, $\dim\mathcal{H}=1$*, a basis is h_1, and $X(\tau)'\beta$ is identified outright.*

PROOF. *Step 1: the non-clearing branch pins d to a single level.* Let $j,k\notin W_m$ lie in the same market. Constancy of (A31) in j gives $d_{p(j)}=d_{p(k)}$. By (D2) the parties reached by non-clearing lists form one component under co-appearance, so d_p takes a common value ζ on all of them.

Step 2: the clearing branch kills a and pins x. Let $j,k\in W_m$ and let τ have positive probability on both. Constancy of (A31) in j gives

$$\text{(A33)} \qquad a(\tau)\big(P_{m,j}-P_{m,k}\big)+\big(x_{p(j)}-x_{p(k)}\big)\;=\;0.$$

Fix τ and take the parties $p\neq p'$ and markets $m\neq m'$ supplied by (D1), with $j_m(p)$ the clearing list of party p in market m as there. Equation (A33) holds in both markets with the same $x_p-x_{p'}$, so $a(\tau)\{P_{m,j_m(p)}-P_{m,j_m(p')}\}=a(\tau)\{P_{m',j_{m'}(p)}-P_{m',j_{m'}(p')}\}$, and the two prize gaps differ by (D1), so $a(\tau)=0$. As (D1) supplies such a configuration for every τ, $a=0$. With $a=0$, (A33) reads $x_{p(j)}=x_{p(k)}$ for every pair of lists clearing in the same market, and (D2) through clearing appearances propagates this to all of $\mathcal{P}$: x_p takes a common value η.

Step 3: the mixed branch ties b to the two levels. Let $j\in W_m$, $k\notin W_m$ and let τ have positive probability on both, as (D3) supplies for every τ. Constancy of (A31) across these two alternatives gives, using Steps 1 and 2,

$$X(\tau)'b+\eta\;=\;\zeta,\qquad\text{that is}\qquad X(\tau)'b=\zeta-\eta\quad\text{for every }\tau\in\mathcal{T},$$

which is $\mathcal{X}b=(\zeta-\eta)\iota_K$. Steps 1–3 show $\mathcal{H}$ is contained in the right side of (A32).

Step 4: sufficiency. Let h belong to the right side of (A32). For $j\in W_m$, (A31) equals $X(\tau)'b+\eta=(\zeta-\eta)+\eta=\zeta$; for $j\notin W_m$ it equals ζ. So $\Delta_h(\cdot,m,\tau)\equiv\zeta$ is constant in j in every market and for every type, and Lemma C.1 gives $\theta\sim\theta+h$. The equivalence is complete and not merely regime-conditional: the choice probabilities are unchanged, hence by (1) so are the shares s_m, hence so are the prizes $P_{m,j}=B_{m,j}/s_{m,j}$ and the winner set (3), so $\theta+h$ supports the same mean-field equilibrium as θ.

Step 5: dimension and basis. By (D4), $\ker\mathcal{X}=\{0\}$, so for each $\kappa\in\mathbb{R}$ the system $\mathcal{X}b=\kappa\,\iota_K$ has at most one solution. If $X(\tau)$ spans a constant it has exactly one, $b=\kappa e$, and (A32) is parametrised bijectively by $(\eta,\zeta)\in\mathbb{R}^2$ through $b=(\zeta-\eta)e$; the images of $(\eta,\zeta)=(1,1)$ and $(0,1)$ are h_1 and h_2, which are linearly independent, so $\dim\mathcal{H}=2$. If $X(\tau)$ does not span a constant the system is solvable only for $\kappa=0$, which forces $\zeta=\eta$ and $b=0$,

so (A32) is the line $\mathbb{R}h_1$ and $\dim \mathcal{H} = 1$; and since every $h \in \mathcal{H}$ then has $b = 0$, $X(\tau)'\beta$ is constant on each equivalence class. □

COROLLARY C.1 (How many normalisations, and of what kind). *Let Assumption C.1 hold and fix parties* $p_0, p_1 \in \mathcal{P}$.

(i) If $X(\tau)$ *spans a constant, each of the restriction pairs*

$$\{\xi_{p_0} = 0,\ \delta_{p_1} = 0\} \qquad \textit{and} \qquad \{\xi_{p_0} = 0,\ \min_p \delta_p = 0\}$$

meets every coset of $\mathcal{H}$ *in exactly one point, and no single affine restriction does.*

(ii) If $X(\tau)$ *does not span a constant, each of the* single *restrictions* $\xi_{p_0} = 0$, $\delta_{p_1} = 0$ *and* $\min_p \delta_p = 0$ *meets every coset in exactly one point. Imposing any two of them at once is not a second normalisation but a restriction on* θ*, and it binds.*

PROOF. Let $h \in \mathcal{H}$, with (η, ζ) as in (A32). Then h shifts every ξ_p by η, every δ_p by ζ, and hence $\min_p \delta_p$ by ζ.

(i) h preserves $\xi_{p_0} = 0$ if and only if $\eta = 0$, and preserves either δ restriction if and only if $\zeta = 0$; together these force $h = 0$, so each coset meets the restriction set at most once. It meets it at least once because, given θ, the element with $\eta = -\xi_{p_0}$ and $\zeta = -\delta_{p_1}$ (respectively $\zeta = -\min_p \delta_p$) carries θ into the set. A single affine restriction cannot suffice because its solution set is a hyperplane in Θ and $\mathcal{H}$ is two-dimensional, so the intersection of $\mathcal{H}$ with the hyperplane's direction space is at least one-dimensional and some $h \neq 0$ preserves it. Nothing is claimed about non-affine restrictions and the argument does not reach them.

(ii) In case (ii) of Proposition C.1, $\mathcal{H} = \mathbb{R}h_1$ and every $h \in \mathcal{H}$ has $\eta = \zeta$, so each of the three displayed scalar conditions is violated by every $h \neq 0$ and is reached from any θ by the element with $\eta = \zeta$ equal to $-\xi_{p_0}$, $-\delta_{p_1}$ or $-\min_p \delta_p$ respectively: one point of each coset, exactly. Two of them together cannot be a normalisation, because the coset of θ meets $\{\xi_{p_0} = 0\} \cap \{\delta_{p_1} = 0\}$ only if $\xi_{p_0} - \delta_{p_1} = 0$ at θ — a condition on the parameter that Assumption C.1 does not supply and that a one-dimensional $\mathcal{H}$ cannot absorb. □

COROLLARY C.2 (What is and is not touched by the convention). *Let Assumption C.1 hold. Along every* $h \in \mathcal{H}$*, with* (η, ζ) *as in* (A32)*:*

(a) every $\alpha(\tau)$ *is unchanged, so no normalisation of* ξ *or* δ *moves responsiveness, at any type, in either level or difference;*

(b) every $\xi_p - \xi_{p'}$ *and every* $\delta_p - \delta_{p'}$ *is unchanged;*

(c) $X(\tau)'\beta$ *changes by* $\zeta - \eta$*, the same scalar for every* τ*. If* $X(\tau)$ *spans a constant, a linear functional* $c'\beta$ *is unchanged if and only if* $c'e = 0$*; in particular if* $X(\tau)$ *carries an explicit constant then every slope of* β *is unchanged and only its intercept moves, one for one with the* δ *convention at a fixed* ξ *convention. If* $X(\tau)$ *does* not *span a constant, then* $\zeta = \eta$ *and* $b = 0$*: every component of* β *is unchanged,* $X(\tau)'\beta$ *is unchanged, and the only object any admissible convention moves is the common level of* ξ *and* δ*.*

PROOF. (a) and (b) are immediate from (A32): $a = 0$, and x and d are constant vectors, so all their within-block differences vanish. For (c), $X(\tau)'b = \zeta - \eta$ by (A32). If $X(\tau)$ spans a constant then $b = (\zeta - \eta)e$ by Step 5 of

Proposition C.1, so $c'b = (\zeta - \eta)\, c'e$; and if $X(\tau) = (1, \tilde{X}(\tau)')'$ then $\mathcal{X}e = \iota_K$ and (D4) give $e = (1, 0, \ldots, 0)'$, so $c'e = c_1$. If $X(\tau)$ does not span a constant then $\mathcal{H} = \mathbb{R}h_1$ by Proposition C.1(ii), along which $\eta = \zeta$ and $b = 0$. □

REMARK C.1 (Which case the estimated design is in). *The type partition is the $K = 32$ cell factorial of gender, higher education, age band and incumbency, but the covariate vector entered against β is not the exhaustive set of those levels. It is $X(\tau) = (\textit{female},\ \textit{higher ed},\ \textit{age z},\ \textit{incumbent})'$ with $n_X = 4$: three binary indicators and age as a single score standardised across types, and no intercept column. That design does not span a constant. Suppose some e had $X(\tau)'e = 1$ at every τ. Two types agreeing on gender, education and incumbency and differing in age band have different standardised ages, the bands being disjoint age intervals; subtracting their equations gives $e_{age} = 0$. Subtracting at two types differing only in gender, only in education, and only in incumbency gives $e_{female} = e_{higher\ ed} = e_{incumbent} = 0$ in turn. Then $X(\tau)'e = 0 \neq 1$. So Proposition C.1(ii) applies:* $\dim \mathcal{H} = 1$, *a basis is h_1, one convention fixes the location, every component of β is invariant to it by Corollary C.2(c), and the only object the convention moves is the common level of ξ and δ. That* $\dim \mathcal{H} = 1$ *holds on the non-incumbent branch too, where the incumbency indicator is identically zero and the design has three live columns rather than four — and (D4) holds there only because that dead column is dropped, since retaining it would leave* $\operatorname{rank}(\mathcal{X}) = 3$ *against $n_X = 4$ and Proposition C.1 would not apply as stated. Case (ii) would* fail, *and the count would be two, on a branch restricted to incumbents: there the incumbency column is identically one across types, ι_K is in the column space, and h_2 reappears.*

C.3. Why the within-branch reason is false, and what replaces it

One might give as the reason for the count at Section 2.3 that only the differences $\xi_p - \xi_{p'}$ and $\delta_p - \delta_{p'}$ appear. That reason would make the invariance space

$$\mathcal{H}^{\mathrm{wb}} \;=\; \big\{\, (0,\ 0,\ \eta\, \iota_{n_\mathcal{P}},\ \zeta\, \iota_{n_\mathcal{P}}) : \eta, \zeta \in \mathbb{R} \,\big\},$$

the two levels moving freely and independently of β. That space is two-dimensional whatever the type design is, whereas $\mathcal{H}$ is two-dimensional only when $X(\tau)$ spans a constant. On the design estimated here it does not (Remark C.1), so $\mathcal{H} = \mathbb{R}h_1$ and the within-branch reason overstated the count by one as well as naming the wrong directions. In general neither space contains the other and they meet in $\mathbb{R}h_1$; in case (ii) that intersection is all of $\mathcal{H}$, so $\mathcal{H} \subsetneq \mathcal{H}^{\mathrm{wb}}$. Equation (12) is the witness: it carries $\xi_{m,j} - \delta_{m,k}$ across the two branches, so the element $(0, 0, 0, \iota_{n_\mathcal{P}}) \in \mathcal{H}^{\mathrm{wb}}$ shifts the mixed log-odds by -1 and is not an invariance, which separates the two spaces in either case; and where $X(\tau)$ does span a constant the element $h_2 \in \mathcal{H}$ moves the intercept of β, which $\mathcal{H}^{\mathrm{wb}}$ holds fixed.

The statement that replaces the reason is (A32) read as an arithmetic: within each branch the levels are free, and across the branches they are not, so a level shift of δ is an invariance only when it is paid for by an equal shift of $X(\tau)'\beta$ — which requires that $X(\tau)$ span a constant, and is impossible otherwise. The number of conventions is then the number of free parameters left in that arithmetic: two where the constant is available, the ξ level and the δ level moving independently with β paying the difference, and one where it is not, the two levels being locked together so that only their common value is free. On the design estimated here the constant is not available, so there is one location to fix, and the price of fixing it is nothing in β: Corollary C.2(c) says every component of

β is invariant and only the common level of ξ and δ moves.

Corollary 1 asserts identification of $\{\alpha(\tau)\}$, of the ξ and δ differences, and of $X(\tau)'\beta$ up to one additive constant. That last clause is what the three displayed contrasts give when each is read at a single type: (12) delivers $X(\tau)'\beta + (\xi_{p_0} - \delta_{p_0})$ and does not separate the two pieces within a type. Read *across* types they separate exactly when $(b, s) \mapsto \mathcal{X}b + s\,\iota_K$ is injective, which is (D4) together with $\iota_K \notin \mathrm{col}(\mathcal{X})$ — the same two conditions that give $\dim \mathcal{H} = 1$. So in case (ii) of Proposition C.1 both $X(\tau)'\beta$ and $\xi_{p_0} - \delta_{p_0}$ are identified outright and the corollary's "up to one additive constant" is conservative rather than wrong. Separating β from the constant across types is also why a design with all $n_{\mathcal{P}}$ of the ξ columns free and one δ column dropped is exactly identified. In case (i) the clause is exact, the constant is $\zeta - \eta$ and is the image of h_2, and h_1 moves only the two levels, which the corollary never claims. The corollary is complete as a statement about the objects it names and incomplete only as a count of conventions, which is Corollary C.1.

Proof of Corollary 1. For the level of $\alpha(\tau)$, take the party pair and the two markets that (i) supplies for τ and read (10) in each. Under (ii), with single-label lists, the term $\xi_p - \xi_{p'}$ is the same in the two markets while the prize gap differs, so the two equations determine $\alpha(\tau)$ and $\xi_p - \xi_{p'}$. Given $\alpha(\tau)$, (10) returns $\xi_p - \xi_{p'}$ for every pair of parties that clear in a common market, and connectedness through clearing appearances extends that to all pairs. Equation (11) carries no prize term and gives $\delta_p - \delta_{p'}$ for parties sharing a market on the non-clearing branch, which the non-clearing half of (iii) extends to all pairs. By (iv) every type has a mixed contrast (12), whose right-hand side is then $X(\tau)'\beta$ plus one unknown constant common to all types, the level of ξ relative to δ. Proposition C.1 shows that under (i)–(v) nothing else is free: the constant is separated from $X(\tau)'\beta$ exactly when $X(\tau)$ does not span a constant, and (v) then gives β. □

remark C.2 (Without condition (ii) the question is not a normalisation). *Proposition C.1 is stated at $\nu_{m,j} = \nu'_{m,j} = 0$. With the local components free, each market contributes its own free direction and (A33) no longer restricts $a(\tau)$, so the level of α is not identified at all; what survives is Proposition 3, which cancels ν along with ξ. That failure is the subject of Section 3.5 and is not repaired by any convention.*

Appendix D. The quotient discontinuity across the coalition ban

Section 6.7 runs the quotient discontinuity on 2012–2016, the pair with proportional coalitions on both sides. This appendix reports the 2016–2020 pair, which straddles the ban. The design is the one described there, unchanged: running variable votes/quotient − 1 in 2016, local linear with separate slopes, uniform kernel, rectangular window, standard errors clustered by municipality, n = 1,372/2,678/5,212 at the three bandwidths over 31,979 party-by-municipality units. The first stage is +0.476 (0.031), +0.459 (0.022) and +0.482 (0.016).

2020 outcome	$h = 0.10$	$h = 0.20$	$h = 0.40$
candidate share	+0.038 (0.041)	−0.007 (0.030)	+0.027 (0.022)
candidate count	+1.254 (0.998)	−0.041 (0.642)	+0.278 (0.442)
wins a seat	+0.031 (0.054)	+0.025 (0.039)	+0.048 (0.028)
placebo: 2016 candidate share	−0.024 (0.050)	−0.057 (0.036)	−0.032 (0.025)

Nothing is significant on any of the twelve cells; the largest $|t|$ is 1.72. The reason to report it is not the

2016–2020 null but an asymmetry with the 2012–2016 arm. The seat persistence that survives on 2012–2016, at $t = 3.03$ in the narrowest window, does not appear here. That is what the coalition ban predicts if the persistence runs through the list rather than through the party: the object that cleared the quotient in 2016 was a coalition, and in 2020 it does not exist. The placebo here, as in Section 6.7, is the same-cycle candidate share — fixed at registration, months before the vote that determines clearing, and therefore pre-treatment with respect to it, but not an outcome from the cycle before the running variable. No 2008 file exists in our data, so neither arm carries a lagged-cycle placebo and neither should be read as claiming one.

Appendix E. A parametric example

This appendix works through one market small enough to write down. It displays three things the argument of Sections 3 and 4 relies on: two lists that clear with *different* prizes, so that the clearing contrast has variation; one that does not, so that the mixed contrast exists; and more than one self-fulfilling clearing-set configuration, so that Section A.5 is a fact about the model rather than a possibility.

Primitives. A council of nine, so $q = 1/9$; three lists and two types, with the payoff (4) throughout.

$$B = (0.30,\ 0.18,\ 0.05), \qquad \rho = (1.40,\ 1.00,\ 0.35),$$
$$\xi = (0,\ -0.20,\ 0), \qquad \delta = (0,\ -1.20,\ -1.90),$$
$$\mu = (0.55,\ 0.45), \qquad X(\tau)'\beta = (0.40,\ -0.30),$$
$$\alpha(\tau) = (1.20,\ 0.60).$$

List 1 is the strong one on every margin: the largest anticipated prize B_j, the best conversion of candidates into votes, and no penalty for standing on it if it fails. List 3 is the weak one on all three.

Where these numbers come from. They are chosen, not estimated. The example has to show three things at once in a market small enough to print, and no observed market provides a comparably transparent illustration of all three at once. Since that is a licence, it is worth spending against it openly.

	example	vereador panel
council seats	9	median 9
lists	3	median 5, tenth percentile 2
prize index B_j/s_j, clearing lists	0.550, 0.445	median 0.125, ninetieth 0.278
its within-market spread	0.105	median 0.092
$\alpha(\tau)$	1.20, 0.60	median 4.77, range 0.13 to 8.27
δ_j	0, −1.20, −1.90	median +0.03, lowest −3.70
vote share, non-clearing list	0.015	median 0.073
self-fulfilling configurations	2 of 8	1.20 at three lists

All panel entries use the party-by-state specification estimated on markets of sixty candidates or more (Section 6.4, Figure 1), so the $\alpha(\tau)$ and δ_j rows refer to the same fit. The α statistics are taken across the

thirty-two demographic types, and the δ statistics across the estimated party-by-state non-clearing effects.

Two departures are deliberate and the third follows from them. The two clearing prize indices are roughly four times the panel median, which is what makes list 3's collapse legible on the page; the *difference* between the two clearing prizes, 0.105, is of the same size as the median within-market spread of 0.092, and it is the difference and not the level that Proposition 2 uses. The non-clearing values δ_j are deliberately dispersed, with list 3 placed in the lower part of the estimated δ range, which is what drives candidates off it and makes its collapse visible on the page. Multiplicity is then the consequence. A three-list market in the data averages 1.20 self-fulfilling configurations, so two is above average and was arranged to be, because a market with one would illustrate nothing.

The panel figures come from Table 5 and from the estimates at which Section 6.8 solves the model. What the example establishes is algebra rather than magnitude: that the clearing–clearing and mixed contrasts of Section 3 hold exactly at a computed equilibrium, and that the multiplicity of Section A.5 is realisable and not merely admissible.

A self-fulfilling configuration. This example has a mean-field equilibrium. For the proposed clearing set $A = \{1, 2\}$, the fixed point of the truncated operator validates the regime. Solving Φ^A to a residual of 9×10^{-15} gives:

list	s_j	vote share	clears	prize index B_j/s_j
1	0.5453	0.6439	yes	0.5502
2	0.4045	0.3412	yes	0.4450
3	0.0502	0.0148	no	—

$\sigma(j \mid \tau)$	1	2	3
type 1	0.5639	0.4069	0.0292
type 2	0.5225	0.4016	0.0758

Two features are worth pausing on. The prizes of the two clearing lists differ, 0.5502 against 0.4450, even though list 1 has the larger B: it also attracts more candidates, and the index falls in the list's share. That is the equilibrium feedback the model is about, and it is why ΔP is not simply ΔB. And list 3 collapses. Its vote share is 0.0148 against a quotient of 0.1111, far below the $0.35/(1.40 + 1.00 + 0.35) \approx 0.13$ it would hold if candidates split evenly. In the $\{1, 2\}$ regime candidates expect list 3 not to clear and therefore receive its low non-clearing value $\delta_3 = -1.90$. That expectation drives candidates away, lowers the list's vote share, and validates the non-clearing branch.

The contrasts available in this market. Section 3 gives three possible log-odds contrasts. This market displays the clearing–clearing and mixed contrasts; the non-clearing contrast requires two non-clearing lists and is supplied by other markets in the panel.

Both clear. $\log[\sigma(1 \mid \tau)/\sigma(2 \mid \tau)] = \alpha(\tau)(P_1 - P_2) + (\xi_1 - \xi_2)$, with $P_1 - P_2 = 0.1052$:

$$\text{type 1: } +0.326270 = 1.20 \times 0.1052 + 0.20, \qquad \text{type 2: } +0.263135 = 0.60 \times 0.1052 + 0.20.$$

The prize difference is common to the two types and α is not, which is what separates them. Differencing across types kills $\xi_1 - \xi_2$ and leaves

$$\alpha(1) - \alpha(2) = \frac{0.326270 - 0.263135}{0.1052} = 0.6000,$$

against a true difference of 0.60. This is Proposition 3 in three numbers, and it is worth being clear about which proposition it is. Proposition 2 says the single-type log-odds equal $\alpha(\tau)\Delta P + (\xi_1 - \xi_2)$; the arithmetic above shows exactly that, with $\xi_1 - \xi_2 = 0.20$ sitting in each line. The double difference across types removes it and returns a difference in α. *No manipulation of this market recovers the level of α without an assumption about ξ*, which is the division Corollary 1 draws and which this example makes concrete.

One clears. $\log[\sigma(1 \mid \tau)/\sigma(3 \mid \tau)] = X(\tau)'\beta + \alpha(\tau)P_1 + \xi_1 - \delta_3$, which evaluates to +2.960219 for type 1 and +1.930110 for type 2, exactly. This contrast is the one that carries $X(\tau)'\beta$, as in Proposition 2, and it does so without an outside option anywhere in the system.

Neither clears. There is only one non-clearing list here, so this contrast is empty in this market. This illustrates why identification uses the panel: relative non-clearing-list values come from markets with two or more non-clearing lists. Pooling the 11,129 markets, Section 6.4 estimates 809 party-by-state non-clearing effects.

Multiplicity, in the same market. Fixing the regime and solving Φ^A for each of the eight subsets:

A	s^A			vote shares			self-fulfilling
$\emptyset$	0.6893	0.2076	0.1031	0.7984	0.1718	0.0299	no
$\{3\}$	0.3992	0.1202	0.4806	0.6596	0.1419	0.1985	no
$\{2\}$	0.4203	0.5169	0.0629	0.5220	0.4585	0.0195	no
$\{2,3\}$	0.2833	0.3791	0.3376	0.4437	0.4241	0.1322	no
$\{1\}$	0.7664	0.1561	0.0775	0.8542	0.1242	0.0216	no
$\{1,3\}$	0.5394	0.0967	0.3639	0.7712	0.0987	0.1301	**yes**
$\{1,2\}$	0.5453	0.4045	0.0502	0.6439	0.3412	0.0148	**yes**
$\{1,2,3\}$	0.4256	0.3100	0.2644	0.5968	0.3105	0.0927	no

Two proposed clearing sets are self-fulfilling, and they disagree about whether list 2 or list 3 clears. List 3 is weaker than list 2 on the prize, on conversion and on the penalty for failing, and it clears in $\{1,3\}$ anyway — because candidates expect it to, join it, and thereby supply the votes. The primitives are the same across the two configurations. What differs is the self-confirming expectation of which list clears, which changes candidate shares and hence vote shares through the list-specific vote productivities.

Each of the eight proposed sets returns the fixed point shown in the table from 2,000 random starting points. That is a numerical diagnostic and not a uniqueness proof, so the two configurations are a lower bound on the number of mean-field equilibria of this market.

The failures are as informative. $\{1,2,3\}$ is not self-fulfilling: with all three paying a prize, list 3 draws 0.2644 of the candidates but only 0.0927 of the vote, short of the quotient, so the regime that assumes it clears contradicts itself. And $\{2,3\}$ fails from the other side, list 1 sitting at 0.4437 of the vote while the regime denies it a prize.

This is the coordination problem Section 6.7 measures on the data: the empirical enumeration finds an average of one self-fulfilling clearing-set configuration in two-list markets and 6.4 in eight-list markets. Here the search finds two such configurations among the eight proposed sets, in a market small enough to inspect directly.

Appendix F. Additional material for the application

This appendix collects supporting analyses for Section 6. Each subsection states the main-text result it supports.

F.1. The mayoral terms in their own units

Section 6.5 adds two mayoral terms to the baseline likelihood, first an indicator that the list carries a mayoral candidate and then, beside it, an indicator that the candidate won, and reports that $\hat{\alpha}$ barely moves. This subsection reports the terms themselves.

The mayoral terms are not negligible. Setting them against $\hat{\alpha}$ itself would compare a utility shift with a coefficient, so they are read in their own units. Since the specification lets the terms differ across branches, both halves are reported:

	clearing	non-clearing
+ *carries a mayoral candidate*		
carries a mayoral candidate	+0.0334 (0.0029)	−0.0571 (0.0091)
+ *and that mayoral candidate won*		
carries a mayoral candidate	+0.0178 (0.0033)	−0.0579 (0.0091)
and that candidate won	+0.0405 (0.0038)	+0.0883 (0.0522)

The magnitudes have to be read off the specification with the winner term, which nests the one without it. The terms enter the index additively, so on a clearing list a losing mayoral candidacy is worth +0.018 and a winning one +0.058. The +0.033 of the specification without the winner term is neither: it is the blend of the two, right only where the winner interaction is absent.

The comparison is best made in the prize's own units. A winning mayoral candidacy buys the same utility as $\Delta P = 0.031$; a mayoral candidacy of any kind buys $\Delta P = 0.009$. Against the median within-cell range of the prize, 0.164 — which the winner-term specification's $\hat{\alpha}$ turns into 0.308 of utility across alternatives — the winning candidacy is about a fifth of what the prize moves and a candidacy of any kind about a seventeenth. The fraction is not robust to how "spread" is defined: against the median within-cell interquartile range, 0.0764, the winning candidacy is about two-fifths rather than a fifth. That is why the prize's own units come first.

The branch contrast is where the evidence on the mayoral term sits. A mayoral candidacy attracts on a list that clears and repels on one that does not, +0.018 against −0.058, a gap several times either coefficient and far outside the standard errors. That is what one expects if the mayoral term is about a prize that is real rather than about a taste for the party.

The winner term runs the other way, +0.088 (0.052) on non-clearing lists against +0.041 (0.004) on clearing ones, and it needs its support printed beside it: 138 lists, 0.3% of the 44,219 alternatives, because a list whose mayoral candidate won clears 98.3% of the time. It is estimated on the residue and is not comparable in precision to the other three. It also attaches to a realised outcome of the same election — the object Section 4.2 spends a page refusing to build the prize from. The unconditional mayoral term, +0.018, does not.

F.2. The first stage of the council-size design

The instrument of Section 6.5 is the step in the art. 29 ceiling on the council from nine seats to eleven at fifteen thousand inhabitants. This subsection reports how large that step is in the data.

We tried to size the step on IBGE population, which is what the law assigns on, and the attempt failed in a way worth reporting. A comparison of means across a window looks decisive: the share seating eleven or more runs from 4.9% just below fifteen thousand to 48.2% just above, in a window of two thousand either side. But a difference of means across a window is not a discontinuity, because council size rises with population throughout. Fitting a local linear regression on each side with a triangular kernel, the jump in the probability of seating eleven or more is

bandwidth	8,000	5,000	3,000	2,000	1,500	1,000
jump in $\Pr(C_m \geq 11)$	0.319	0.229	0.159	0.123	0.088	0.044
(s.e.)	(0.028)	(0.035)	(0.046)	(0.058)	(0.067)	(0.083)

It falls toward zero as the window narrows, and at the narrowest it cannot be told from nothing. A real discontinuity does the opposite: the estimate settles as the bandwidth shrinks and the bias from the trend disappears. What this pattern says is that most of the 48.2% was slope, and that we cannot demonstrate a local jump at fifteen thousand at all.

We do not think this shows the law is not followed. The likelier explanation is that the running variable is wrong: the population files we hold are for 2014, 2018 and 2022, while a council's size is fixed from the IBGE estimate current when the *Lei Orgânica* last set it. Error in an RD's running variable blurs the assignment and drives the estimated jump toward zero, which is exactly the pattern above. A density test in the McCrary spirit is correspondingly uninformative — a jump in the log density of +0.26 (0.11) at a window of two thousand, falling to +0.04 (0.13) at five thousand.

So the honest position is this. That the statute is a ceiling and that a quarter of the eleven-seat band sits below it are facts about the law and the counts, and they stand. That the council-size design has a strong first stage at the threshold is something we assert on the vote proxy below and cannot confirm on population with the files we hold. Obtaining the population vintage each *Lei Orgânica* actually referenced is what the design needs, and until then the results that lean on it — the instrument of Section 6.5 and the counterfactual check of Section 6.9 — should be read as suggestive.

The table below uses the municipal vote total as the running variable — at Brazilian turnout fifteen thousand inhabitants is close to nine thousand votes — because the vote total is the variable the estimation frame is built on and is measured in the same files as everything else. The population version above is the cleaner assignment and gives the same picture; a full design on population, with a data-driven bandwidth and the density test of Cattaneo, Jansson, and Ma (2020), is the version this section should eventually carry. On the vote proxy, the share of markets with eleven seats or more runs

municipal votes	< 7,000	7,000–8,500	8,500–9,250	9,250–10,000	10,000–11,500	11,500–14,000
markets	5,588	808	338	324	524	742
mean council	9.00	9.21	9.50	9.86	10.48	10.64
share with 11+	0.3%	10.5%	25.4%	42.9%	73.5%	80.2%

The markets with nine seats have a ninetieth percentile of 9,174 votes and those with eleven a tenth percentile of 9,295: the two are nearly disjoint.

F.3. Limits of the council-size design

Is the threshold manipulated? A municipality that crossed fifteen thousand inhabitants would gain two council seats, so the population distribution is worth inspecting. We ran two tests and neither settles the question, for the same reason the first stage of Section F.2 does not.

The first is in the spirit of McCrary (2008) rather than the procedure itself: the population distribution slopes down everywhere, so a raw below-against-above comparison is negative at any point, and we ask instead whether the drop at a statutory threshold exceeds the drop at nearby placebo cutoffs. At 15,000, 30,000 and 50,000 inhabitants the standardised excess drops are −0.06, −0.15 and +1.43 for 2016 and +0.29, +0.62 and +1.16 for 2020. Two of the six are unremarkable, three are mildly positive, and the largest sits at 50,000 — which is the one threshold of the three we would not want to read, since art. 29, VI caps a councillor's *subsídio* as a share of the state deputy's at a breakpoint there as well, so any bunching would have two motives. (Art. 29-A, which caps the legislature's total spending, steps at a hundred thousand and above and does not bear on these windows.)

The second fits a local linear regression to the binned log density on each side of 15,000. The jump is +0.26 (0.11) in a window of two thousand, +0.16 (0.11) at three thousand and +0.04 (0.13) at five thousand: a suggestion of excess mass just above the cutoff that does not survive widening the window.

We draw no conclusion from either. Both are density tests on a population series whose vintage does not match the one the *Lei Orgânica* referenced, and a running variable measured with error will move mass across a cutoff on its own. A version of this test worth reporting needs the right vintage and the estimator of Cattaneo, Jansson, and Ma (2020), and we have neither.

Two things this does not do. *The narrow-window version is underpowered, which is the trade-off every bandwidth choice in this literature makes (Calonico, Cattaneo, and Titiunik 2014; Cattaneo, Idrobo, and Titiunik 2020).* As the window tightens the standard error on α, with γ, on all candidates runs 0.1379, 0.3003, 0.5080, withheld, 0.8677. By the narrowest window with a usable fit the estimate is 1.0394 against a standard error of 0.8677 and rejects nothing — including the 0.5432 of the 60+ sample, which it sits well above. Nor is the estimate a local average treatment effect at the cutoff in the sense of Lee (2008): the identifying variation is the interaction of C_m with $z_{m,j}$ across markets, and the window only makes C_m plausibly unrelated to what else differs between towns.

Identification is not confined to the threshold. In the full sample and the wide windows some of the variation in C_m is large councils in large municipalities rather than the nine-to-eleven step; councils run to fifty-five seats. The council-size design cannot bound α away from zero on the non-incumbent object; Section 6.5 reports what it does and does not deliver.

F.4. Three candidate selection rules

Section 6.7 enumerates the self-fulfilling regimes of each market and imposes no selection rule. This subsection reports the three candidate rules it summarises: stability, history and an extremal rule.

Stability. A natural refinement discards equilibria to which no adjustment process converges. We examine it on the ten equilibria of the numerical illustration in Section A.5, a four-list market, and not on the estimation sample. At those ten equilibria the derivative of the best-response operator on the tangent space has spectral radius between 1.15 and 2.06, so every one of them fails the undamped best-response test.

Every dominant eigenvalue is real and negative, however, so the instability reflects overshooting: the prize B_j/s_j falls in the share, reaction functions slope steeply down, and undamped tâtonnement oscillates past the fixed point. Under the damped dynamic $s \leftarrow (1-\psi)s + \psi\Phi(s)$ a real eigenvalue λ of the undamped map becomes $1 - \psi + \psi\lambda$. For $\lambda < 1$ that has modulus below one once ψ is small enough; for $\lambda > 1$ it exceeds one at every $\psi > 0$, and convex damping repairs nothing. Each of the ten equilibria is accordingly locally attracting once ψ falls below a threshold between 0.65 and 0.93, and $\psi \leq 0.65$ serves all ten at once. This is the Cournot phenomenon, and Cournot equilibrium is not discarded because tâtonnement cycles when reaction functions are steep.

The undamped criterion is therefore informative about the adjustment rule, and not by itself about whether an equilibrium can be reached under a slower dynamic. The exercise is an illustration: it shows that undamped instability need not eliminate any equilibrium, because the instability can arise entirely from overshooting. For the equilibria examined, stability under damped best-response dynamics does not refine the equilibrium correspondence.

History. The next candidate is coordination on the previous cycle: candidates sort toward the lists that cleared last time. A plain persistence regression cannot test this, since lagged success predicts current success through party strength and organisation, and conditioning on current votes removes the outcome variation because clearing is a deterministic function of votes and the quotient.

The quotient supplies a discontinuity instead. A list clears if and only if its votes reach the quotient, so lists just above and just below have nearly identical fundamentals and opposite regime membership. Taking the running variable to be votes/quotient − 1 in the earlier cycle, the first stage is strong: clearing raises the probability of taking a seat that cycle by about 0.5, short of one only because a list below the cutoff can still take a leftover seat by highest average.

We run this on 2012–2016, which has coalitions on both sides. The 2016–2020 pair straddles the abolition of proportional coalitions, so the object that cleared in 2016 does not exist in 2020, and a null there would be uninformative; Section D reports it. The placebo below is the 2012 candidate share: registrations close in August and the vote is in October, so that share is fixed before the vote that determines clearing and cannot respond to it. It is a same-cycle pre-vote placebo rather than a lagged-cycle one, since no 2008 file exists in our data.

2016 outcome	$h = 0.10$	$h = 0.20$	$h = 0.40$
candidate share	+0.027 (0.042)	+0.006 (0.031)	+0.017 (0.021)
candidate count	+0.690 (0.710)	−0.125 (0.536)	+0.080 (0.396)
wins a seat	+0.135 (0.045)	+0.068 (0.032)	+0.034 (0.022)
placebo: 2012 candidate share	+0.055 (0.042)	+0.026 (0.029)	+0.023 (0.021)

We detect no discontinuity in candidate entry, on the candidate share or the candidate count, at any bandwidth. Entry is the margin the coordination account requires: if clearing were a coordination device, it would attract

candidates in the following cycle. The estimates are imprecise enough that they do not rule out every substantively relevant effect, but they give the history-based account no support.

Winning a seat does persist, and we do not rely on it. The seat estimate is sensitive to the bandwidth, declining from 0.135 at $h = 0.10$ to 0.034 at $h = 0.40$, and the placebo, an outcome fixed months before any vote was cast, has the same sign throughout: it is two fifths of the seat estimate in the narrowest window and two thirds of the seat estimate in the widest. A discontinuity of that size in a predetermined outcome makes the seat pattern difficult to read as persistence caused by having cleared the quotient. It is more readily read as organisational strength working through votes.

An extremal rule. The remaining possibility is that the realised regime is extremal under an ordering the analyst can state. The natural ordering is the total candidate share that a regime's members hold at that regime's fixed point; no market has a tie for the largest. Among the self-fulfilling regimes of each market, the observed regime is the largest in 73.5% of markets. That figure is inflated by markets with a single self-fulfilling regime, where the rule has nothing to choose between. Among the 4,106 markets with more than one self-fulfilling regime the observed regime is the largest in 52.4%, and among the 512 such markets with eight lists in 34.8%.

The benchmark is a uniform random draw from the self-fulfilling regimes of each market, scored the same way and computed market by market. Such a draw would select the observed regime in 29.8% of the 4,106 markets, and in 14.4% of the 512 with eight lists. The extremal rule raises those rates to 52.4% and 34.8%, but it remains too inaccurate to serve as a selection rule, and we do not adopt it.[6]

F.5. Does the binding registration cap censor the estimating sample?

This subsection supports Section 6.6. The question should be asked, because the two halves of this paper appear to disagree. If the cap binds, a list at the ceiling holds the number of candidates the law allowed and not the number free choice would produce, so $\hat{\sigma}_m(j \mid \tau)$ for that list is censored rather than a choice probability — and Section 6.4 treats it as one.

The censoring is neither small nor random. Counting a list as at the ceiling when its ratio reaches the multiple the statute gives it — 200% for a coalition in a municipality of a hundred thousand voters or fewer, 150% otherwise — 22.5% of 2016 lists and 12.0% of 2020 lists qualify; in the estimating sample of markets with sixty candidates or more that is 18.2% of lists and at least one in 68.4% of markets. Applying the 200% multiple to every 2016 list, solo lists included, gives the narrower 17.9%, 15.7% and 63.8%, and that is the definition the robustness row below uses. And a list at the ceiling is a large list, so its prize is small: the anticipated prize averages 0.107 at the ceiling against 0.166 elsewhere. The censored observations sit where the regressor is low.

It bites little.

[6] The benchmark is low partly because the observed regime is not always self-fulfilling: it is in 53.3% of the 512 eight-list markets. The chance that a uniform draw lands on any one given self-fulfilling regime, the mean of $1/K_m$ with K_m the number of self-fulfilling regimes in market m, is 36.7% over the 4,106 markets and 25.2% over the 512.

sample	markets	α	(s.e.)
all markets of 60+	5,089	1.9210	0.0739
drop capped lists and their candidates	5,078	**1.8335**	0.0772
drop markets containing a capped list	1,838	1.8663	0.1155
keep only those markets	3,251	2.0477	0.0894

The second row is the test. Removing every list at the ceiling, and every candidate who stood on one, takes 8.0% of the cells out of the sample and lowers the estimate by 0.088 — about a twentieth of it, and a little more than one standard error of either row. The two rows share most of their observations, so that is not a test of their difference; it bounds what the cap's censoring can be doing to the comparison across lists within a cell, which is what identifies α.

The last two rows are a different comparison and should not be read as evidence about censoring. Markets that contain a capped list are larger and more contested than markets that do not, so splitting on that is splitting on market type; the gap between 1.87 and 2.05 is heterogeneity across markets of different kinds, and the first two rows hold the markets fixed.

None of this speaks to the other half of the entry problem. The cap constrains *how many* a list may register; parties also choose *whom*, and that we do not test.

F.6. An ex ante vote productivity

This subsection supports Section 6.8. The winner rule (3) needs ρ, the votes a list draws per candidate, and the fixed-point exercises of that subsection set it to what the election actually delivered. Evaluated at the observed shares that is an identity — $\rho_j s_j$ is then the list's vote share — so the near-perfect agreement it produces there is definitional. In the fixed point it is not an identity, since the shares are solved from $\hat{\theta}$, but it still hands the model the vote conversion that occurred.

The test that is not rigged replaces ρ with one that exists in August. For a list fielding parties $P(j)$ we sum those parties' shares of the municipal vote last cycle and divide by the candidates the list fields, recovering coalition membership from the candidate-level party code so that a 2016 alliance carries its members' lags rather than the lead party's alone. The result is a failure, and we report it as one.

	at the observed shares		fixed point	
W_m from	lists	whole regime	clearing set	1 − SSE/SST
candidate share $\geq q$	0.828	0.527	—	—
vote share, ρ realised	0.999	0.996	0.910	+0.334
vote share, ρ lagged	0.686	0.247	0.650	−0.186
everything clears	0.655	—	0.655	—

An ex ante ρ predicts the clearing set no better than the constant rule that says every list clears, and the shares it solves out are worse than uniform. Restricting to the two markets in five in which *every* list carries a lagged vote does not repair it: agreement of 0.716 against a base rate of 0.777, and 1 − SSE/SST of −0.269. The reason

is visible upstream, where lagged and realised ρ correlate at 0.26. A party's municipal vote four years earlier is a poor forecast of what its candidates will personally draw, which is the quantity an open list makes decisive, and coalitions re-form from one cycle to the next.

Two things follow and only the first is comfortable. The estimates of Section 6.4 are untouched, because the likelihood conditions on the published clearing set and never evaluates (3): no ρ enters them. What is touched is the reach of the fixed point. The 0.910 agreement between the solved clearing set and the published one, the multiplicity counts of Section 6.7 and the counterfactual of Section 6.9 are all conditional on ρ, and are therefore statements about what the model does *given* the vote conversion rather than about what could have been predicted before any vote was cast. We know of no way to close that gap with these data.

The same gap appears from a second direction, which is why we think it is a property of the exercise rather than of this particular measurement. Section A.4 replaces the sharp winner rule with one that pays a list with a probability rising through the quotient, and calibrates the bandwidth to what a candidate could actually forecast in August. The calibrated value is about as large as the quotient itself, most lists are then genuinely uncertain of clearing, and the model's fit *falls*: the agreement with the published clearing set drops from 0.899 at a near-sharp rule to 0.878 at the calibrated one. Both routes say the same thing. The model agrees with the data best when it is given information the candidates did not have.

F.7. The prize by type

This subsection supports the by-type estimates of Section 6.4. The prize is not the same quantity for every type on a list, and that limits how the spread of $\hat{\alpha}(\tau)$ can be read. The model hands every type on list j the same $B_{m,j}/s_{m,j}$. The institution does not. On a clearing list an incumbent takes 0.596 seats per candidate and a non-incumbent 0.108, while incumbents are an eighth of the candidates and take 41.6% of the seats. Write κ_τ for the ratio of a type's seat rate to its list's average: it runs from 0.17 to 4.43 across the thirty-two types, and the list average captures only 23.2% of the variance in cell-level seat rates.

If candidates respond to their own chance, the regressor should be $\kappa_\tau B_{m,j}/s_{m,j}$ and the estimator would return $\hat{\alpha}(\tau) = \alpha\kappa_\tau$. This diagnostic runs on the whole panel with party effects at the party rather than the party-by-state level — a second fit, whose $\hat{\alpha}(\tau)$ has median 1.97 and runs from −0.45 to 4.03 (the second fit's own median and range, both at the vintage this section carried before its most recent revision) rather than the 4.66 and −1.02 to 9.00 of Figure 1. This is because κ_τ is a seat rate and needs the markets the headline sample drops. Across the thirty-two types $\hat{\alpha}(\tau)$ and κ_τ correlate at +0.845, which looks like that story confirmed. It is not, for two reasons. Dividing $\hat{\alpha}(\tau)$ by κ_τ makes the estimates *more* dispersed rather than less, the coefficient of variation rising from 0.728 to 1.254; and the correlation lives entirely between the incumbent and non-incumbent blocks, being −0.418 within the sixteen incumbent types and −0.083 within the sixteen non-incumbent ones.

What that leaves is narrower than the correlation suggests and still substantial. The largest single feature of Figure 1 — that incumbent types return roughly twice what non-incumbent types do — lies along exactly the dimension on which the prize is most mismeasured. We therefore do not read that gap as responsiveness. The variation *within* each block is not explained by κ and we have no account of it.

The headline is unaffected and for a reason worth stating, since it was not the reason the restriction was made. The headline column drops incumbents because their prize is built from the lagged vote share of a party that won here with them on the list, not because of any leave-out problem in the prize; that restriction also confines the

estimate to the block where κ is below one and the incumbency mismeasurement does not operate. κ is built from realised seats and so cannot be substituted into the model as a regressor — it is a diagnostic on what the existing estimates track, not a correction.

F.8. The attenuation formula against the simulation

This subsection supports Section 5.6.

The attenuation factor $\lambda = [\mathrm{Var}(\Delta P) - \mathrm{Var}(\text{noise})]/\mathrm{Var}(\Delta P)$ can be computed from the multinomial variance of the share without any simulation, and it predicts the bias closely *here*: at $I = 138$ over nine lists the simulation's λ is 0.600 and its recovery ratio is 0.595, agreeing to within 0.005 at the scale the application runs on. Across $I \in \{55, 86, 200, 500\}$ at six lists the pairs are $(0.27, 0.31)$, $(0.57, 0.55)$, $(0.82, 0.76)$ and $(0.94, 0.94)$.

Computed on the panel, the same formula gives λ of -0.09, 0.24, 0.32, 0.50, 0.70 and 0.89 across market-size bands; $\hat{\alpha}$ rises together with it over the first three bands — a pattern, not a threshold the data handed us, since no band-restricted test of those three bands was run. Above sixty candidates, fitting $\hat{\alpha}_b = \lambda_b \alpha$ through the origin on the four bands above sixty rejects proportionality on every banding (Q between 26.0 and 39.9 on 3 df across the four-band fits under both fixed-effect specifications, and $Q = 27.6$ on 2 df when the gate-stopped 150–299 band is dropped; $p < 10^{-5}$ throughout); $\hat{\alpha}$ sits above the proportional line in the smallest of these bands and below it in the largest, which is the opposite of what the correction below would require.

We do not apply the correction, for two reasons. The formula returns $\lambda = -0.09$ in the smallest bucket, which is impossible as a variance ratio and shows it overstates the noise where markets are small. And dividing by it over-corrects monotonically, $\hat{\alpha}/\lambda$ falling from 3.59 to 1.24 across the bands above sixty, over which $\hat{\alpha}$ does not move proportionally with λ rather than holding constant.

The closed-form factor is therefore not used as a correction. The split-sample estimator of Section 5.7 re-estimates and does not divide, and on the panel it moves the baseline coefficient by the ratio the simulation predicts, 0.63 against 0.60 (Section 6.4); the failure of proportionality across market-size bands under the closed-form factor is not explained by that result and remains open. The estimates of Section 6.4 are the split-sample ones.

F.9. Order conditions

For a market–type cell, let c_{clear} and c_{non} denote the numbers of clearing and non-clearing lists on which the type has a candidate. Such a cell supplies $c_{\text{clear}} + c_{\text{non}} - 1$ independent log-odds in all. The clearing block takes $c_{\text{clear}} - 1$ of them and the non-clearing block $c_{\text{non}} - 1$, so exactly *one* is left for the mixed block, whatever the two counts are: any other mixed pair is the sum of that one and two within-kind contrasts. Counting $\min\{c_{\text{clear}}, c_{\text{non}}\}$ mixed contrasts would double-count the block. On the vereador panel:

block	contribution	cells	contrasts
clearing × clearing	prize contrast for $\alpha(\tau)$	102,808	216,967
non-clearing × non-clearing	relative $\delta_{m,j}$	29,505	83,334
mixed	$X(\tau)'\beta$, given the list terms	56,073	56,073

Against a scalar α or a short vector $\{\alpha(\tau)\}$, an n_X-vector β, and the δ's, none of the three blocks is close to binding. The model indexes δ by list, which on this panel means the thirty-six parties the non-clearing lists reach; Section 6.4 estimates the finer party-by-state version, 809 of them, and 83,334 non-clearing contrasts leave that comfortable too.

62.3% of markets contain at least one list of each kind and so reach the mixed block; 35.0% hold a cell reaching all three, the binding requirement being two non-clearing lists in one cell rather than two anywhere. And 99.0% and 99.4% of municipalities in the two cycles have at least two clearing lists, which is what the block carrying α needs.

The connectedness Corollary 1 asks for holds on this panel with national party labels: the thirty-six parties reached by non-clearing lists form a single component, linked through the 29,505 cells that hold two or more of them. With the party-within-state labels of Section 6.4 it cannot hold across states, since no label is shared between them, and the invariance is then one common shift of the party effects per state.

The rank condition and the cross-type supports. The within-cell design QZ of Proposition 6, built as the estimator builds it — coalition lists with their composition weights, party-within-state labels, one δ fixed — has 1,534 columns and rank 1,509 on the markets of sixty or more candidates with a prize coefficient for each type, and 1,503 columns and rank 1,478 with a common coefficient. The rank is read from the eigenvalues of $Z'QZ$, with columns in their natural units: the largest is 1.7×10^5, the smallest counted as non-zero is 1.0, and the tolerance is 5.7×10^{-6}, so the count does not depend on where the tolerance is set. The null space is identified with a known set of directions and not only counted. For each state, take the vector that is one on every ξ and δ column of that state and zero elsewhere. For each of the 25 states other than the one holding the fixed δ, QZ annihilates the vector, with a squared residual below 10^{-14} of the columns' own mass; the 25 vectors have rank 25, which is the dimension of the null space; and the largest component of any null direction on an α or β column is below 10^{-13}. The same is found in each cycle taken alone. The estimator fixes one δ and imposes no convention on the other 25 directions: its iterations start at zero and take minimum-norm steps, which have no component in them. The reported party effects are that representative, and only differences of party effects within a state have an interpretation.

A market–type cell with candidates on c_{clear} clearing and c_{non} non-clearing lists supplies $c_{\text{clear}} + c_{\text{non}} - 1$ non-redundant empirical log-odds; when both counts are positive these can be arranged as $c_{\text{clear}} - 1$ within-clearing, $c_{\text{non}} - 1$ within-non-clearing and one mixed, and when one count is zero the cell supplies no mixed contrast. They are the comparisons that can be formed from non-empty sample cells, which is fewer than the model's population probabilities would support. On the same markets, 5,088 of 5,089 hold two clearing lists with different anticipated prize indices, and the graph linking types that are present together in such a market is connected; 4,251 of them also hold a non-clearing list, and the differences $X(\tau) - X(\tau')$ among types present together there have rank four. On the 2020 markets of sixty or more, where every list is one party, (D1) holds for all 32 types with at least 57 witnessing party pairs, (D3) holds for all 32, and $X(\tau)$ has rank four and does not span a constant.

Condition (D1) on the anticipated prize. The estimated design uses the anticipated prize of Section 6.3, so (D1) is checked on that quantity, with the clearing branch read as the headline estimates read it and a type counted as present on a list when it has a candidate there. On all markets, condition (D1) of Assumption C.1 holds for all 32 of the 32 types: for every type there are two markets and a party pair clearing in both whose prize gaps differ. The

median type has 470 witnessing party pairs, and the weakest has 25, with a best co-clearing count of 23 markets. On the markets of sixty candidates or more the condition again holds for all 32 types, but support is thin for two of them, which have 4 and 11 witnessing pairs; the median is again 470. Of the 525 party pairs that co-clear in two or more markets, 520 have differing prize gaps and five do not. Read at the realised prize seats/n_{list} instead, the condition also holds for all 32 types on both samples, with the same weakest count of 25 pairs.

Cells touching a single list of a given kind contribute nothing to that block. They are not discarded, since a cell with one clearing and one non-clearing list still contributes a mixed contrast, but the estimating sample differs across blocks and should be described that way.

F.10. What changed

	single market	panel of markets
unit of equilibrium	the nation	municipality × cycle
type	absorbs geography	demographic profile
winner set	$\arg\max_j \hat{s}_j$, a singleton	$\{j$: took a seat$\}$, observed
prize	$B_{j^\star}$ normalised to 1	anticipated seats per candidate
prize regressor	a constant column	varies across and within markets
outside option	required, constructed	**absent from every block**
loser payoff	normalised to zero	δ_p, estimated
$\alpha(\tau)$	not identified	identified per type
β	from the outside option	from mixed contrasts
estimating equations	one	three, stacked
observations	K types	217 + 83 + 56 thousand rows, ~5,500 clusters

The row that matters is the fifth. Everything the single-market estimand was criticised for — that $\hat{\alpha}\hat{B}$ equalled the average log-odds rescaled by the aligned share, and moved with the classification rule rather than with behaviour — followed from the prize entering as one number. It enters here as a variable, and the criticism does not transfer.

F.11. The prize is an average, and lists are not exchangeable

This subsection supports Section 6.3. The quantity we build is the list's seats over its candidates, which is what every member would face if personal votes were spread evenly. They are not: on this panel the candidates who take a clearing list's seats hold 44.5% of its votes while being 16.7% of its candidates, a ratio of 2.6 at the median and above two for 78.2% of clearing lists. Writing top_j for the share of list j's nominal votes held by the seats_j candidates who actually took its seats, and comparing it with the $\text{seats}_j/n_{m,j}$ that even spreading would give, the ratio has a median of 2.70 across the 47,678 clearing lists with a vote record, a tenth percentile of 1.74 and a ninetieth of 4.70.

Whether that breaks the estimate is a separate question from whether it is true, and it is answerable. Splitting markets of sixty or more candidates by the median ratio among their clearing lists:

markets of 60+, by within-list concentration	$\hat{\alpha}$	(s.e.)
all	1.9210	(0.0739)
least concentrated third	2.2672	(0.1318)
middle third	2.2623	(0.1348)
most concentrated third	1.8284	(0.1272)

The least concentrated third and the middle third give the same coefficient, 2.27 and 2.26, and the most concentrated third a lower one, 1.83. The least-to-most difference is −0.44 against a standard error on the difference of 0.18, treating the thirds as independent. The coefficient is lower where a list's seats are held by fewer of its candidates. That is the direction to expect from the approximation: where a few candidates hold most of a list's votes, the list's seats per candidate overstate the opportunity the rest of its candidates face, and they respond less to it. The gap is about a fifth of the estimate. The per-capita prize is a worse description of the institution than the model implies, and the estimate registers it.

One feature of the table is worth flagging rather than explaining away: two of the three thirds lie above the pooled estimate. The pooled coefficient is not an average of the three, and with party-by-state effects there are 872 of them, each estimated on fewer observations once the sample is split. That is a reason to read the *comparison* across the three and not their levels.

The test has a real limit and it is not the endogeneity of the splitting variable, which never enters the estimate. It is range. The tercile cuts fall at 2.89 and 3.55, so the comparison is between lists that are quite concentrated and lists that are very concentrated; *there is no exchangeable region in these data to compare against*. What the table shows is that the approximation moves the magnitude by about a fifth across the variation Brazil supplies and leaves the sign alone. It does not establish what a candidate-level prize would give, and we do not claim it.

Appendix G. A registered design for a cross-rule test

The estimate this paper reports is read off choices made under one payment rule. Whether it is a fact about the people choosing is a different question from whether it fits the cycle it was fitted to, and nothing inside a single rule regime separates those two questions. This section sets out a test that does, states in advance what would count as passing it, what would count as a failure worth reporting and what would count as no result at all, discloses where the text we lodged in advance falls short of the rule that was enacted, and says why this paper does not report the answer.

What the test asks. Whether the caring is a property of candidates or an artefact of the rule used to measure it. Proposition 2 recovers $\alpha(\tau)$ by comparing two paid lists inside one market, and that comparison exists because the rule pays several lists at once. A parameter identified by a feature of the payment rule may be a property of the people doing the choosing or a property of the rule doing the identifying. Separating them takes a second rule.

The design. Calibrate the model on one cycle's rule. Freeze the parameters and write the predictions down before any choice made under the second rule is read. Then score the frozen predictions against those choices. The freeze is what makes the exercise a test rather than a fit: parameters re-estimated on the cycle being scored answer a different question, and a prediction written after the outcome is read answers none.

Brazil supplies the second rule, because the rule for who competes for the leftover seats changed twice. In 2016, only lists above the quotient competed for them. Lei 13.488/2017 opened that round to every party that contested the election, and that is the rule in force in 2020; Section 6.1 sets out both of those rules and what the change does to the model's own branch. In 2024 the round was split in two. A party reaching 80% of the electoral quotient and holding an unelected candidate with 20% of the quotient competed in the greatest-averages round; when no party satisfied both conditions at once, the seats still unfilled went to a final round open to every party and every candidate, with the candidate bar not applied. We take that description from the enacted text of the electoral court's 2024 resolution in the official gazette (Res.-TSE 23.734/2024), whose amended leftover-seat provisions carry a constitutional-challenge cross-reference on their face. We characterise no holding: the applied rule is what the electoral administration was bound to execute, and it is the applied rule a scored prediction meets.

Which cycle is frozen, and why it has to be that one. The model pays a list when its vote share reaches the quotient; the enacted rule pays a list that takes a seat. The two rules disagree about 2 of 25,653 lists in 2016 and part company in 2020, where 4,691 of 40,117 lists — 11.7% — took a seat without reaching the quotient (Section 6.1). So the freeze cycle is 2016. The disagreement between the modeller's rule and the enacted rule runs to two lists in the whole registry there, against one list in eight in 2020, and a two-list disagreement bounds what a failure of the frozen prediction could charge to the rules at the point of calibration.

What counts as a pass, and what counts as a failure worth reporting. Two outcomes are results. If the frozen parameters predict the choices made under the scoring cycle's rule within the tolerance written down in advance, the caring travelled across a change in the rule that measured it, and it is a property of the people. If they do not, and the error has a pattern the design can describe, the exercise is a documented failure of a structural model across an enacted change in the payoff rule — an *informative* failure. Documented failures of that kind are scarce, in a literature where the exercise that could produce one — an ex ante forecast scored against an enacted reform — is itself uncommon (Pathak and Shi 2021). Scarcity is not the same as absence: a prediction scored after the fact can be rewritten, and a model refitted on the cycle being scored cannot fail. Writing the prediction down first is what makes the failure reportable, because it removes our own room to bury it.

Two failures that yield no result, and we name them before running it. It does not follow that any outcome is publishable. Two failure modes produce nothing, and naming them in advance is what keeps the design two-sided rather than self-serving.

The first is *uninformative* failure. The scales may not be comparable across the two regimes, so that no score can be computed at all; or a score is computed, the prediction is wrong, and the error has no pattern that any statement distinguishes from noise. Either way nothing is learned about the question, and what we would have to report is an attempt rather than a finding.

The second is *unattributable* failure, and this paper's own machinery is what creates it. The test varies Brazil's rule. There is a second rule in the exercise — ours, the one written into (9) — and it has never been measured against anything. If the frozen prediction fails, a failure caused by our winner rule diverging from the enacted allocation is not distinguishable, on the evidence this design collects, from a failure caused by behaviour. Freezing on 2016 narrows the channel, because the two rules disagree about only two lists there, so almost all of the divergence enters through the cycle being scored. It does not close the channel and we do not claim it

does. Bounding it is therefore not a matter of sharpening an estimate; it is what protects the informative-failure outcome, since a failure nobody can attribute is not a documented failure of anything.

What was lodged in advance does not match what was enacted, and the mismatch is printed here. A design lodged in advance is worth what its text says, and ours does not yet say the right thing. Its target for the scoring cycle is a quotient-only paid branch: a list is paid when the quotient step awards it a seat. The enacted rule pays below-quotient lists in two distinct places, and the conditions deciding which lists are paid sit largely outside our text. The candidate condition at the greatest-averages round is absent. The condition that carries the allocation on from that round to the final one is absent. The party-eligibility condition at the final round — every list that contested — is absent, and so is the candidate condition there, which the enacted text does not impose at all. And the procedure the enacted text supplies for a municipality in which no list reaches the quotient is absent; worse, the branch we did lodge returns *unpaid* for every list in exactly those municipalities, where the enacted rule seats a full council. Five absences — four conditions and a procedure — and one branch wrong where it is not missing.

We print the mismatch rather than amend the text and present the amended version as what was lodged. An amendment is what the mismatch needs and it is specified and on the record as owed. A pre-registration section that hides the distance between what was written and what was enacted is worse than no section at all, because it turns a device for disciplining the analyst into a device for reassuring the reader.

One question we cannot settle, converted into a check. Whether Brazilian constitutional electoral anteriority permitted a change decided in February 2024 to govern an election held that October is a question of doctrine and we do not answer it. What the electoral administration was bound to execute is a different object and that one is established: the resolution was published inside the statutory window for instructions applicable to the next election, and it is the instruction the electoral boards ran. The residual is that a rule we cannot formally exclude — the rule as it stood before the amendment — may award a different paid set.

The amendment converts that residual into a check rather than an argument. It registers, before any datum from the scoring cycle is touched, a statistic measuring how often the two candidate maps from votes to paid lists award a different paid set, computed on the 2016 and 2020 votes rather than on the cycle being scored. Whether the two maps disagree, and how often, is what that statistic is there to establish. And it registers the reading in advance. A specificity failure in the reproduction of the enacted allocation — our rule paying lists the allocation did not pay — with the disagreements concentrated on lists paid below the quotient, is met by reporting the branch under both maps and recording the comparison as the registered verdict, not by adopting whichever map the data prefer. A legal question we cannot settle becomes a comparison we can run, and what gets reported is settled before the answer is visible.

Why this paper does not report the result. The design is here and the result is not, and what withholds the result is the design's own logic rather than any gap in the design or in the argument for it. A frozen prediction is evidence in proportion to how firmly the freeze can be dated. A timestamp we put on a file of our own is a claim about our own conduct, and a reader has no way to audit it; a date a journal puts on a published design is a date neither author controls. Publishing the design first is therefore not a delay in reporting the test. It is the step that makes the test worth reporting.

Two things must also happen before the exam can be sat, and neither is a matter of waiting. The lodged text must be amended to name the enacted rule, for the reasons just given. And what this paper leaves open bears directly on whether a failure could be attributed: the room left by multiplicity (Section 6.7) and the distance between the model's winner rule and the enacted allocation are the two channels through which an unattributable failure would arrive, and a design that cannot bound them cannot deliver the informative-failure outcome. Section 7 states what this paper settles. What it does not settle is whether the number travels, and we would rather put that question with the terms of its answer already published.